ON DERIVED GALOIS DEFORMATION RINGS IN CHARACTERISTIC 0 AND A CONJECTURE OF VENKATESH

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ABSTRACT. Let Π be a regular algebraic cuspidal automorphic representation of GL_n over a CM field F, and ρ its associated Galois representation in characteristic 0. With mild assumptions, we use the Taylor–Wiles method to define a free action of the potentially semistable derived deformation ring of ρ on the Π-part of the p-adic cohomology of GL_n/F, confirming a prediction of Venkatesh. As a key step, we establish an equivalence between the derived deformation ring of ρ with the completion at the point corresponding to ρ of the derived deformation ring of the mod p reduction $\bar{\rho}$, generalising well-known results by Kisin.

CONTENTS

1. Introduction 2
Outline 5
Notation and conventions 5
Acknowledgements 6
2. Preliminaries 6
2.1. Locally symmetric spaces 6
2.2. Animated algebras 7
2.3. Bloch–Kato Selmer groups 8
2.4. Poitou–Tate duality and local conditions 9
3. Derived deformation theory 12
3.1. Artinian algebras 12
3.2. Integral models of artinian E-algebras 13
3.3. Topologies on animated algebras 17
3.4. Moduli stacks of Galois representations 18
3.5. Derived deformation functors 19
3.6. Tangent complexes and representability 22
3.7. Localisation at a characteristic 0 point 24
4. Patching and Venkatesh's conjecture 29
4.1. Taylor–Wiles primes 30
4.2. Patching 33
4.3. Tangent complex calculations 36
4.4. Proof of the main theorem 38
References 43

1. Introduction

Langlands philosophy posits a correspondence between suitable subclasses of automorphic representations and Galois representations. This correspondence is expected to be compatible with p-adic deformations in the sense that there should exist a homomorphism of rings $R \to T$ where R is a deformation ring parametrising p-adic deformations of a Galois representation and T is a p-adic Hecke algebra which acts on p-adic automorphic forms. Restriction of scalars along $R \to T$ defines an action of R on p-adic automorphic forms, and this article is concerned with defining a derived enhancement of this action.

To explain the motivation for this question, let us describe what we will mean by p-adic automorphic forms. Fix a prime p, a CM field F with maximal totally real subfield F^+, an isomorphism $\iota\colon \mathbb{C} \cong \overline{\mathbb{Q}}_p$ and a finite extension $E/\mathbb{Q}_p$ assumed large enough. Let $\mathbf{G} = \mathrm{GL}_n/F$ and suppose Π is a regular algebraic cuspidal automorphic representation of $\mathbf{G}(\mathbb{A}_F)$ of weight $\lambda \in (\mathbb{Z}^n_+)^{\mathrm{Hom}(F,E)}$. Fix an open compact subgroup $K \subset \mathbf{G}(\mathbb{A}_F^\infty)$ and consider the locally symmetric space X_K of level K, which is a smooth manifold of dimension $\dim X_K = [F^+\colon \mathbb{Q}]n^2 - 1$ equipped with a locally constant sheaf of E-vector spaces $\mathcal{V}_\lambda$ determined by the weight λ. The cohomology group $H_*(X_K, \mathcal{V}_\lambda)$ is a finite-dimensional E-vector space equipped with an action of Hecke operators and serves as our space of p-adic automorphic forms of level K and weight λ. The system of eigenvalues associated to Π defines an eigenspace

$$H_*(X_K, \mathcal{V}_\lambda)_\Pi \subset H_*(X_K, \mathcal{V}_\lambda),$$

which is non-zero in degrees $[q_0, q_0 + l_0]$, where $l_0 = [F^+\colon \mathbb{Q}]n - 1 \in \mathbb{N}$ is the defect of $\mathbf{G}$ and $2q_0 + l_0 = \dim X_K$. The dimension of the Π-eigenspace in degree $q_0 + i$ is $\binom{l_0}{i}$ times a fixed factor depending on the prime-to-p level K^p and not on i. This suggestive numerology underlies the following conjecture.

Conjecture. [PV21, Conj. 1.2.1] *$H_*(X_K, \mathcal{V}_\lambda)_\Pi$ is a free graded module over $\wedge^* V_\Pi$, where V_Π is the dual adjoint Bloch–Kato Selmer group attached to Π, a vector space of dimension l_0.*

Properly speaking, this is the p-adic realisation of a conjecture for cohomology with $\mathbb{Q}$-coefficients where the role of V_Π is played by a conjectural motivic cohomology group, whose p-adic realisation is V_Π. To explain how the dual adjoint Bloch–Kato Selmer group arises, recall that by work of Harris–Lan–Taylor–Thorne [HLTT16] and Scholze [Sch15], one can attach to Π a continuous Galois representation

$$\rho\colon \Gamma_F = \mathrm{Gal}(\overline{F}/F) \to \mathrm{GL}_n(E),$$

uniquely characterised up to isomorphism by its values at Frobenius elements matching the Hecke eigenvalues of Π in a precise way. This representation is ramified at a finite set of places S, and we work throughout under the assumption that the Weil–Deligne representations associated to the restrictions $\rho|_{\Gamma_{F_v}}$ are generic, or equivalently that the local characteristic 0 deformation rings are formally smooth E-algebras. To simplify the exposition in the introduction, we ignore the places in S not above p.

The representation ρ and its Hodge–Tate weights $\mathrm{HT}(\rho)$ define a (classical) deformation functor $D_\rho^{\mathrm{ss}}\colon \mathrm{Art}_E \to \mathrm{Set}$ where Art_E is the category of local artinian E-algebras, given by

$$B \mapsto \{r_B\colon \Gamma_{F,S} \to \mathrm{GL}_n(B) \mid r_B \equiv \rho \bmod \mathfrak{m}_B, r_B \text{ potentially semistable, } \mathrm{HT}(r_B) = \mathrm{HT}(\rho)\}/\sim$$

and $\sim$ denotes strict equivalence. When ρ is absolutely irreducible, D_ρ^{ss} is pro-represented by a complete noetherian local E-algebra R_ρ^{ss}, whose spectrum we view as an intersection

$$(\text{deformations of } \rho) \cap (\text{deformations of } \prod_{v\in S} \rho_v \text{ w. local conditions}) \subset (\text{ deformations of } \prod_{v\in S} \rho_v),$$

and which is pro-represented by the completed tensor product

$$R_\rho^{\mathrm{ss}} \cong R_\rho \hat{\otimes}_{R_\rho^{\mathrm{loc}}} R_\rho^{\mathrm{loc,ss}}.$$

By [All16, Prop. 1.3.12], the tangent space of R_ρ^{ss} is given by the Bloch–Kato Selmer group $H^1_g(\Gamma_{F,S}, \operatorname{ad}\rho)$, which is known to vanish (as the Bloch–Kato conjecture predicts) in many cases by the main result of [A'C24]. In other words, $\operatorname{Spec} R_\rho^{\mathrm{ss}}$ is a point. Now, tangent-obstruction theory predicts the dimension of R_ρ^{ss} to be given by an Euler characteristic which comes out to $-l_0 < 0$. This suggests that the intersection above is non-transverse and that R_ρ^{ss} underlies a 'derived object' which encodes this fact. A natural candidate is given by replacing $\otimes$ with $\otimes^{\mathbf{L}}$, but the resulting object only has good formal properties when the rings involved are complete intersections of the expected dimension, which is hard to verify for R_ρ.

To circumvent this issue, a more flexible framework has been developed by Galatius–Venkatesh [GV18], who construct a derived enhancement of classical deformation rings for mod p representations. Later, Zhu [Zhu21] gave a more general construction which also works in characteristic 0. In both setups, a *derived deformation ring* is an animated (or simplicial) ring whose underlying static ring is canonically isomorphic to a classical deformation ring, e.g. $\pi_0(\mathcal{R}_\rho) \cong R_\rho$ where $\mathcal{R}_\rho$ is the unrestricted derived deformation ring of ρ.

The benefit of the derived framework can be explained in terms of tangent complexes; to both $\mathcal{R}_\rho$ and R_ρ we can associate a complex of E-vector spaces, denoted $\mathrm{T}_E^\bullet(\mathcal{R}_\rho)$ and $\mathrm{T}_E^\bullet(R_\rho)$, respectively. For both rings, the 0th cohomology is Galois cohomology:

$$\mathrm{T}_E^0(\mathcal{R}_\rho) \cong \mathrm{T}_E^0(R_\rho) \cong H^1(\Gamma_{F,S}, \operatorname{ad}\rho).$$

However, whereas the cohomology groups $\mathrm{T}_E^i(R_\rho)$ for $i \geq 0$ lack a clear-cut description in terms of Galois cohomology, for the derived deformation ring we have

$$\mathrm{T}_E^i(\mathcal{R}_\rho) \cong H^{i+1}(\Gamma_{F,S}, \operatorname{ad}\rho)$$

for every i. In this sense, $\mathcal{R}_\rho$ is a more natural object to consider; in particular, this makes the derived framework more amenable to computations.

In this article, we introduce a derived version D_ρ^{ss} of the functor D_ρ^{ss} above and obtain a corresponding derived deformation ring $\mathcal{R}_\rho^{\mathrm{ss}}$ such that

$$\pi_0(\mathcal{R}_\rho^{\mathrm{ss}}) \cong R_\rho^{\mathrm{ss}}.$$

Whereas R_ρ^{ss} is predicted by the Bloch–Kato conjecture to be a field and therefore to have a trivial tangent complex $\mathrm{T}_E^\bullet(R_\rho^{\mathrm{ss}}) \simeq 0$, the derived deformation ring satisfies

$$\mathrm{T}_E^*(\mathcal{R}_\rho^{\mathrm{ss}}) = \mathrm{T}_E^1(\mathcal{R}_\rho^{\mathrm{ss}}) \cong H^1_g(\Gamma_{F,S}, \operatorname{ad}\rho(1))^\vee,$$

which is l_0-dimensional. In this way, the derived deformation ring witnesses obstructions not captured by its underlying classical deformation ring. Thus, in hope of explaining the dimension formula for the Π-contribution, we formulate two goals: to define a free graded action

$$\pi_*(\mathcal{R}_\rho^{\mathrm{ss}}) \circlearrowright H_*(X_K, \mathcal{V}_\lambda)_\Pi,$$

and to prove that $\pi_*(\mathcal{R}_\rho^{\mathrm{ss}})$ is an exterior algebra on $H^1_g(\Gamma_{F,S}, \operatorname{ad}\rho(1))$. The latter fact is Corollary 3.7.10 and follows from general properties of quasi-smooth animated rings and our calculation of $\mathrm{T}^*_E(\mathcal{R}_\rho^{\mathrm{ss}})$. Without further ado, let us state a simplified version of our main result.

Theorem 1.0.1 (Theorem 4.0.1)**.** *With notation as above, and under various assumptions on F and ρ, there exists a free graded action of $\pi_*(\mathcal{R}_\rho^{\mathrm{ss}})$ on $H_*(X_K, \mathcal{V}_\lambda)_\Pi$.*

The proof of Theorem 1.0.1 relies on the Taylor–Wiles method as applied to derived deformation rings by Galatius–Venkatesh. As usual, the method concerns mod p deformations, so to relate deformations of ρ to those of $\bar\rho$, we prove derived analogues of Kisin's widely referenced results [Kis09, Lem. 2.3.3, Prop. 2.3.5], which relate the two notions in the classical setting. For example, we obtain the following result as a consequence of Proposition 3.7.2 and Theorem 3.7.3, which are applicable in broad generality.

Corollary 1.0.2. *Let $\mathcal{R}_{\bar\rho} \in \mathcal{CNL}_k$ and $\mathcal{R}_\rho \in \mathcal{CNL}_E$ denote the derived deformation rings of $\bar\rho$ and ρ, respectively, and let $\xi\colon \mathcal{R}_{\bar\rho} \to \mathcal{O}$ be a point corresponding to an $\mathcal{O}$-lattice $\xi \subset \rho$. Then the derived completion*

$$(\mathcal{R}_{\bar\rho} \otimes_{\mathcal{O}} E)^\wedge_\xi \coloneqq \lim_r (\mathcal{R}_{\bar\rho} \otimes_{\mathcal{O}} E)/^{\mathbf{L}} I_\xi^r$$

of $\mathcal{R}_{\bar\rho} \otimes_{\mathcal{O}} E$ at $I_\xi = \ker(\pi_0(\mathcal{R}_{\bar\rho}) \otimes_{\mathcal{O}} E \to E)$ is equivalent to $\mathcal{R}_\rho$.

Using the same results, and under our technical assumptions, we prove an equivalence

$$\mathcal{R}_\rho^{\mathrm{ss}} \simeq (\mathcal{R}_{\mathbf{S}} \otimes_{\mathcal{O}} E)^\wedge_\xi,$$

where $\mathcal{R}_{\mathbf{S}}$ represents a mod p deformation problem $\mathbf{S}$ defined using Kisin's potentially semistable deformation rings (see Corollary 3.7.9). This equivalence allows us to carry out patching arguments in the mod p setting and then pass to characteristic 0.

In [GV18], a free action of the crystalline version of $\pi_*(\mathcal{R}_{\mathbf{S}})$ on the *integral* cohomology is defined, in the Fontaine–Laffaille setting and assuming the Calegari–Geraghty vanishing conjecture for mod p cohomology [CG18, Conj. B]. Inverting p then yields the desired action. In a general setting, this strategy is not applicable, and the naive generalisation of the integral result is not expected to always hold. Our construction of the characteristic 0 action takes a modified approach which requires milder assumptions.

Let us describe in slightly simplified terms the Taylor–Wiles argument involved in proving Theorem 1.0.1. The first step, following [GV18], is to approximate the mod p deformation ring $\mathcal{R} \coloneqq \mathcal{R}_{\mathbf{S}}$ by a derived tensor product of static rings using the Taylor–Wiles method. For every $N \geq 1$, one has an augmented deformation problem $\mathbf{S}_N$ of Taylor–Wiles level N and a corresponding derived deformation ring $\mathcal{R}_N \coloneqq \mathcal{R}_{\mathbf{S}_N}$ with underlying static ring $R_N \coloneqq \pi_0(\mathcal{R}_N)$. Descending from Taylor–Wiles level yields a weak equivalence

$$\mathcal{R} \simeq \mathcal{R}_N \otimes^{\mathbf{L}}_{S_N} \mathcal{O},$$

where S_N is the $\mathcal{O}$-group ring of a finite p-group. Via this equivalence, we construct a morphism

$$\mathcal{R} \simeq \mathcal{R}_N \otimes^{\mathbf{L}}_{S_N} \mathcal{O} \to R_N \otimes^{\mathbf{L}}_{S_N} \mathcal{O} \to \overline{R}_N \otimes^{\mathbf{L}}_{\overline{S}_N} \mathcal{O}_N =: \mathcal{C}_N$$

where $R_N \twoheadrightarrow \overline{R}_N$ and $S_N \twoheadrightarrow \overline{S}_N$ are artinian quotients and $\mathcal{O}_N \coloneqq \mathcal{O}/\varpi^N$. We think of $\mathcal{C}_N$ as a 'mod ϖ^N approximation' of $\mathcal{R}$. A compactness argument allows us to extract a compatible system of approximations, yielding a map

$$g\colon \mathcal{R} \to \mathcal{C}_\infty = \lim_N \mathcal{C}_N$$

for some sequence of N. This induces a morphism on localised and completed rings

$$(\mathcal{R}_{\mathbf{S}} \otimes_{\mathcal{O}} E)^{\wedge}_{\xi} \simeq \mathcal{R}^{\mathrm{ss}}_{\rho} \to (\mathcal{C}_{\infty} \otimes_{\mathcal{O}} E)^{\wedge}_{\xi}.$$

We prove that this morphism is an equivalence by comparing tangent complexes. Carrying out this non-constructive argument while retaining sufficient control over the induced morphism of tangent complexes is the main technical difficulty in the proof. Applying an '$R = T$' theorem from [A'C24], we identify $(\mathcal{C}_{\infty} \otimes_{\mathcal{O}} E)^{\wedge}_{\xi}$ with a derived tensor product of power series rings:

$$(\mathcal{C}_{\infty} \otimes_{\mathcal{O}} E)^{\wedge}_{\xi} \simeq E[[X_1, \dots, X_s]] \otimes^{\mathbf{L}}_{E[[X_1,\dots,X_{s+l_0}]]} E$$

A'Campo proves that $E[[X_1, \dots, s]]$ acts freely on a patched homology module $(M_{\infty})^{\wedge}_{\xi}$, and descending from Taylor–Wiles level, we obtain a free action

$$\pi_*(\mathcal{R}^{\mathrm{ss}}_{\rho}) \circlearrowleft (M_{\infty})^{\wedge}_{\xi} \otimes^{\mathbf{L}}_{E[[X_1,\dots,X_{s+l_0}]]} E \cong H_*(X_K, \mathcal{V}_{\lambda})_{\Pi}.$$

A priori, this action could depend on the non-canonical choice made in defining the morphism g above. In [GV18], this possibility is ruled out by establishing a type of compatibility with the action of the derived Hecke algebra introduced in [Ven19]. Thus, so far the naturality part of Conjecture 1 is only known in the setting of [GV18], though we expect it to hold in general.

Outline. Let us outline the content of the sections below. In Section 2, we recall generalities concerning the cohomology of locally symmetric spaces and animated rings. Section 3 discusses the derived deformation theory of Galois representations: we define various derived deformation functors and prove all their necessary properties. In particular, Section 3.7 concerns the link between deformations of a characteristic 0 representation and its mod p reduction. In Section 4, we use the patching and the results of the preceding sections to prove the main result, Theorem 4.0.1.

Notation and conventions. The following notation will be used throughout the article. F is a CM field with ring of integers $\mathcal{O}_F$. When S is a set of places of F, we use the notation S_p for $\{v \in S \colon v \mid p\}$, and the completion of $\mathcal{O}_F$ at a place v is $\mathcal{O}_{F_v}$. Our field of coefficients is denoted E and is a finite extension of $\mathbb{Q}_p$ with ring of integers $\mathcal{O}$, uniformiser ϖ and residue field $k := \mathcal{O}/\varpi$. The extension E is tacitly assumed to be large enough, so that all of our Galois representations are defined over E and E contains all embeddings $F \hookrightarrow \overline{\mathbb{Q}}_p$. For any $r \geq 1$, we let $\mathcal{O}_r := \mathcal{O}/\varpi^r$. Galois groups are denoted by Γ, e.g. $\Gamma_{F,S} = \mathrm{Gal}(\overline{F}_S/F)$ and $\Gamma_{F_v} = \mathrm{Gal}(\overline{F}_v/F_v)$. When $\bar{\rho}$ or ρ is a representation of a global Galois group such as $\Gamma_{F,S}$, we denote restrictions by subscripts, e.g. $\bar{\rho}_v = \bar{\rho}|_{\Gamma_{F_v}}$.

This article uses the language of quasicategories (from now on, ∞-categories). Generally, we use calligraphic letters to denote ∞-categories, so that e.g. $\mathcal{A}lg_{\mathcal{O}/}$ is the animation of the category $\mathrm{Alg}_{\mathcal{O}}$ of $\mathcal{O}$-algebras (see Section 2.2). When we are interested in both an animated object and its underlying static object, we sometimes distinguish them by fonts; this is always explicitly declared at some point in the text, and if so then e.g. $R = \pi_0(\mathcal{R})$. Unless otherwise specified, limits and colimits are taken in the ∞-categorical sense. We use the symbol $\varprojlim$ to denote the 1-categorical inverse limit whereas the undecorated $\lim$ always denotes a limit in the sense of ∞-categories. We make no typographic distinction between $\varinjlim$ and colim; their conflation is harmless as they yield the same object. Finally, we recall a couple of constructions. If Y is an object of an ∞-category $\mathcal{C}$, we denote by $\mathcal{C}_{/Y}$ the over-∞-category with objects $(X \to Y)$ [Lur09, §1.2.9]. Given an ∞-category $\mathcal{C}$ and a full subcategory $h'\mathcal{C} \hookrightarrow h\mathcal{C}$ of its homotopy category, the pullback of $N(h'\mathcal{C}) \to N(h\mathcal{C}) \leftarrow \mathcal{C}$, where $N(\cdot)$ denotes the nerve, is the ∞-subcategory of $\mathcal{C}$ spanned by the vertices lying in $h'\mathcal{C}$ [Cis19, Ex. 3.9.5].

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2. Preliminaries

The goal of this section is to describe our general setup and collect some useful results from the literature for easy reference.

2.1. **Locally symmetric spaces.** Let $\mathbf{G} = \mathrm{GL}_n /F$. We begin by recalling how automorphic representations of $\mathbf{G}(\mathbb{A}_F)$ contribute to the cohomology of locally symmetric spaces, following the exposition in [Hev24, §2.1]. We let $X^{\mathbf{G}}$ be the symmetric space of $\mathrm{Res}_{F/\mathbb{Q}}\,\mathbf{G}$ as in [BS73, §2], which is a homogeneous $\mathbf{G}(F\otimes_{\mathbb{Q}}\mathbb{R})$-space.

An open compact subgroup $K\subset\mathbf{G}(\mathbb{A}_F^\infty)$ is *good* if $K=\prod_v K_v$ with $K_v\subseteq\mathbf{G}(\mathcal{O}_{F_v})$ and K is neat. If S is a finite set of places of F, we write $K=K^S\times K_S\coloneqq\prod_{v\in S}K_v\times\prod_{v\notin S}K_v$. Given such a K, we define the locally symmetric space of level K as

$$X_K=\mathbf{G}(F)\backslash(X^{\mathbf{G}}\times\mathbf{G}(\mathbb{A}_F^\infty)/K),$$

a smooth manifold of dimension $[F^+\colon\mathbb{Q}](n^2-1)$. The defect of $\mathbf{G}$ is the positive integer

$$l_0\coloneqq[F^+\colon\mathbb{Q}]n-1.$$

Let $\lambda=(\lambda_\iota)_{\iota\in\mathrm{Hom}(F,E)}\in(\mathbb{Z}^n_+)^{\mathrm{Hom}(F,E)}$ be a dominant weight and suppose we are given representations $\tau_v\colon I_{F_v}\to\mathrm{GL}_n(E)$ with open kernel for $v\mid p$. Then [SZ99] associate to each τ_v an irreducible smooth $\mathcal{O}$-representation $\sigma^\circ(\tau_v)$ of $\mathrm{GL}_n(\mathcal{O}_{F_v})$, and we let $\sigma^\circ(\tau)=\bigotimes_{v|p}\sigma^\circ(\tau_v)$. From the pair (λ,τ) we then obtain a locally constant sheaf of $\mathcal{O}$-modules $\mathcal{V}^\circ_{\lambda,\tau}=\mathcal{V}^\circ_\lambda\otimes\sigma^\circ(\tau)\in\mathrm{Sh}(X_K)$ where $\mathcal{V}^\circ_\lambda$ is an $\mathcal{O}$-lattice in the highest weight representation of $(\mathrm{GL}_n)_E^{\mathrm{Hom}(F,E)}$ of highest weight λ. We define $\sigma(\tau)\coloneqq\sigma^\circ(\tau)\otimes_{\mathcal{O}}E$ and $\mathcal{V}_{\lambda,\tau}\coloneqq\mathcal{V}^\circ_{\lambda,\tau}\otimes_{\mathcal{O}}E$. The complex

$$R\Gamma(X_K,\mathcal{V}^\circ_{\lambda,\tau})\in\mathcal{D}(\mathcal{O})$$

carries an action of the *abstract Hecke algebra* $\mathbb{T}^S\coloneqq\mathcal{H}(\mathbf{G}(\mathbf{A}_F^{\infty,S}),K^S)$ where S is a finite set of places such that $K_v=\mathrm{GL}_n(\mathcal{O}_{F_v})$ for $v\notin S$. If Π is a regular algebraic cuspidal representation of $\mathbf{G}$, we let $\mathfrak{p}_{\Pi,\iota}=\ker(\mathbb{T}^S\to\mathrm{End}_{\mathbb{C}}(\otimes'_{v\notin S}\Pi_v^{K_v}))$ and define the Π-part as the localisation

$$H_*(X_K,\mathcal{V}_{\lambda,\tau})_\Pi\coloneqq H_*(X_K,\mathcal{V}_{\lambda,\tau})_{\mathfrak{p}_{\Pi,\iota}},$$

i.e. the eigenspace (over E) corresponding to the system of Hecke eigenvalues of Π. We write

$$\mathbb{T}^S\big(R\Gamma(X_K,\mathcal{V}^\circ_{\lambda,\tau})\big)\subset\mathrm{End}_{\mathcal{D}(\mathcal{O})}\big(R\Gamma(X_K,\mathcal{V}^\circ_{\lambda,\tau})\big)$$

for the corresponding faithful Hecke algebra. There exists an open compact subgroup $K_\tau\subset K$ such that

$$R\Gamma(X_K,\mathcal{V}_{\lambda,\tau})_\Pi=R\Gamma\big(K/K_\tau,R\Gamma(X_{K_\tau},\mathcal{V}_\lambda)\otimes_E\sigma(\tau)\big),$$

and since taking invariants is an exact functor in characteristic 0, we have

$$H_*(X_K,\mathcal{V}_{\lambda,\tau})_\Pi\cong\big(H_*(X_{K_\tau},\mathcal{V}_\lambda)_\Pi\big)_{K/K_\tau}.$$

Proposition 2.1.1. [ACC+23, Thm. 2.4.10] *Let $2q_0 = \dim X_K - l_0$, fix a regular algebraic cuspidal representation Π of weight λ and inertial type $\tau = (\tau_v)_{v|p} = (\mathrm{LL}(\Pi_v)|_{I_{F_v}})_{v|p}$. Let m_Π denote $\frac{1}{\dim \sigma(\tau)} \dim_{\mathbb{C}}(\Pi^\infty)^{K^p}$. Then $H_i(X_K, \mathcal{V}_{\lambda,\tau})_\Pi = 0$ for $i \notin [q_0, q_0 + l_0]$ and for $i = 0, \dots, l_0$ we have*

$$\dim H_{q_0+i}(X_K, \mathcal{V}_{\lambda,\tau})_\Pi = m_\Pi \binom{l_0}{i}.$$

2.2. **Animated algebras.** We begin by recalling the definition of animated algebras and its relation to simplicial rings, following the presentation in [ČS24, §5.1]. Let $\mathrm{Alg}_{\mathcal{O}}$ denote the ordinary category of commutative $\mathcal{O}$-algebras and let

$$\mathcal{A}lg_{\mathcal{O}/} \coloneqq \mathcal{A}ni(\mathrm{Alg}_{\mathcal{O}})$$

denote the ∞-category of *animated $\mathcal{O}$-algebras*, i.e. the ∞-category generated under sifted colimits by the subcategory $\mathrm{Alg}_{\mathcal{O}}^{\mathrm{ffg}} \subset \mathrm{Alg}_{\mathcal{O}}$ of finite free $\mathcal{O}$-algebras; it is the ∞-category underlying the simplicial model category $\mathrm{Fun}(\Delta^{\mathrm{op}}, \mathrm{Alg}_{\mathcal{O}})$ of simplicial $\mathcal{O}$-algebras [Lur09, Cor. 5.5.9.3]. If A is an animated $\mathcal{O}$-algebra, an *A-module* is a module over the underlying $\mathbb{E}_1$-ring of A (see [Lur04, §2.4]), whereas an *animated A-module* is a connective such module. When $A = \pi_0(A)$ is static, the category

$$\mathcal{A}ni(\mathrm{Mod}_A)$$

of animated A-modules is equivalent to the derived category $D^{\geq 0}(A)$ of connective complexes of A-modules, which is equivalent to the localisation of the simplicial model category $\mathrm{Fun}(\Delta^{\mathrm{op}}, \mathrm{Mod}_A)$ of simplicial A-modules at weak equivalences.

Definition 2.2.1. [Lur04, Def. 2.5.9] An animated $\mathcal{O}$-algebra $A \in \mathcal{A}lg_{\mathcal{O}/}$ is:

(1) *Noetherian* if $\pi_0(A)$ is noetherian and $\pi_i(A)$ is a finitely generated $\pi_0(A)$-module for every $i \geq 0$.
(2) *Local* if $\pi_0(A)$ is local.

The content of the following proposition is found immediately below [Lur04, Rem. 3.1.4].

Proposition 2.2.2. [Lur04] *Let $A \in \mathcal{A}lg_{\mathcal{O}/}$ and suppose B lies in the smallest subcategory of $\mathcal{A}lg_{A/}$ containing $A[X]$ which is closed under finite colimits. Then $\mathrm{Map}_A(B, -)$ commutes with filtered colimits.*

If the assumption of Proposition 2.2.2 holds, we say that B is a *finitely presented A-algebra.*

Proposition 2.2.3. [Lur04, Prop. 3.1.5] *Let $A \in \mathcal{A}lg_{\mathcal{O}/}$ be noetherian, $j \geq 0$ and suppose B is an A-algebra. The following are equivalent:*

(1) *The functor $\mathrm{Map}_A(B, -)$ commutes with filtered colimits when restricted to the subcategory $\tau_{\leq j}(\mathcal{A}lg_A)$ of j-truncated A-algebras.*
(2) *The A-algebra $\tau_{\leq j}B$ is noetherian and $\pi_0(B)$ is a finitely presented $\pi_0(A)$-algebra.*

If the equivalent conditions in Proposition 2.2.3 hold, we say that B is *of finite presentation to order j as an A-algebra*. If the conditions hold for every $j \geq 0$, we say that B is *almost of finite presentation over A*. Note that if A is a static noetherian ring and $A \twoheadrightarrow B$, then B is almost of finite presentation over A by (2) above.

Given an animated $\mathcal{O}$-algebra $A \in \mathcal{A}lg_{\mathcal{O}/}$, an animated A-module $M \in \mathcal{M}od_{\mathcal{O}}$ and an element $a \colon \{*\} \to A$, we obtain a map $\mathcal{O}[t] \to A$ and define

$$M/^{\mathbf{L}}a \coloneqq M \otimes^{\mathbf{L}}_{\mathcal{O}[t]} \mathcal{O},$$

where the map $\mathcal{O}[t] \to \mathcal{O}$ corresponds to $0 \in \mathcal{O}$. More generally, given $a_1, \dots, a_m \in A$, we define

$$M/^{\mathbf{L}}(a_1, \dots, a_m) \coloneqq M \otimes^{\mathbf{L}}_{\mathcal{O}[t_1,\dots,t_m]} \mathcal{O} \simeq M/^{\mathbf{L}}a_1/^{\mathbf{L}} \dots /^{\mathbf{L}}a_m.$$

There are natural maps

$$M/^{\mathbf{L}}(a_1^{r+1}, \dots, a_m^{r+1}) \to M/^{\mathbf{L}}(a_1^r, \dots, a_m^r)$$

forming an inverse system of animated $\mathcal{O}$-modules. If $M \in \mathcal{A}lg_{\mathcal{O}/}$, this is an inverse system of animated $\mathcal{O}$-algebras and the limit depends only on the closed subspace of $\operatorname{Spec} \pi_0(A)$ defined by $(a_1, \dots, a_m)$, as the following result shows.

Proposition 2.2.4. [Lur04, Prop. 6.1.1] *Let $A \in \mathcal{A}lg_{\mathcal{O}/}$ and let $I = (a_1, \dots, a_m) \subset \pi_0(A)$ be a finitely generated ideal. Then, for any $B \in \mathcal{A}lg_{\mathcal{O}/}$, the natural map*

$$\operatorname*{colim}_r \operatorname{Map}_{\mathcal{A}lg_{\mathcal{O}/}}(A/^{\mathbf{L}}(a_1^r, \dots a_m^r), B) \to \operatorname{Map}_{\mathcal{A}lg_{\mathcal{O}/}}(A, B)$$

identifies the left-hand side with the subspace spanned by morphisms $f\colon A \to B$ such that the induced morphism $\pi_0(A) \to \pi_0(B)$ factors through $\pi_0(A)/I^N$ for some $N \geq 1$.

Definition 2.2.5. [ČS24, 5.6.1] Let $A \in \mathcal{A}lg_{\mathcal{O}/}$ be noetherian and let $I = (a_1, \dots, a_m) \subset \pi_0(A)$. Then A is *derived I-adically complete* if the natural map $A \to \lim_r A/^{\mathbf{L}}(a_1^r, \dots, a_m^r)$ is a weak equivalence.

Definition 2.2.6. A local noetherian animated $\mathcal{O}$-algebra A is *complete* if A is derived $\mathfrak{m}_{\pi_0(A)}$-adically complete. We let $\mathcal{CNL}_k \subset (\mathcal{A}lg_{\mathcal{O}/})_{/k}$ and $\mathcal{CNL}_E \subset (\mathcal{A}lg_{\mathcal{O}/})_{/E}$ denote the full subcategories of complete noetherian local animated $\mathcal{O}$-algebras with residue field $\pi_0(A)/\mathfrak{m}_{\pi_0(A)}$ equal to k and E, respectively.

Lemma 2.2.7. [BS15, Lem. 3.4.13] *Let $A \in \mathcal{M}od_{\mathcal{O}}$ and suppose $\pi_i(A)$ is ϖ-adically complete in the usual sense for every $i \geq 0$. Then A is derived ϖ-adically complete.*

2.3. **Bloch–Kato Selmer groups.** In this section, we recall the definition of the Bloch–Kato Selmer group mentioned in the introduction, and prove a dimension formula. We also define a complex whose cohomology is given by Bloch–Kato Selmer groups, which will be related to derived deformation rings in Corollary 3.7.9. See [Mil86] for a complete reference on Galois cohomology.

Let S be a finite set of places of F containing S_p and suppose $\rho\colon \Gamma_{F,S} \to \mathrm{GL}_n(E)$ is a continuous representation which is de Rham at all places above p. Let $v \in S$. If $v \mid p$, then

$$H^1_g(\Gamma_{F_v}, \operatorname{ad}\rho) \coloneqq \ker\left(H^1(\Gamma_{F_v}, \operatorname{ad}\rho) \to H^1(\Gamma_{F_v}, \operatorname{ad}\rho \otimes_{\mathbb{Q}_p} B_{\mathrm{dR}})\right),$$

where the map on the right is the natural one induced by the tensor product and B_{dR} is Fontaine's ring of periods. If $v \nmid p$, let $I_{F_v} \subset \Gamma_{F_v}$ denote the inertia subgroup and set

$$H^1_g(\Gamma_{F_v}, \operatorname{ad}\rho) \coloneqq \ker\left(H^1(\Gamma_{F_v}, \operatorname{ad}\rho) \to H^1(I_{F_v}, \operatorname{ad}\rho)\right).$$

where the map on the right is the one induced by the inclusion $I_{F_v} \subset \Gamma_{F_v}$. The (geometric adjoint) *Bloch–Kato Selmer group* of ρ is the space of global classes with restrictions lying in the groups defined above, i.e.

$$H^1_g(\Gamma_{F,S}, \operatorname{ad}\rho) \coloneqq \ker\left(H^1(\Gamma_{F,S}, \operatorname{ad}\rho) \to \prod_{v \in S} \frac{H^1(\Gamma_{F_v}, \operatorname{ad}\rho)}{H^1_g(\Gamma_{F_v}, \operatorname{ad}\rho)}\right).$$

We use lowercase letters to denote the dimension of cohomology groups or subspaces thereof, e.g.

$$h^1_g(\Gamma_{F,S}, \operatorname{ad}\rho) \coloneqq \dim_E H^1_g(\Gamma_{F,S}, \operatorname{ad}\rho).$$

Proposition 2.3.1. *Let $\rho\colon \Gamma_{F,S} \to \mathrm{GL}_n(E)$ be a continuous representation, such that:*

(1) *ρ is absolutely irreducible.*
(2) *For every $v \in S$, the Weil-Deligne representation $\mathrm{WD}(\rho_v)$ is generic.*
(3) *For every $v \in S_p$, ρ_v is potentially semistable with distinct Hodge-Tate weights.*

Then, with $l_0 = [F^+\colon \mathbb{Q}]n - 1$,

$$h^1_g(\Gamma_{F,S}, \mathrm{ad}\,\rho) = h^1_g(\Gamma_{F,S}, \mathrm{ad}\,\rho(1)) - l_0.$$

Proof. By the Greenberg–Wiles formula,

$$h^1_g(\Gamma_{F,S}, \mathrm{ad}\,\rho) - h^1_g(\Gamma_{F,S}, \mathrm{ad}\,\rho(1)) =$$
$$\sum_{v\in S} \big(h^1_g(\Gamma_{F_v}, \mathrm{ad}\,\rho) - h^0(\Gamma_{F_v}, \mathrm{ad}\,\rho)\big) + h^0(\Gamma_{F,S}, \mathrm{ad}\,\rho) - h^0(\Gamma_{F,S}, \mathrm{ad}\,\rho(1)) - \sum_{v|\infty} h^0(\Gamma_{F_v}, \mathrm{ad}\,\rho).$$

Assumption (1) implies $h^0(\Gamma_{F,S}, \mathrm{ad}\,\rho) = 1$ and $h^0(\Gamma_{F,S}, \mathrm{ad}\,\rho(1)) = 0$, and assumption (2) implies that the local terms coming from $v \in S \setminus S_p$ vanish. Thus, we have

$$h^1_g(\Gamma_{F,S}, \mathrm{ad}\,\rho) - h^1_g(\Gamma_{F,S}, \mathrm{ad}\,\rho(1)) = \sum_{v\in S_p} \big(h^1_g(\Gamma_{F_v}, \mathrm{ad}\,\rho) - h^0(\Gamma_{F_v}, \mathrm{ad}\,\rho)\big) + 1 - [F^+\colon \mathbb{Q}]n^2.$$

Now, when $\mathrm{WD}(\rho_v)$ is generic, the sum over $v \in S_p$ above equals the sum of total multiplicities of the negative Hodge-Tate weights of $\mathrm{ad}\,\rho$ [BK90, Cor. 3.8.4], and by assumption (3) this number is $[F^+\colon \mathbb{Q}]n(n-1)/2$. Since $l_0 = [F^+\colon \mathbb{Q}]n - 1$, the result follows. □

2.4. **Poitou–Tate duality and local conditions.** In this section, we construct complexes with local conditions whose cohomology are given by Bloch–Kato Selmer groups. These will be used later on to relate Bloch–Kato Selmer groups to derived deformation rings. In the mod p setting, such complexes are constructed in [GV18, §B]. In fact, the authors give a more general construction than we shall describe here; we restrict to the case where the local conditions come from a Selmer system (in degree 1). The arguments are essentially the same as those of *loc.cit.*; we include details for the reader's convenience.

Throughout this section, we assume that $\rho\colon \Gamma_{F,S} \to \mathrm{GL}_n(E)$ is potentially semistable at all places above p. We let $C^\bullet(\Gamma_{F,S}, \mathrm{ad}\,\rho)$ denote the usual complex of continuous cochains, whose cohomology is $H^*(\Gamma_{F,S}, \mathrm{ad}\,\rho)$, and similarly for Γ_{F_v} and $\mathrm{ad}\,\rho(1)$. Given a Selmer system $\mathcal{L} = (\mathcal{L}_v)_{v\in S}$, i.e. a collection of subspaces

$$\mathcal{L}_v \subseteq H^1(\Gamma_{F_v}, \mathrm{ad}\,\rho),$$

we let $\mathcal{L}_v^\perp \subseteq H^1(\Gamma_{F_v}, \mathrm{ad}\,\rho(1))$ be the orthogonal complement of $\mathcal{L}_v$ with respect to the cup product, and $\mathcal{L}^\perp = (\mathcal{L}_v^\perp)_{v\in S}$. As in [GV18, §B.3], the Selmer system $\mathcal{L}$ defines subcomplexes

$$C^\bullet_{\mathcal{L}_v}(\Gamma_{F_v}, \mathrm{ad}\,\rho) \subseteq C^\bullet(\Gamma_{F_v}, \mathrm{ad}\,\rho)$$

for $v \in S$, where

- $C^0_{\mathcal{L}_v}(\Gamma_{F_v}, \mathrm{ad}\,\rho) = C^0(\Gamma_{F_v}, \mathrm{ad}\,\rho)$.
- $C^1_{\mathcal{L}_v}(\Gamma_{F_v}, \mathrm{ad}\,\rho) = \ker\big(Z^1(\Gamma_{F_v}, \mathrm{ad}\,\rho) \to H^1(\Gamma_{F_v}, \mathrm{ad}\,\rho)/\mathcal{L}_v\big)$.
- $C^j_{\mathcal{L}_v}(\Gamma_{F_v}, \mathrm{ad}\,\rho) = 0$, for $j \geq 2$.

Here, $Z^1(\Gamma_{F_v}, \operatorname{ad}\rho) \subset C^1(\Gamma_{F_v}, \operatorname{ad}\rho)$ denotes the cocycles. We define

$$C^\bullet_{\mathcal{L}}(\Gamma_{F,S}, \operatorname{ad}\rho) = \operatorname{fib}\left(C^\bullet(\Gamma_{F,S}, \operatorname{ad}\rho) \to \bigoplus_{v\in S} \frac{C^\bullet(\Gamma_{F_v}, \operatorname{ad}\rho)}{C^\bullet_{\mathcal{L}_v}(\Gamma_{F_v}, \operatorname{ad}\rho)} \right),$$

i.e. the cone given in degree i by

$$C^i_{\mathcal{L}}(\Gamma_{F,S}, \operatorname{ad}\rho) = C^i(\Gamma_{F,S}, \operatorname{ad}\rho) \oplus \bigoplus_{v\in S} \frac{C^{i-1}(\Gamma_{F_v}, \operatorname{ad}\rho)}{C^{i-1}_{\mathcal{L}_v}(\Gamma_{F_v}, \operatorname{ad}\rho)}$$

whose cohomology we denote by

$$H^*_{\mathcal{L}}(\Gamma_{F,S}, \operatorname{ad}\rho) = H^*\big(C^\bullet_{\mathcal{L}}(\Gamma_{F,S}, \operatorname{ad}\rho)\big).$$

Note that the degree-wise quotient $C^\bullet(\Gamma_{F_v}, \operatorname{ad}\rho)/C^\bullet_{\mathcal{L}_v}(\Gamma_{F_v}, \operatorname{ad}\rho)$ is the complex

$$0 \to C^1(\Gamma_{F_v}, \operatorname{ad}\rho)/C^1_{\mathcal{L}_v}(\Gamma_{F_v}, \operatorname{ad}\rho) \to C^2(\Gamma_{F_v}, \operatorname{ad}\rho) \to \dots,$$

whose cohomology groups are given by

$$0, \quad H^1(\Gamma_{F_v}, \operatorname{ad}\rho)/\mathcal{L}_v, \quad H^2(\Gamma_{F_v}, \operatorname{ad}\rho).$$

It follows that we have a long exact sequence

$$\begin{aligned}
0 &\longrightarrow H^0_{\mathcal{L}}(\Gamma_{F,S}, \operatorname{ad}\rho) \longrightarrow H^0(\Gamma_{F,S}, \operatorname{ad}\rho) \longrightarrow 0 \\
&\longrightarrow H^1_{\mathcal{L}}(\Gamma_{F,S}, \operatorname{ad}\rho) \longrightarrow H^1(\Gamma_{F,S}, \operatorname{ad}\rho) \longrightarrow \textstyle\bigoplus_{v\in S} H^1(\Gamma_{F_v}, \operatorname{ad}\rho)/\mathcal{L}_v \\
&\longrightarrow H^2_{\mathcal{L}}(\Gamma_{F,S}, \operatorname{ad}\rho) \longrightarrow H^2(\Gamma_{F,S}, \operatorname{ad}\rho) \longrightarrow \textstyle\bigoplus_{v\in S} H^2(\Gamma_{F_v}, \operatorname{ad}\rho) \\
&\longrightarrow H^3_{\mathcal{L}}(\Gamma_{F,S}, \operatorname{ad}\rho).
\end{aligned}$$

Similarly, we define $H^*_{\mathcal{L}^\perp}(\Gamma_{F,S}, \operatorname{ad}\rho(1))$ as the cohomology of the complex

$$C^\bullet_{\mathcal{L}^\perp}(\Gamma_{F,S}, \operatorname{ad}\rho(1)) = \operatorname{fib}\left(C^\bullet(\Gamma_{F,S}, \operatorname{ad}\rho(1)) \to \bigoplus_{v\in S} \frac{C^\bullet(\Gamma_{F_v}, \operatorname{ad}\rho(1))}{C^\bullet_{\mathcal{L}^\perp_v}(\Gamma_{F_v}, \operatorname{ad}\rho(1))} \right),$$

where $C^\bullet_{\mathcal{L}^\perp_v}(\Gamma_{F_v}, \operatorname{ad}\rho(1))$ denotes the complex

$$0 \to C^0(\Gamma_{F_v}, \operatorname{ad}\rho(1)) \to \ker\left(Z^1(\Gamma_{F_v}, \operatorname{ad}\rho) \to H^1(\Gamma_{F_v}, \operatorname{ad}\rho)/\mathcal{L}_v\right) \to 0 \to \dots.$$

We now prove a duality statement for these cohomology groups. The following theorem is a characteristic 0 version of [GV18, Thm. B.1] (in the special case where the local conditions come from a Selmer system in degree 1).

Theorem 2.4.1. *With notation as above, there is a natural perfect pairing*

$$H^i_{\mathcal{L}}(\Gamma_{F,S}, \operatorname{ad}\rho) \times H^{3-i}_{\mathcal{L}^\perp}(\Gamma_{F,S}, \operatorname{ad}\rho(1)) \to E.$$

Proof. The case $i = 0$ follows from Poitou–Tate duality: we have $H^0_{\mathcal{L}}(\Gamma_{F,S}, \operatorname{ad}\rho) \cong H^0(\Gamma_{F,S}, \operatorname{ad}\rho)$, and, by Poitou–Tate duality,

$$H^3_{\mathcal{L}^\perp}(\Gamma_{F,S}, \operatorname{ad}\rho(1)) \cong \operatorname{coker}\left(H^2(\Gamma_{F,S}, \operatorname{ad}\rho(1)) \to \bigoplus_{v\in S} H^2(\Gamma_{F_v}, \operatorname{ad}\rho(1)) \right) \cong H^0(\Gamma_{F,S}, \operatorname{ad}\rho)^\vee,$$

see e.g. [NSW00, §6, (8.6.10)] for a statement with torsion coefficients, from which the characteristic 0 statement follows from passage to the limit.

It remains to verify the statement for $i = 1$, i.e. to construct a perfect pairing

$$H^1_{\mathcal{L}}(\Gamma_{F,S}, \operatorname{ad}\rho) \times H^2_{\mathcal{L}^\perp}(\Gamma_{F,S}, \operatorname{ad}\rho(1)) \to E.$$

We follow the argument of [GV18, §B.5]. By the long exact sequence above,

$$H^1_{\mathcal{L}}(\Gamma_{F,S}, \operatorname{ad}\rho) \cong \ker\left(Z^1(\Gamma_{F,S}, \operatorname{ad}\rho) \to \prod_{v\in S} H^1(\Gamma_{F_v}, \operatorname{ad}\rho)/\mathcal{L}_v\right),$$

whereas an element of $H^2_{\mathcal{L}^\perp}(\Gamma_{F,S}, \operatorname{ad}\rho(1))$ is represented by a tuple

$$(x', (y'_v)_{v\in S}) \in Z^2(\Gamma_{F,S}, \operatorname{ad}\rho(1)) \times \prod_{v\in S} C^1(\Gamma_{F_v}, \operatorname{ad}\rho(1))$$

such that $x'_v + \mathrm{d}y'_v \in \mathcal{L}_v^\perp$ for every $v \in S$, where $(\cdot)_v$ denotes the restriction map at v, and d is the differential. Given x and $(x', (y'_v)_{v\in S})$ as above, we choose an arbitrary $z \in C^2(\Gamma_{F,S}, E(1))$ such that $\mathrm{d}z = x \smile x'$ and obtain elements

$$-x_v \smile y'_v + z_v \in C^2(\Gamma_{F_v}, E(1)).$$

One verifies that this expression is closed, and hence we obtain elements of $H^2(\Gamma_{F_v}, E(1)) \cong E$. Taking the sum over v produces an element of E, which we define to be $\langle x, (x', (y'_v)_{v\in S})\rangle$.

This pairing is independent of choice of z since $\sum[z_v] = 0$. Furthermore, replacing y'_v by $y'_v + w'_v$ for some $w'_v \in C^1_{\mathcal{L}^\perp}(\Gamma_{F_v}, \operatorname{ad}\rho(1))$ changes the local contribution at v by $x_v \smile w'_v$, which is cohomologically trivial since x_v and w'_v define orthogonal classes. Lastly, if we replace x by $x - \mathrm{d}a$, one may choose $z - a \smile x'$ in place of z, and similarly for x'. Thus, this is a well-defined pairing and it remains to prove that it is non-degenerate.

To see this, we consider the following two-part filtrations:

$$Ш^1(\Gamma_{F,S}, \operatorname{ad}\rho) := \{[x] \in H^1(\Gamma_{F,S}, \operatorname{ad}\rho) \mid \forall v \in S\colon [x_v] = 0\ \} \subseteq H^1_{\mathcal{L}}(\Gamma_{F,S}, \operatorname{ad}\rho),$$
$$Q^1_{\mathcal{L}^\perp} := \operatorname{coker}\left(H^1(\Gamma_{F,S}, \operatorname{ad}\rho(1)) \to \prod_{v\in S} H^1(\Gamma_{F,S}, \operatorname{ad}\rho(1))/\mathcal{L}_v^\perp\right) \subseteq H^1_{\mathcal{L}^\perp}(\Gamma_{F,S}, \operatorname{ad}\rho(1)).$$

The subspace $Q^1_{\mathcal{L}^\perp}$ is identified with the classes represented by tuples $(x, (y'_v)_{v\in S})$ such that

$$x'_v = x'_v + \mathrm{d}y'_v = 0$$

for all $v \in S$. It follows that the pairing respects this filtration; i.e. the restriction

$$Ш^1(\Gamma_{F,S}, \operatorname{ad}\rho) \times Q^1 \to E$$

is identically 0. Moreover, we have an isomorphism

$$H^1_{\mathcal{L}^\perp}(\Gamma_{F,S}, \operatorname{ad}\rho(1))/Q^1 \cong Ш^2(\Gamma_{F,S}, \operatorname{ad}\rho(1)).$$

The problem thus reduces to verifying that the induced pairings

$$Ш^1(\Gamma_{F,S}, \operatorname{ad}\rho) \times Ш^2(\Gamma_{F,S}, \operatorname{ad}\rho(1)) \to E$$
$$H^1_{\mathcal{L}}(\Gamma_{F,S}, \operatorname{ad}\rho)/Ш^1(\Gamma_{F,S}, \operatorname{ad}\rho) \times Q^1_{\mathcal{L}^\perp} \to E$$

In the first case, the obtained pairing is the Tate pairing ([NSW00, Chap. VIII, Thm. 8.6.7]). In the second case, we choose a representative $(x', (y'_v)_{v\in S}) = (0, (y'_v)_{v\in S})$ for the right-hand class such that $\mathrm{d}y'_v = 0$, and $z = 0$. The local contribution at v of the pairing reduces to $-x_v \smile y'_v$, which is the usual local pairing between in degree 1 (up to a minus sign). □

We record the following special case which will be referenced later (Corollary 3.7.9).

Corollary 2.4.2. *Let* $\mathcal{L} = \left(H^1_g(\Gamma_{F_v}, \operatorname{ad}\rho)\right)_{v\in S_p} \cup \left(H^1(\Gamma_{F_v}, \operatorname{ad}\rho)\right)_{v\in S\setminus S_p}$*, and suppose* $\mathrm{WD}(\rho_v)$ *is generic for* $v \in S_p$*. Then, with notation as above, there is a natural isomorphism*

$$H^i_{\mathcal{L}}(\Gamma_{F,S}, \operatorname{ad}\rho) \cong H^{3-i}_{\mathcal{L}^\perp}(\Gamma_{F,S}, \operatorname{ad}\rho(1))^\vee.$$

In particular, $H^2_{\mathcal{L}}(\Gamma_{F,S}, \operatorname{ad}\rho) \cong H^1_g(\Gamma_{F,S}, \operatorname{ad}\rho(1))^\vee$.

Proof. The first statement follows from Theorem 2.4.1. Now, if $v \in S_p$ and $\mathrm{WD}(\rho_v)$ is generic, the inclusion

$$H^1_f(\Gamma_{F,S}, \operatorname{ad}\rho) \subseteq H^1_g(\Gamma_{F,S}, \operatorname{ad}\rho)$$

is an equality by [All16, Lem. 1.1.7], where H^1_f denotes the Bloch–Kato Selmer group defined using the crystalline period ring B_{cris} in place of B_{dR}. Since the orthogonal of $H^1_f(\Gamma_{F,S}, \operatorname{ad}\rho)$ is $H^1_f(\Gamma_{F,S}, \operatorname{ad}\rho(1))$ (see [BK90, Prop. 3.8]), the second statement follows. □

3. Derived deformation theory

3.1. **Artinian algebras.** In this section, we recall the notion of artinian animated algebras from [Lur04].

Definition 3.1.1. The category $\mathcal{A}rt_k$ (resp. $\mathcal{A}rt_E$) of *artinian animated* $\mathcal{O}$*-algebras over* k (resp. over E) is defined as the sub-∞-category of $(\mathcal{A}lg_{\mathcal{O}/})_{/k}$ (resp. $(\mathcal{A}lg_{\mathcal{O}/})_{/E}$) spanned by $(B \to k)$ (resp. $(B \to E)$) such that

(1) $\pi_0(B)$ is an artinian ring with residue field k (resp. E).
(2) $\pi_*(B)$ is a finitely generated $\pi_0(B)$-module.

Any artinian animated $\mathcal{O}$-algebra is truncated. Note that if $B \in \mathcal{A}rt_E$ then $\pi_0(B)$ is an E-algebra and $\pi_*(B)$ is a finite-dimensional E-vector space.

Proposition 3.1.2. *Let* $A \in \mathcal{CNL}_k$ *and* $B \in \mathcal{A}rt_k$*. Then, for any ideal* $I \subset \pi_0(A)$*, the natural morphism*

$$\operatorname*{colim}_s \operatorname{Map}_{/k}(A/^{\mathbf{L}}I^s, B) \to \operatorname{Map}_{/k}(A, B)$$

is an equivalence.

Proof. The maximal ideal of $\pi_0(B)$ is nilpotent, hence given any map $A \to B$, the induced morphism $\pi_0(A) \to \pi_0(B)$ factors through a power of $I \subset \pi_0(A)$. We conclude using Proposition 2.2.4. □

A useful description of artinian algebras allowing for proofs by induction is given in terms of small extensions.

Definition 3.1.3. [Lur04, Def. 3.3.1] Let $B \in \mathcal{A}lg_{\mathcal{O}/}$ and $M \in \mathcal{M}od_B$ a connective B-module. A *small extension* of B by M is a pullback

$$\begin{array}{ccc} B' & \longrightarrow & B \\ \downarrow & & \downarrow{\scriptstyle s} \\ B & \xrightarrow[(\mathrm{id},0)]{} & B \oplus M[1] \end{array}$$

where $B \oplus M[1]$ is the trivial square-zero extension (see below [Lur04, Prop. 3.1.5]) and s is a section of the projection $B \oplus M[1] \to B$.

Lemma 3.1.4. [Lur04, Lem. 6.2.6] *Let $(B \to E) \in \mathcal{A}rt_E$. Then there exists a factorisation*

$$B \simeq B_m \to B_{m-1} \to \cdots \to B_0 \simeq E$$

in $\mathcal{A}rt_E$ such that B_j is a small extension of B_{j-1} by $E[n_j]$ for an increasing sequence $n_j \geq 0$ and $\dim_E(\pi_*(B_j)) = j+1$. The analogous statement is true for $(B \to k) \in \mathcal{A}rt_k$.

Lemma 3.1.5. *If $\mathcal{R} \in \mathcal{CNL}_k$ and $B \in \mathcal{A}rt_k$, then $\mathrm{Map}_{/k}(\mathcal{R}, B)$ has finitely many connected components and finite homotopy groups.*

Proof. We argue by induction on m, the length of a series of small extensions for B as in Lemma 3.1.4. For $m = 0$, we have $B \simeq k$, and $\mathrm{Map}_{/k}(\mathcal{R}, k)$ is contractible. For the induction step, suppose the claim holds for every $B' \in \mathcal{A}rt_k$ given by series of length $m' \leq m$. By Lemma 3.1.4, there is an algebra $B' \in \mathcal{A}rt_k$ and an equivalence

$$B \simeq B' \times_{B' \oplus k[n+1]} B'$$

for some $n \geq 0$, where B' is given by a series of length $m' < m$. Applying $\mathrm{Map}_{/k}(\mathcal{R}, -)$ gives a pullback

$$\mathrm{Map}_{/k}(\mathcal{R}, B) \simeq \mathrm{Map}_{/k}(\mathcal{R}, B') \times_{\mathrm{Map}_{/k}(\mathcal{R}, B' \oplus k[n+1])} \mathrm{Map}_{/k}(\mathcal{R}, B')$$

and the associated long exact sequence of homotopy groups

$$\cdots \to \pi_{i+1}\big(\mathrm{Map}_{/k}(\mathcal{R}, B' \oplus k[n+1])\big) \to \pi_i\big(\mathrm{Map}_{/k}(\mathcal{R}, B)\big) \to \big(\pi_i\big(\mathrm{Map}_{/k}(\mathcal{R}, B')\big)\big)^2 \to \cdots.$$

By the induction hypothesis, $\pi_i\big(\mathrm{Map}_{/k}(\mathcal{R}, B')\big)$ is finite. To complete the proof, it therefore suffices to prove the same statement for $\pi_{i+1}\big(\mathrm{Map}_{/k}(\mathcal{R}, B' \oplus k[n+1])\big)$. Now, we have a map

$$\mathrm{Map}_{/k}(\mathcal{R}, B' \oplus k[n+1]) \longrightarrow \mathrm{Map}_{/k}(\mathcal{R}, B')$$

induced by the projection $B' \oplus k[n+1] \to B'$, whose fibre over a map f equals

$$\mathrm{Map}_{/B'}(\mathcal{R}, B' \oplus k[n+1]) \simeq \mathrm{Map}_{/k}(\mathcal{R}, k \oplus k[n+1]).$$

The homotopy groups of the fibre are

$$\pi_i \mathrm{Map}_{/k}(\mathcal{R}, k \oplus k[n+1]) \cong \mathrm{T}_k^{n+1-i}(\mathcal{R}),$$

which is finite by Proposition 3.6.4 and the noetherianity assumption on $\mathcal{R}$. The corresponding fibre sequence therefore implies the sought statement, which completes the induction step. □

3.2. **Integral models of artinian E-algebras.** In this section, we introduce a notion of integral models of artinian E-algebras. The definition is made in analogy with [Kis09, 2.3.5] and will be used in Definition 3.7.1. Note that we have an equivalence of ∞-categories

$$((\mathcal{A}lg_{\mathcal{O}/})_{/E})_{/(B \to E)} \simeq (\mathcal{A}lg_{\mathcal{O}/})_{/B}.$$

Definition 3.2.1. Let $(B \to E) \in \mathcal{A}rt_E$. The ∞-category $\mathcal{I}ntB$ of *integral models of B* is defined as the sub-∞-category of $(\mathcal{A}lg_{\mathcal{O}/})_{/B}$ spanned by all $(A \to B)$ such that:

(1) $\pi_*(A)$ is a finitely generated $\mathcal{O}$-module.
(2) The induced map $\pi_*(A) \to \pi_*(B)$ is injective.
(3) The induced map $A \otimes^{\mathbf{L}}_{\mathcal{O}} E \to B \otimes^{\mathbf{L}}_{\mathcal{O}} E$ is a weak equivalence.

By Proposition 2.2.3(2) and Definition 3.2.1(1), any $A \in \mathcal{I}ntB$ is derived ϖ-adically complete and almost of finite presentation over $\mathcal{O}$. When B is clear from context, we sometimes denote objects of $\mathcal{I}ntB$ simply by A. The key property of integral models we shall need is the following.

Proposition 3.2.2. *For any $B \in \mathcal{A}rt_E$, the ∞-category $\mathcal{I}ntB$ is filtered.*

Before embarking on the proof of Proposition 3.2.2, we establish some preliminary results about integral models.

Lemma 3.2.3. *Let $(B \to E) \in \mathcal{A}rt_E$ and $(A \to B) \in \mathcal{I}ntB$. The natural map*

$$\mathrm{Map}_{\mathcal{A}lg_{\mathcal{O}/}}(A, B \times_E \mathcal{O}) \to \mathrm{Map}_{\mathcal{A}lg_{\mathcal{O}/}}(A, B)$$

is a weak equivalence.

Proof. Since $\mathcal{O}$ and E are static and $\pi_0(A)$ is a finitely generated $\mathcal{O}$-module, we have equivalences of mapping spaces

$$\mathrm{Map}_{\mathcal{A}lg_{\mathcal{O}/}}(A, E) \simeq \mathrm{Map}_{\mathcal{A}lg_{\mathcal{O}/}}(\pi_0(A), E) \simeq \mathrm{Map}_{\mathcal{A}lg_{\mathcal{O}/}}(\pi_0(A), \mathcal{O}) \simeq \mathrm{Map}_{\mathcal{A}lg_{\mathcal{O}/}}(A, \mathcal{O}).$$

Consequently,

$$\mathrm{Map}_{\mathcal{A}lg_{\mathcal{O}/}}(A, B \times_E \mathcal{O}) \simeq \mathrm{Map}_{\mathcal{A}lg_{\mathcal{O}/}}(A, B) \times_{\mathrm{Map}_{\mathcal{A}lg_{\mathcal{O}/}}(A,E)} \mathrm{Map}_{\mathcal{A}lg_{\mathcal{O}/}}(A, \mathcal{O}) \simeq \mathrm{Map}_{\mathcal{A}lg_{\mathcal{O}/}}(A, B).$$

□

To prove Proposition 3.2.2, we will show that integral models can be constructed via small extensions, in the following sense.

Lemma 3.2.4. *Let $B \in \mathcal{A}rt_E$ be given by a sequence of small extensions*

$$B \simeq B_m \to B_{m-1} \to \cdots \to B_0 \simeq E$$

as in Lemma 3.1.4. Then, for any $A \in \mathcal{I}ntB$, the morphism $A \to \mathcal{O}$ factors as

$$A \simeq A_m \to A_{m-1} \to \cdots \to A_0 \simeq \mathcal{O},$$

where, for every $j \geq 1$, $A_j \in \mathcal{I}ntB_j$ is a small extension of A_{j-1} by $\varpi^{-r_j}\mathcal{O}[n_j]$ for a non-decreasing sequence $0 \leq n_1 \leq \cdots \leq n_m$ and some $r_j \geq 0$.

Before embarking on the proof of the Lemma, let us describe the homotopy groups of a small extension. Let $A \in \mathcal{A}lg_{\mathcal{O}/}$ be n-truncated and suppose $M \in \mathcal{M}od_A$ has homotopy concentrated in degree $n-1$, so that $M[1]$ has homotopy concentrated in degree n. Given a small extension $A' = A \times_{A \oplus M[1]} A$, we obtain a long exact sequence

$$\cdots \to \pi_{i+1}(A) \to \pi_{i-1}(M) \to \pi_i(A') \to \pi_i(A) \to \pi_{i-2}(M) \to \dots$$

which implies that $\pi_i(A') \cong \pi_i(A)$ for every $i \neq n$ and that

$$\pi_{n-1}(M) \cong \ker(\pi_n(A') \to \pi_n(A)).$$

Thus, a single small extension of the form considered in Lemma 3.2.4 only changes the homotopy groups in the top degree by adding a module which is free of rank 1 over $\mathcal{O}$.

Proof of Lemma 3.2.4. Suppose we have a factorisation $B \simeq B_m \to \cdots \to B_0$. We prove the lemma by induction on m. The case $m = 0$ is trivial since $\mathcal{I}ntE \simeq \{\mathcal{O}\}$. For the induction step, suppose the statement holds for any length $m-1$ sequence $B'_{m-1} \to \cdots \to B'_0$. Let $n = n_m$ denote the largest integer such that $\pi_n(B) \neq 0$, and let $A \in \mathcal{I}ntB_m$. The kernel

$$I = \ker\left(\pi_*(A) \hookrightarrow \pi_*(B_m) \to \pi_*(B_{m-1})\right) \subset \pi_n(A)$$

is a $\pi_0(A)$-submodule of $\pi_*(A)$, which is free of rank 1 as an $\mathcal{O}$-module. Using [Lur04, Prop. 3.3.3, Prop. 3.3.5], we construct a morphism $A \to A_{m-1}$ in $\mathcal{A}lg_{\mathcal{O}/}$, where A_{m-1} ('A/I' in the notation of *loc. cit.*) satisfies the following properties:

(1) For any $Y \in \mathcal{A}lg_{\mathcal{O}/}$, the map $A \to A_{m-1}$ induces an identification
$$\mathrm{Map}_{\mathcal{A}lg_{\mathcal{O}/}}(A_{m-1}, Y) \hookrightarrow \mathrm{Map}_{\mathcal{A}lg_{\mathcal{O}/}}(A, Y)$$
with the subspace spanned by maps $A \to Y$ for which $I \subset \ker(\pi_*(A) \to \pi_*(Y))$.
(2) The induced map $\pi_*(A) \to \pi_*(A_{m-1})$ is an isomorphism in degree $i \neq n$ and for $i = n$ is the map $\pi_n(A) \to \pi_n(A)/I$.
(3) A is a small extension of A_{m-1} by $I[n]$, i.e. there is an equivalence
$$A \simeq A_m = A_{m-1} \times_{A_{m-1} \oplus I[n+1]} A_{m-1}.$$

From (1)-(3), we deduce that the map $A_{m-1} \to B_{m-1}$ defines an object of $\mathcal{I}ntB_{m-1}$ and that the square
$$\begin{array}{ccc} A & \longrightarrow & A_{m-1} \\ \downarrow & & \downarrow \\ B & \longrightarrow & B_{m-1} \end{array}$$
commutes up to homotopy. By the induction hypothesis, $A_{m-1} \to \mathcal{O}$ factors as a series of small extensions of the required form, and we are done. □

Lemma 3.2.5. *Let $B \in \mathcal{A}rt_E$. Then $B \times_E \mathcal{O}$ is equivalent in $(\mathcal{A}lg_{\mathcal{O}/})_{/\mathcal{O}}$ to a filtered colimit of algebras of the form of A_m in Lemma 3.2.4.*

Proof. To prove the second claim, we use induction on m again. The case $m = 0$ is trivial since $E \times_E \mathcal{O} \simeq E0$. Suppose the statement is true for any B' given by a length $m-1$ sequence of small extensions. Let $n = n_m + 1$ and note that
$$E[n] = \operatorname*{colim}_{r \geq 0} \varpi^{-r}\mathcal{O}[n].$$
Since filtered colimits commute with finite limits, under the induction hypothesis we have
$$\begin{aligned} B_m \times_E \mathcal{O} &\simeq (B_{m-1} \times_{B_{m-1} \oplus E[n]} B_{m-1}) \times_E \mathcal{O} \\ &\simeq (B_{m-1} \times_E \mathcal{O}) \times_{(B_{m-1} \times_E \mathcal{O}) \oplus E[n]} (B_{m-1} \times_E \mathcal{O}) \\ &\simeq \operatorname*{colim}_{A_{m-1}} (A_{m-1} \times_{A_{m-1} \oplus E[n]} A_{m-1}) \end{aligned}$$
where the colimit runs over a filtered diagram of $A_{m-1} \in \mathcal{I}ntB_{m-1}$ of the sought form. We have equivalences
$$\begin{aligned} A_{m-1} \oplus E[n] &\simeq A_{m-1} \oplus \operatorname*{colim}_{r \geq 0} \varpi^{-r}\mathcal{O}[n] \\ &\simeq \operatorname*{colim}_{r \geq 0} A_{m-1} \oplus \varpi^{-r}\mathcal{O}[n]. \end{aligned}$$
Indeed, A_{m-1} preserves $\mathcal{O}[n]$ inside $E[n]$, since if $\mathcal{E}nd$ denotes the derived endomorphism ring, we have
$$A_{m-1} \to \mathcal{E}nd_{\mathcal{O}}(E[n]) \simeq \pi_0\big(\mathcal{E}nd_{\mathcal{O}}(E)\big) \cong E,$$
and hence the action map $A_{m-1} \to E$ factors through $\pi_0(A_{m-1})$ and lands in $\mathcal{O}$. It follows that
$$\operatorname*{colim}_{A_{m-1}} (A_{m-1} \times_{A_{m-1} \oplus E[n]} A_{m-1}) \cong \operatorname*{colim}_{A_{m-1}} \operatorname*{colim}_{r \geq 0} (A_{m-1} \times_{A_{m-1} \oplus \varpi^{-r}\mathcal{O}[n]} A_{m-1}),$$
and thus B_m is a filtered colimit of the sought form. □

We are now in a position to prove the main result of this section, Proposition 3.2.2.

Proof of Proposition 3.2.2. By the local characterisation of filtered ∞-categories [Lur18, Tag 02PS], it suffices to prove the following statement: Given $A, A' \in \mathcal{I}ntB$ and a morphism of anima

$$\gamma\colon \partial\Delta^n \to \mathrm{Map}_{/B}(A, A')$$

for some $n \geq 0$, there exists an $A'' \in \mathcal{I}ntB$ and a morphism $g\colon A' \to A''$ such that the composition

$$g \circ \gamma\colon \partial\Delta^n \to \mathrm{Map}_{/B}(A, A') \to \mathrm{Map}_{/B}(A, A'')$$

is null-homotopic. In other words, we would like to define the dotted arrow in the diagram

$$\begin{array}{ccc} \partial\Delta^n & \xrightarrow{\gamma} & \mathrm{Map}_{/B}(A, A') \\ \downarrow & & \downarrow{\scriptstyle g_*} \\ \Delta^n & \dashrightarrow & \mathrm{Map}_{/B}(A, A'') \end{array}$$

Since B is final in $(\mathcal{A}lg_{\mathcal{O}/})_{/B}$ and by Lemma 3.2.5, we have

$$\{*\} \simeq \mathrm{Map}_{/B}(A, B) \simeq \mathrm{Map}_{/B}(A, B \times_E \mathcal{O}) \simeq \mathrm{Map}_{/B}(A, \operatorname*{colim}_{A''} A''),$$

where A'' runs over a filtered subcategory of $\mathcal{I}ntB$. Thus, the composition

$$\partial\Delta^n \to \mathrm{Map}_{/B}(A, A') \to \mathrm{Map}_{/B}(A, \operatorname*{colim}_{A''} A'')$$

is null-homotopic. Now, since $A' \in \mathcal{I}ntB$ is a finitely presented $\mathcal{O}$-algebra by Proposition 2.2.3, the map $A' \to \operatorname{colim} A''$ factors through some A''. It follows that there exists a $g\colon A' \to A''$ such that the composition $g \circ \gamma$ is null-homotopic. □

Lemma 3.2.6. *Let $(B \to E) \in \mathcal{A}rt_E$. We have the following:*

1. *There is a natural weak equivalence $\operatorname*{colim}_{A \in \mathcal{I}ntB} A \to B \times_E \mathcal{O}$.*
2. *There is a canonical isomorphism of graded rings $\pi_*(B \times_E \mathcal{O}) \cong \pi_*(B) \times_E \mathcal{O}$.*
3. *Any $A \in \mathcal{I}ntB$ is derived ϖ-adically complete.*
4. *For any $A \in \mathcal{I}ntB$ and $r \geq 1$, $(A/^{\mathbf{L}}\varpi^r \to k) \in \mathcal{A}rt_k$.*

Proof. (1) By Lemma 3.2.3, any $(A \to B) \in \mathcal{I}ntB$ defines a map $A \to B \times_E \mathcal{O}$. Taking the colimit over $\mathcal{I}ntB$, we produce

$$\operatorname*{colim}_{A \in \mathcal{I}ntB} A \to B \times_E \mathcal{O},$$

and we claim that this map is a weak equivalence. Since $\mathcal{I}ntB$ is filtered (Proposition 3.2.2) and filtered colimits commute with π_*, it is enough to prove that the map

$$\operatorname*{colim}_{A \in \mathcal{I}ntB} \pi_*(A) \to \pi_*(B \times_E \mathcal{O})$$

is an isomorphism. Since $\pi_*(A) \to \pi_*(B)$ is injective for every $A \in \mathcal{I}ntB$ (Definition 3.2.1(3)), injectivity follows. To see that the map is surjective, we use the isomorphism provided by Lemma 3.2.5,

$$\pi_*(B \times_E \mathcal{O}) \cong \operatorname*{colim}_{A''} \pi_*(A''),$$

a filtered colimit over objects $A'' \in \mathcal{I}ntB$. Surjectivity follows.

(2) There is a Mayer-Vietoris sequence

$$\cdots \to \pi_n(B \times_E \mathcal{O}) \to \pi_n(B) \times \pi_n(\mathcal{O}) \to \pi_n(E) \to \dots,$$

and hence for every $n \geq 1$, we have $\pi_n(B \times_E \mathcal{O}) \cong \pi_n(B)$ since $\pi_n(\mathcal{O}) = \pi_n(E) = 0$. For $n = 0$, the sequence implies $\pi_0(B \times_E \mathcal{O}) \cong \ker(\pi_0(B) \times \mathcal{O} \to E) \cong \pi_0(B) \times_E \mathcal{O}$.

(3) Since $A \in \mathcal{I}ntB$, $\pi_*(A)$ is a finitely generated $\mathcal{O}$-module and hence ϖ-adically complete. Thus, A is derived ϖ-adically complete by Lemma 2.2.7.

(4) Let $A \in \mathcal{I}ntB$ and $r \geq 1$. From Quillen's spectral sequence [Qui67, Thm. 2.6]

$$E^2_{i,j}\colon \operatorname{Tor}_i^{\mathcal{O}[t]}\left(\pi_j(A), \mathcal{O}\right) \implies \pi_{i+j}(A/^{\mathbf{L}}\varpi^r),$$

we deduce that $\pi_0(A/^{\mathbf{L}}\varpi^r) \cong \pi_0(A)/\varpi^r$, which is artinian since $\pi_0(A)$ is finitely generated and free as an $\mathcal{O}$-module. Moreover, for any $n \geq 1$, the spectral sequence induces a short exact sequence

$$0 \to \pi_n(A)/\varpi^r \to \pi_n(A/^{\mathbf{L}}\varpi^r) \to \pi_{n-1}(A)[\varpi^r] \to 0.$$

Since A is truncated, $\pi_*(A/^{\mathbf{L}}\varpi^r)$ is finitely generated over $\pi_0(A)$, hence over $\pi_0(A/^{\mathbf{L}}\varpi^r)$. □

Lemma 3.2.7. *Let $(A \to B) \in \mathcal{I}ntB$. Then the map $A \to E$ factors through $\mathcal{O} \to E$, and we have a colimit-reflecting lifting*

$$\begin{array}{ccc} \mathcal{I}ntB & \dashrightarrow & (\mathcal{A}lg_{\mathcal{O}/})_{/O} \\ \downarrow & & \downarrow \\ (\mathcal{A}lg_{\mathcal{O}/})_{/B} & \longrightarrow & (\mathcal{A}lg_{\mathcal{O}/})_{/E}. \end{array}$$

Proof. The functor $\mathcal{I}ntB \to (\mathcal{A}lg_{\mathcal{O}/})_{/B}$ factors through

$$(\mathcal{A}lg_{\mathcal{O}/})_{/\operatorname*{colim}_{A\in\mathcal{I}ntB} A} \simeq (\mathcal{A}lg_{\mathcal{O}/})_{/B\times_E\mathcal{O}} \simeq (\mathcal{A}lg_{\mathcal{O}/})_{/B} \times_{(\mathcal{A}lg_{\mathcal{O}/})_{/E}} (\mathcal{A}lg_{\mathcal{O}/})_{/\mathcal{O}},$$

where we have used Lemma 3.2.6(1) in the first equivalence. Thus, we obtain the sought lifting. Reflection of colimits follows from the fact that colimits in an over-∞-category $\mathcal{C}_{/X}$ are computed in $\mathcal{C}$ [Lur09, Prop. 1.2.13.8]. □

3.3. **Topologies on animated algebras.** In this section, we discuss the *ind-ϖ-adic topology* in the derived setting which we, following [Zhu21, §2.4], will use to define derived moduli stacks of continuous representations. Let $\mathrm{Set}^{\mathrm{cp}}$ denote the category of finite sets. The starting point is the observation that any static $\mathcal{O}$-module M defines an ind-profinite set, also denoted M:

$$M = \{\{M'/\varpi^r\}_{r\geq 1}\}_{M'\subset M} \in \mathrm{Ind}(\mathrm{Pro}(\mathrm{Set}^{\mathrm{cp}})),$$

where M' runs over the filtered category of finitely generated submodules of M. In particular, if M is finitely generated then its associated ind-profinite set is isomorphic to a profinite set in the subcategory $\mathrm{Pro}(\mathrm{Set}^{\mathrm{cp}}) \subset \mathrm{Ind}(\mathrm{Pro}(\mathrm{Set}^{\mathrm{cp}}))$. The following result of Zhu allows us to generalise these observations from static $\mathcal{O}$-modules to animated $\mathcal{O}$-modules, thus encoding a notion of ind-ϖ-adic topology in the derived setting.

Lemma 3.3.1. [Zhu21, 2.4.18] *Let Γ be a profinite set and define*

$$\begin{aligned} C_{\mathrm{sc}}(\Gamma, -)\colon \mathrm{Mod}_{\mathcal{O}} &\to \mathrm{Mod}_{\mathcal{O}} \\ M &\mapsto \mathrm{Map}_{\mathrm{Ind}(\mathrm{Pro}(\mathrm{Set}^{\mathrm{cp}}))}(\Gamma, M). \end{aligned}$$

Then $C_{\mathrm{sc}}(\Gamma, -)$ is exact and lax symmetric monoidal and therefore extends to a t-exact functor $\mathcal{M}od_{\mathcal{O}} \to \mathcal{M}od_{\mathcal{O}}$ which preserves colimits. Moreover, $C_{\mathrm{sc}}(\Gamma, -)$ lifts to a functor $\mathcal{A}lg_{\mathcal{O}/} \to \mathcal{A}lg_{\mathcal{O}/}$ which preserves finite limits and sifted colimits.

Lemma 3.3.2. *Let $A \in \mathcal{M}od_{\mathcal{O}}$ and suppose that $\pi_*(A)$ is a finitely generated $\mathcal{O}$-module. Then there is a natural weak equivalence*

$$C_{\mathrm{sc}}(\Gamma, A) \simeq \lim_{r \geq 1} C_{\mathrm{sc}}(\Gamma, A/^{\mathbf{L}}\varpi^r)$$

which respects the monoidal structure. Hence, if $A \in \mathcal{A}lg_{\mathcal{O}/}$ then the equivalence holds in $\mathcal{A}lg_{\mathcal{O}/}$.

Proof. For any $A \in \mathcal{M}od_{\mathcal{O}} \simeq D^{\geq 0}(\mathcal{O})$, we have natural morphisms (of $\mathcal{O}$-algebras, if $A \in \mathcal{A}lg_{\mathcal{O}/}$)

$$C_{\mathrm{sc}}(\Gamma, A) \to C_{\mathrm{sc}}(\Gamma, \lim A/^{\mathbf{L}}\varpi^r) \to \lim C_{\mathrm{sc}}(\Gamma, A/^{\mathbf{L}}\varpi^r).$$

If A is derived ϖ-adically complete – in particular, if $\pi_*(A)$ is finitely generated over $\mathcal{O}$ (Lemma 2.2.7) – the first map is an equivalence. Let $P(A)$ be the property that the second map is a weak equivalence. Then $P(\mathcal{O}[0])$ holds, $P(A)$ implies $P(A[1])$ and for any fibre sequence $A \to B \to C \to A[1]$, if P holds for two of A, B, C then it holds for the third by the five lemma. Moreover, for any A and B, $P(A \oplus B)$ is implied by $P(A)$ and $P(B)$ since $C_{\mathrm{sc}}(\Gamma, -)$ commutes with colimits (Lemma 3.3.1). Finally, any retract $A \to B \to A$ induces a commutative diagram

$$\begin{array}{ccccc} A & \longrightarrow & B & \longrightarrow & A \\ \downarrow & & \downarrow & & \downarrow \\ \lim_r C_{\mathrm{sc}}(\Gamma, A/^{\mathbf{L}}\varpi^r) & \longrightarrow & \lim_r C_{\mathrm{sc}}(\Gamma, B/^{\mathbf{L}}\varpi^r) & \longrightarrow & \lim_r C_{\mathrm{sc}}(\Gamma, A/^{\mathbf{L}}\varpi^r) \end{array}$$

where the composition of horizontal arrows is the identity, so that if $P(B)$ holds then the induced map $\pi_*(A) \to \pi_*(\lim_r C_{\mathrm{sc}}(\Gamma, B/^{\mathbf{L}}\varpi^r))$ is both injective and surjective, i.e. $P(A)$ holds. It follows that $P(A)$ holds for any perfect animated $\mathcal{O}$-module, since the category $\mathcal{D}_{\mathrm{perf}}(\mathcal{O})$ of perfect complexes (i.e. complexes C with finitely generated homology $H_*(C)$) is the smallest strictly full, saturated, triangulated subcategory of $\mathcal{D}(\mathcal{O})$ containing $\mathcal{O}[0]$ ([Sta23, Tag 0ATI]). □

3.4. **Moduli stacks of Galois representations.** In this section, we discuss moduli stacks of representations of a profinite group Γ valued in a smooth affine group scheme $\mathbf{G}/\mathcal{O}$ of finite type, following [Zhu21, §2.4]. We will only consider what is referred to as *strongly continuous representations* in *loc. cit.*

For any $n \in \mathbb{N}$, we let $\mathcal{O}[\mathbf{G}^n]$ denote the coordinate ring of $\mathbf{G}^n$. With boundary maps defined using the group law on $\mathbf{G}$ and degeneracies defined by inclusions $\mathbf{G}^{n_1} \times \{\mathrm{id}\} \times \mathbf{G}^{n_2} \to \mathbf{G}^{n_1+n_2+1}$, we obtain a cosimplicial animated $\mathcal{O}$-algebra $\mathcal{O}[\mathbf{G}^\bullet] \in \mathcal{F}un(\Delta, \mathcal{A}lg_{\mathcal{O}/})$ which is finitely presented in the following sense.

Lemma 3.4.1. *The functor $\mathrm{Map}_{\mathcal{F}un(\Delta, \mathcal{A}lg_{\mathcal{O}/})}(\mathcal{O}[\mathbf{G}^\bullet], -)$ commutes with filtered colimits.*

Proof. Since $\mathcal{O}[\mathbf{G}^n] \cong \mathcal{O}[\mathbf{G}] \otimes_{\mathcal{O}} \cdots \otimes_{\mathcal{O}} \mathcal{O}[\mathbf{G}]$ and $\mathcal{O}[\mathbf{G}]$ is a finitely presented $\mathcal{O}$-algebra, the statement follows from Proposition 2.2.2 and [Lur09, Prop. 5.3.4.13]. □

Given $A \in \mathcal{A}lg_{\mathcal{O}/}$, there is a natural equivalence (see [Zhu21, Rem. 2.2.2]):

$$\mathrm{Map}_{\mathcal{A}ni(\mathrm{Grp})}(\Gamma, \mathbf{G}(A)) \simeq \mathrm{Map}_{\mathcal{F}un(\Delta, \mathcal{A}lg_{\mathcal{O}/})}(\mathcal{O}[\mathbf{G}^\bullet], C_{\mathrm{disc}}(\Gamma^\bullet, A)),$$

where for any $[n] \in \Delta$, $C_{\mathrm{disc}}(\Gamma^n, -)$ is the animation of the functor which maps a static $\mathcal{O}$-module M to the set of all functions $\Gamma^n \to M$ (cf. the definition of $C_{\mathrm{sc}}(\Gamma^\bullet, -)$ in Lemma 3.3.1). Replacing $C_{\mathrm{disc}}(\Gamma^\bullet, -)$ by $C_{\mathrm{sc}}(\Gamma^\bullet, -)$, we obtain the following definition.

Definition 3.4.2. [Zhu21, Def. 2.4.21] Let Γ be a profinite group and $\mathbf{G}/\mathcal{O}$ an affine group scheme of finite type. The *derived moduli stack of framed* $\mathbf{G}$*-valued (strongly continuous)* Γ*-representations* is the functor

$$\begin{aligned}\mathcal{X}^{\square}_{\Gamma,\mathbf{G}}\colon \mathcal{A}lg_{\mathcal{O}/} &\to \mathcal{A}ni\\ A &\mapsto \mathrm{Map}_{\mathcal{F}un(\Delta,\mathcal{A}lg_{\mathcal{O}/})}\big(\mathcal{O}[\mathbf{G}^\bullet], C_{\mathrm{sc}}(\Gamma^\bullet, A)\big).\end{aligned}$$

Lemma 3.4.3. [Zhu21, Lem. 2.4.22] *If $A \in \mathcal{A}lg_{\mathcal{O}/}$ is static and $\mathbf{G} = \mathrm{GL}_n$, then $\mathcal{X}^{\square}_{\Gamma,\mathbf{G}}(A)$ is the set of homomorphisms $\rho_A\colon \Gamma \to \mathrm{GL}_n(A)$ such that A^n is a filtered colimit of Γ-stable finitely generated $\mathcal{O}$-modules with continuous Γ-action.*

Lemma 3.4.4. *The functor $\mathcal{X}^{\square}_{\Gamma,\mathbf{G}}\colon \mathcal{A}lg_{\mathcal{O}/} \to \mathcal{A}ni$ of Definition 3.4.2 commutes with filtered colimits and finite limits. If $A = \lim_r A/^{\mathbf{L}}\varpi^r \in \mathcal{A}lg_{\mathcal{O}/}$ is perfect as an animated $\mathcal{O}$-module, then we have a natural equivalence*

$$\mathcal{X}^{\square}_{\Gamma,\mathbf{G}}(\lim_r A/^{\mathbf{L}}\varpi^r) \simeq \lim_r \mathcal{X}^{\square}_{\Gamma,\mathbf{G}}(A/^{\mathbf{L}}\varpi^r).$$

Proof. By definition, $\mathcal{X}^{\square}_{\Gamma,\mathbf{G}}$ is given by the composition

$$\mathrm{Hom}_{\mathcal{F}un(\Delta,\mathcal{A}lg_{\mathcal{O}/})}(\mathcal{O}[\mathbf{G}^\bullet], -) \circ C_{\mathrm{sc}}(\Gamma, -) \circ \iota,$$

where $\iota\colon \mathcal{A}lg_{\mathcal{O}/} \to \mathcal{F}un(\Delta, \mathcal{A}lg_{\mathcal{O}/})$ is the inclusion of constant cosimplicial objects. Being accessible and a right adjoint, ι preserves filtered colimits and all limits. The functor $C_{\mathrm{sc}}(\Gamma^\bullet, -)$ preserves filtered colimits, finite limits and limits of the form $A = \lim A/^{\mathbf{L}}\varpi^r$ for A as in the statement by Lemma 3.3.1. Finally, $\mathrm{Hom}_{\mathcal{F}un(\Delta,\mathcal{A}lg_{\mathcal{O}/})}(\mathcal{O}[\mathbf{G}^\bullet], -)$ commutes with all limits and filtered colimits (Lemma 3.4.1). This completes the proof. □

3.5. **Derived deformation functors.** In this section, we define various derived deformation functors of Galois representations in terms of the stack $\mathcal{X}^{\square}_{\Gamma,\mathbf{G}}$ of Definition 3.4.2. From now on, we will only consider the case $\mathbf{G} = \mathrm{GL}_n/\mathcal{O}$.

Definition 3.5.1. Let $\mathcal{X}\colon \mathcal{A}lg_{\mathcal{O}/} \to \mathcal{A}ni$ be a functor and fix $\bar\rho \in \mathcal{X}(k)$. We define

$$\begin{aligned}\mathcal{X}_{\bar\rho}\colon \mathcal{A}rt_k &\to \mathcal{A}ni\\ (B \to k) &\mapsto \mathcal{X}(B) \times_{\mathcal{X}(k)} \{\bar\rho\},\end{aligned}$$

which we extend to $\mathcal{P}ro(\mathcal{A}rt_k)$ by setting $\mathcal{X}_{\bar\rho}(\{A_i\}) := \lim_i \mathcal{X}_{\bar\rho}(A_i)$. Similarly, if $\rho \in \mathcal{X}(E)$, we obtain a functor $\mathcal{X}_\rho\colon \mathcal{A}rt_E \to \mathcal{A}ni$ given by $\mathcal{X}(B \to E) = \mathcal{X}(B) \times_{\mathcal{X}(E)} \{\rho\}$.

Definition 3.5.2. Let Γ be a profinite group and suppose $\bar\rho \in \mathcal{X}^{\square}_{\Gamma,\mathbf{G}}(k)$. The *unrestricted framed derived deformation functor* of $\bar\rho$ is the functor

$$\mathcal{D}^{\square}_{\bar\rho} := (\mathcal{X}^{\square}_{\Gamma,\mathbf{G}})_{\bar\rho}\colon \mathcal{A}rt_k \to \mathcal{A}ni.$$

Similarly, if $\rho \in \mathcal{X}^{\square}_{\Gamma,\mathbf{G}}(E)$, we let

$$\mathcal{D}^{\square}_{\rho} := (\mathcal{X}^{\square}_{\Gamma_{F,S},\mathbf{G}})_{\rho}\colon \mathcal{A}rt_E \to \mathcal{A}ni.$$

Remark 3.5.3. [Zhu21, Eq. 2.6] The derived deformation functors introduced above for a mod p representation coincide with the ones introduced by Galatius–Venkatesh in [GV18, Def. 5.4].

The functors in Definition 3.5.2 correspond to framed deformations. The unframed deformation functors will be given by derived quotients with respect to the conjugation action. To this end, let $\mathbf{Z} \subset \mathbf{G}$ be the center of $\mathbf{G}$ and define $\mathbf{PG} = \mathbf{G}/\mathbf{Z}$: $\mathcal{A}lg_{\mathcal{O}/} \to \mathcal{A}ni$ (as will be made clear later on, we consider conjugation by $\mathbf{PG}$ instead of $\mathbf{G}$ for representability reasons). We define functors $\widehat{\mathbf{PG}}_k, \widehat{\mathbf{G}}_k \colon \mathcal{A}rt_k \to \mathcal{A}ni$ by

$$\widehat{\mathbf{PG}}_k(B \to k) \coloneqq \mathbf{PG}(B) \times_{\mathbf{PG}(k)} \{\mathrm{id}\},$$

and similarly for $\widehat{\mathbf{G}}_k$. Similarly, we define $\widehat{\mathbf{G}}_E, \widehat{\mathbf{PG}}_E \colon \mathcal{A}rt_E \to \mathcal{A}ni$. Defined as above, the groups $\widehat{\mathbf{G}}_k$ and $\widehat{\mathbf{PG}}_k$ act on e.g. $\mathcal{D}^{\square}_{\bar\rho}$ by conjugation, i.e. there is a simplicial diagram given by the bar construction:

$$(\widehat{\mathbf{PG}}_k^{\bullet} \times \mathcal{D}^{\square}_{\bar\rho}) = \Big(\dots \; \widehat{\mathbf{PG}}_k^{2} \times \mathcal{D}^{\square}_{\bar\rho} \;\substack{\to\\ \to\\ \to}\; \widehat{\mathbf{PG}}_k \times \mathcal{D}^{\square}_{\bar\rho} \;\rightrightarrows\; \mathcal{D}^{\square}_{\bar\rho} \Big) \in \mathcal{F}un\big(\Delta^{\mathrm{op}}, \mathcal{F}un(\mathcal{A}rt_k, \mathcal{A}ni)\big).$$

By taking a colimit, we obtain the functor of unframed deformations of $\bar\rho$.

Definition 3.5.4. With notation as above, the *functor of derived (unframed) deformations* is defined as the geometric realisation (i.e. colimit)

$$|\widehat{\mathbf{PG}}_k^{\bullet} \times \mathcal{D}^{\square}_{\bar\rho}| \coloneqq \operatorname*{colim}_{\Delta^{\mathrm{op}}} (\widehat{\mathbf{PG}}_k^{\bullet} \times \mathcal{D}^{\square}_{\bar\rho}) \in \mathcal{F}un(\mathcal{A}rt_k, \mathcal{A}ni).$$

The functor $\mathcal{D}_\rho$ of derived deformations of a characteristic 0 representation ρ is defined analogously.

Remark 3.5.5. This definition is modelled after classical Galois deformation theory. An alternative way of defining unframed deformation functors is passing to the quotient at the level of stacks by forming $\mathcal{X}_{\Gamma,\mathbf{G}} \coloneqq |\mathcal{X}^{\square}_{\Gamma,\mathbf{G}} \times \mathbf{PG}^{\bullet}|$ as in [Zhu21, Def. 2.2.14], then defining $\mathcal{D}_{\bar\rho} \coloneqq (\mathcal{X}_{\Gamma,\mathbf{G}})_{\bar\rho}$ for $\bar\rho \in \mathcal{X}_{\Gamma,\mathbf{G}}(k)$. This yields the same functor up to equivalence, which can be proved by considering the natural map $(\mathcal{X}^{\square}_{\Gamma,\mathbf{G}})_{\bar\rho} \to (\mathcal{X}_{\Gamma,\mathbf{G}})_{\bar\rho}$ and comparing tangent complexes (see Section 3.6).

Definition 3.5.6. A (mod p) *derived deformation problem* is a tuple

$$\mathbf{S} = (\bar\rho, S, \{\mathcal{D}_v\}_{v\in S})$$

where $\bar\rho \colon \Gamma_{F,S} \to \mathrm{GL}_n(k)$, S is a finite set of finite places of F and $\{\mathcal{D}_v\}_{v\in S}$ is a set of functors $\mathcal{D}_v \in \mathcal{F}un(\mathcal{A}rt_k, \mathcal{A}ni)$ equipped with actions of $\widehat{\mathbf{PG}}_k$ and equivariant maps $\mathcal{D}_v \to \mathcal{D}^{\square}_{\bar\rho_v}$. Similarly, a *characteristic 0 deformation problem* is a tuple $(\rho, S, \{\mathcal{D}_v\}_{v\in S})$ where $\mathcal{D}_v \to \mathcal{D}^{\square}_{\rho_v}$.

Definition 3.5.7. Let $\mathbf{S} = (\bar\rho, S, \{\mathcal{D}_v\}_{v\in S})$ be a deformation problem. We define a functor

$$\mathcal{D}^{\square}_{\mathbf{S}} \coloneqq \mathcal{D}^{\square}_{\bar\rho} \times_{\prod_{v\in S} \mathcal{D}^{\square}_{\bar\rho_v}} \prod_{v\in S} \mathcal{D}_v \colon \mathcal{A}rt_k \to \mathcal{A}ni$$

which we call the *functor of derived framed deformations of type* $\mathbf{S}$. Since the morphisms $\mathcal{D}_v \to \mathcal{D}^{\square}_{\bar\rho_v}$ are equivariant under conjugation, we may form the geometric realisation

$$\mathcal{D}_{\mathbf{S}} = |\widehat{\mathbf{PG}}_k^{\bullet} \times \mathcal{D}^{\square}_{\mathbf{S}}|,$$

which we call the *functor of derived (unframed) deformations of type* $\mathbf{S}$. We make the same definitions for a characteristic 0 deformation problem $\mathbf{S}$.

Example 3.5.8. Although Definition 3.5.7 is quite general, we will only consider local conditions arising from classical deformation rings in the following way. Let $v \in S_p$ and denote by $R^{\square}_{\bar\rho_v}$ the usual framed deformation ring of a representation $\bar\rho_v \colon \Gamma_{F_v} \to \mathrm{GL}_n(k)$. Then $\mathcal{D}^{\square}_{\bar\rho_v}$ is represented by $R^{\square}_{\bar\rho_v}$ by

[BIP23, Thm. 1.1] and [GV18, Lem. 7.5]. Any conjugation-invariant quotient $R^{\square}_{\bar\rho_v} \twoheadrightarrow R^{D}_{\bar\rho_v}$ defines a local condition $\mathcal{D}_v$ via

$$\mathcal{D}_v \simeq \mathrm{Map}_{/k}(R^{D}_{\bar\rho_v}, -) \to \mathrm{Map}_{/k}(R^{\square}_{\bar\rho_v}, -) \simeq \mathcal{D}^{\square}_{\bar\rho_v}.$$

Definition 3.5.9. Let $\mathbf{S} = (\bar\rho, S, \{\mathcal{D}_v\}_{v\in S})$ be a deformation problem and suppose $T \subseteq S$ is a subset. We define the *functor of derived T-framed liftings of type* $\mathbf{S}$ as

$$\mathcal{D}^{\square,T}_{\mathbf{S}} = \mathcal{D}^{\square}_{\mathbf{S}} \times \prod_{v\in T} \widehat{\mathbf{G}}_k.$$

There exists an action of $\widehat{\mathbf{G}}_k$ on $\mathcal{D}^{\square,T}_{\mathbf{S}}$ given by

$$\gamma\colon (\rho_A, \{\alpha_v\}_{v\in T}) \mapsto (\gamma\rho_A\gamma^{-1}, \{\gamma\alpha_v\}_{v\in T}).$$

We define the *functor of derived T-framed type* $\mathbf{S}$ *deformations* as the geometric realisation

$$\mathcal{D}^{T}_{\mathbf{S}} \coloneqq |\mathcal{D}^{\square,T}_{\mathbf{S}} \times \widehat{\mathbf{G}}^{\bullet}_k| \in \mathcal{F}un(\mathcal{A}rt_k, \mathcal{A}ni).$$

Proposition 3.5.10. *Let $\mathbf{S} = (\bar\rho, S, \{\mathcal{D}_v\}_{v\in S})$ be a deformation problem and $T \subset S$. Then the natural morphism $\mathcal{D}^{T}_{\mathbf{S}} \to \mathcal{D}_{\mathbf{S}}$ is formally smooth of relative dimension $n^2|T| - 1$.*

Proof. See e.g. [A'C24, Lem. 5.1.5]. □

At $v \in S \setminus S_p$, our local conditions will be trivial, defined by the usual local deformation rings $R^{\square}_{\bar\rho_v}$.

Proposition 3.5.11. [All16, Prop. 1.2.2] *Let $v \nmid p$. Then $\dim R^{\square}_{\bar\rho_v}[1/\varpi] = n^2$, and if $\rho_v\colon \Gamma_{F_v} \to E$ is a representation such that the Weil–Deligne representation $\mathrm{WD}(\rho_v)$ is generic, the corresponding point of $\mathrm{Spec}\, R^{\square}_{\bar\rho_v}[1/\varpi]$ is smooth.*

At the places dividing p, we will consider local conditions defined by a variant of Kisin's [Kis08] potentially semistable deformation rings, introduced in [Hev24, §5.1], defined by imposing conditions on Hodge–Tate weights and the action of inertia.

Definition 3.5.12. Let $v \mid p$. A Weil–Deligne inertial type of F_v is an isomorphism class of pairs $\tau = (\rho_\tau, N_\tau)$ where $\rho_\tau\colon I_{F_v} \to \mathrm{GL}_n(E)$ is a representation with open kernel and $N_\tau \in M_n(E)$ is a nilpotent matrix such that τ is isomorphic to the restriction $(r, N)|_{I_{F_v}}$ of a Weil–Deligne representation of Γ_{F_v}.

Hevesi defines a partial order $\preceq$ on Weil–Deligne inertial types, where $(\rho_\tau, N_\tau) \preceq (\rho_{\tau'}, N_{\tau'})$ if $\rho_\tau \cong \rho_{\tau'}$ and, roughly, the nilpotence of N_τ is bounded from below by that of $N_{\tau'}$; see [Hev24, Def. 2.11] for the precise definition.

Theorem 3.5.13. *Let $v \mid p$. Fix a dominant weight $\lambda \in (\mathbb{Z}^n)^{\mathrm{Hom}(F,E)}$ and a Weil–Deligne inertial type τ of F_v. Suppose $\bar\rho_v\colon \Gamma_{F_v} \to \mathrm{GL}_n(k)$. Then we have the following:*

1. [Hev24, Thm. 5.2] *There exists a local condition $\mathcal{D}^{\lambda_v,\preceq\tau_v}_{\bar\rho_v} \to \mathcal{D}^{\square}_{\bar\rho_v} \simeq \mathrm{Map}_{/k}(R^{\square}_{\bar\rho_v}, -)$ represented by an $\mathcal{O}$-flat reduced quotient $R^{\square}_{\bar\rho_v} \twoheadrightarrow R^{\lambda_v,\preceq\tau_v}_{\bar\rho_v}$ such that a morphism $R^{\square}_{\bar\rho_v} \to E$ factors through $R^{\lambda_v,\preceq\tau_v}_{\bar\rho_v}$ if and only if the corresponding representation is potentially semistable with labelled Hodge–Tate weights $(\lambda_{\iota,1}+n-1 > \cdots > \lambda_{\iota,n})_{\iota\in\mathrm{Hom}(F,E)}$ and $\mathrm{WD}(\bar\rho_v)^{\mathrm{ss}}|_{I_{F_v}} \preceq \tau_v$.*
2. [Kis08, Thm. 3.3.8] $\dim R^{\lambda_v,\preceq\tau_v}_{\bar\rho_v}[1/\varpi] \le n^2 + \frac{1}{2}n(n-1)[F_v\colon \mathbb{Q}_p]$.
3. [All16, Prop. 1.3.12] *If ρ_v is a characteristic 0 lift of $\bar\rho_v$ such that the attached Weil–Deligne representation $\mathrm{WD}(\rho_v)$ is generic, the corresponding point of $\mathrm{Spec}\, R^{\lambda_v,\preceq\tau_v}_{\bar\rho_v}[1/\varpi]$ is smooth with tangent space $H^1_g(\Gamma_{F_v}, \mathrm{ad}\,\rho)$.*

3.6. **Tangent complexes and representability.** In this section, we recall the definition of the tangent complex and Lurie's representability theorem for formally cohesive functors, [Lur04, §6.2].

Definition 3.6.1. [Lur04, Def. 6.2.1] A functor $\mathcal{D}\colon \mathcal{A}rt_k \to \mathcal{A}ni$ (resp. $\mathcal{D}\colon \mathcal{A}rt_E \to \mathcal{A}ni$) is *formally cohesive* if the following hold:

(1) $\mathcal{D}(k)$ is contractible (resp. $\mathcal{D}(E)$ is contractible).
(2) If $B' \to B$ and $B'' \to B$ are surjective (on π_0) morphisms in $\mathcal{A}rt_k$ (resp. $\mathcal{A}rt_E$), then

$$\mathcal{D}(B' \times_B B'') \simeq \mathcal{D}(B') \times_{\mathcal{D}(B)} \mathcal{D}(B'').$$

Lemma 3.6.2. *The derived deformation functors introduced in Section 3.5 are formally cohesive.*

Proof. Property (1) follows from Lemma 3.4.3. To prove (2), let $\mathcal{X}^{\square}\colon \mathcal{A}lg_{\mathcal{O}/} \to \mathcal{A}ni$ be a functor satisfying Lemma 3.4.4, and let $\mathcal{D}^{\square} \coloneqq \mathcal{X}^{\square}_{\bar{\rho}}$ be the deformation functor at a point $\bar{\rho} \in \mathcal{X}^{\square}(k)$. Then if $B' \to B \leftarrow B''$ is a diagram in $\mathcal{A}rt_k$, we have

$$\mathcal{D}(B' \times_B B'') = \mathcal{X}^{\square}(B' \times_B B'') \times_{\mathcal{X}^{\square}(k)} \{\bar{\rho}\} \simeq \mathcal{X}^{\square}_{\bar{\rho}}(B') \times_{\mathcal{X}^{\square}_{\bar{\rho}}(B)} \mathcal{X}^{\square}_{\bar{\rho}}(B'') \simeq \mathcal{D}(B') \times_{\mathcal{D}(B)} \mathcal{D}(B''),$$

as needed. This equivalence is equivariant for the action of

$$\widehat{\mathbf{PG}}(B' \times_B B'') = \widehat{\mathbf{PG}}(B') \times_{\widehat{\mathbf{PG}}(B)} \widehat{\mathbf{PG}}(B''),$$

which implies the claim for the unframed version $|\widehat{\mathbf{PG}}^{\bullet} \times \mathcal{D}^{\square}|$. The proof for a characteristic 0 deformation functor is identical. □

Given a formally cohesive functor $\mathcal{D}\colon \mathcal{A}rt_k \to \mathcal{A}ni$, $r \geq 1$ and $\xi_r \in \mathcal{D}(\mathcal{O}_r)$ where $\mathcal{O}_r = \mathcal{O}/\varpi^r$, one can define a functor on perfect connective $\mathcal{O}_r$-modules by

$$V \mapsto \mathcal{D}(\mathcal{O}_r \oplus V).$$

By [Lur04, Prop. 2.5.5, Lem. 6.2.2], this functor has a unique filtered colimit-preserving and finite limit-preserving extension to the category $\mathcal{M}od_{\mathcal{O}_r}$ of all $\mathcal{O}_r$-modules, which is representable in the sense that there exists an $\mathcal{O}_r$-module $\mathrm{T}^{\bullet}_{\mathcal{O}_r}(\mathcal{D}) \in \mathcal{M}od_{\mathcal{O}_r}$ such that

$$V \mapsto \big(V \otimes_{\mathcal{O}_r} \mathrm{T}^{\bullet}_{\mathcal{O}_r}(\mathcal{D})\big)_0,$$

the 0:th space of the spectrum $V \otimes_{\mathcal{O}_r} \mathrm{T}^{\bullet}_{\mathcal{O}_r}(\mathcal{D})$. Concretely, the homotopy groups of $\mathrm{T}^{\bullet}_A(\mathcal{D})$ are given by

$$\mathrm{T}^i_A(\mathcal{D}) \coloneqq \pi_i\big(\mathrm{T}^{\bullet}_A(\mathcal{D})\big) \cong \pi_0\big(\mathcal{D}(A \oplus A[i]) \times_{\mathcal{D}(A)} \{x\}\big).$$

The $\mathcal{O}_r$-module $\mathrm{T}^{\bullet}_{\mathcal{O}_r}(\mathcal{D}) \in \mathcal{M}od_{\mathcal{O}_r}$ is called the *tangent complex of $\mathcal{D}$ at $\mathcal{O}_r$*. Similarly, for a formally cohesive functor $\mathcal{D}\colon \mathcal{A}rt_E \to \mathcal{A}ni$, we define its tangent complex $\mathrm{T}^{\bullet}_E(\mathcal{D})$ at E in terms of the functor mapping a perfect connective $V \in \mathcal{M}od_E$ to $\mathcal{D}(E \oplus V)$. The construction of the tangent complex is functorial in $\mathcal{D}$ and $\mathcal{O}_r$, and we define

$$\mathrm{T}^{\bullet}_{\mathcal{O}}(\mathcal{D}) \coloneqq \lim_r \mathrm{T}^{\bullet}_{\mathcal{O}_r}(\mathcal{D}).$$

If $\mathcal{D}$ is a formally cohesive functor represented by $\mathcal{R}$, we write $\mathrm{T}^{\bullet}_{?}(\mathcal{R}) \coloneqq \mathrm{T}^{\bullet}_{?}(\mathcal{D})$. The tangent complex of $\mathcal{R}$ at x is given by

$$\mathrm{T}^{\bullet}_A(\mathcal{R}) = \mathrm{Map}_{\mathcal{M}od_{\mathcal{R}}}(\mathrm{L}_{\mathcal{R}/\mathcal{O}}, A) \cong \mathrm{Map}_{\mathcal{M}od_A}(\mathrm{L}_{\mathcal{R}/\mathcal{O}} \otimes^{\mathbf{L}}_{\mathcal{R}} A, A),$$

where $\mathrm{L}_{\mathcal{R}/\mathcal{O}}$ denote the algebraic cotangent complex of $\mathcal{R}$. If $A \to B \to \mathcal{O}$ is a morphism in $\mathcal{A}lg_{\mathcal{O}/}$, we denote by $\mathrm{T}^{\bullet}_{\mathcal{O}}(B/A)$ the fibre of the induced map $\mathrm{T}^{\bullet}_{\mathcal{O}}(B) \to \mathrm{T}^{\bullet}_{\mathcal{O}}(A)$.

Theorem 3.6.3. [Lur04, Cor. 6.2.14] *Let $\mathcal{D}\colon \mathcal{A}rt_k \to \mathcal{A}ni$ be a formally cohesive functor. The following are equivalent:*

(1) *There exists an $\mathcal{R} \in \mathcal{CNL}_k$ (Definition 2.2.6) and an equivalence of functors*

$$\mathcal{D} \simeq \mathrm{Hom}_{(\mathcal{A}lg_{\mathcal{O}/})_{/k}}(\mathcal{R}, -).$$

(2) *The k-vector spaces $\mathrm{T}^i_k(\mathcal{D})$ vanish for $i < 0$ and are finite dimensional for $i \geq 0$.*

The analogous statement holds for a formally cohesive functor $\mathcal{D}\colon \mathcal{A}rt_E \to \mathcal{A}ni$.

Given $\mathcal{R} \in \mathcal{CNL}_k$ with maximal ideal $\mathfrak{m} \subset \pi_0(\mathcal{R})$ and $B \in \mathcal{A}rt_k$, we have natural equivalences

$$\mathrm{Map}_{/k}(\mathcal{R}, B) \simeq \mathrm{Map}_{/k}(\lim_r \mathcal{R}/^{\mathbf{L}}\mathfrak{m}^r, B) \simeq \operatorname*{colim}_r \mathrm{Map}_{/k}(\mathcal{R}/^{\mathbf{L}}\mathfrak{m}^r, B).$$

The first equivalence holds because $\mathcal{R}$ is derived $\mathfrak{m}$-adically complete and the second holds by Proposition 2.2.4 and the artinian assumption on B. Since $\mathcal{R}$ is noetherian, by [Lur04, Prop. 6.1.8] we have

$$\pi_0(\lim \mathcal{R}/^{\mathbf{L}}\mathfrak{m}^r) \cong \varprojlim \pi_0(\mathcal{R}).$$

The quotient $\mathcal{R}/^{\mathbf{L}}\mathfrak{m}^r$ need not be truncated, but one can rephrase Theorem 3.6.3(1) in terms of pro-representability using the pro-artinian completion $\{\tau_{\leq i}(\mathcal{R}/^{\mathbf{L}}\mathfrak{m}^r)\}_{i,r} \in \mathcal{P}ro(\mathcal{A}rt_k)$, as in [GV18].

Proposition 3.6.4. *Let $\mathcal{R} \in \mathcal{CNL}_k$ and let $\mathcal{R} \to \mathcal{O}_r = \mathcal{O}/\varpi^r$ be a morphism over k. Then, for every $i \geq 0$, $\mathrm{T}^i_{\mathcal{O}_r}(\mathcal{R})$ is a finitely generated $\mathcal{O}_r$-module.*

Proof. The case $r = 1$ is Theorem 3.6.3. By considering the short exact sequence

$$0 \to \mathcal{O}_r \to \mathcal{O}_{r+1} \to k \to 0$$

we deduce the analogous statement for $\mathcal{O}_r$-coefficients using induction. □

Lemma 3.6.5. *Suppose we have morphisms $A \to B \to \mathcal{O}$ in $\mathcal{A}lg_{\mathcal{O}/}$ and that B is almost of finite presentation over A. Then, if $\mathrm{T}^i_{\mathcal{O}}(A)$ is a finitely generated $\mathcal{O}$-module for every $i \geq 0$, the same holds for $\mathrm{T}^i_{\mathcal{O}}(B)$.*

Proof. By [Lur04, Prop. 3.2.14], the relative cotangent complex $\mathrm{L}_{B/A}$ is an almost perfect animated B-module, i.e. $\pi_i(\mathrm{L}_{B/A})$ is a finitely generated $\pi_0(B)$-module for every $i \geq 0$. It follows that $\mathrm{L}_{B/A} \otimes^{\mathbf{L}}_B \mathcal{O}$ is an almost perfect animated $\mathcal{O}$-module. Dualising the fibre sequence

$$\mathrm{L}_{A/\mathcal{O}} \otimes^{\mathbf{L}}_A \mathcal{O} \to \mathrm{L}_{B/\mathcal{O}} \otimes^{\mathbf{L}}_B \mathcal{O} \to \mathrm{L}_{B/A} \otimes^{\mathbf{L}}_B \mathcal{O} \overset{+1}{\to}$$

we obtain a fibre sequence of tangent complexes with $\mathcal{O}$-coefficients

$$\mathrm{T}^\bullet_{\mathcal{O}}(B/A) \to \mathrm{T}^\bullet_{\mathcal{O}}(B) \to \mathrm{T}^\bullet_{\mathcal{O}}(A) \overset{+1}{\to}.$$

By assumption, $\mathrm{T}^i_{\mathcal{O}}(A)$ is finitely generated over $\mathcal{O}$ for every i. The same now follows for $\mathrm{T}^\bullet_{\mathcal{O}}(B)$. □

The tangent complexes of the deformation functors introduced in Section 3.5 are given by continuous cohomology in the usual sense.

Proposition 3.6.6. [Zhu21, Prop. 2.4.24] *Let Γ be a profinite group, $\mathbf{G} = \mathrm{GL}_n/\mathcal{O}$ and $\bar\rho \in \mathcal{X}^{\square}_{\Gamma,\mathbf{G}}(k)$. The tangent complex of $\mathcal{D}_{\bar\rho}$ at k is given by an animated k-module denoted $C^{\bullet}_{\mathrm{sc}}(\Gamma, \mathrm{ad}\,\bar\rho)[1]$ such that*

$$\mathrm{T}^i_k(\mathcal{D}_{\bar\rho}) \cong \pi_{-i}\big(C^{\bullet}_{\mathrm{sc}}(\Gamma, \mathrm{ad}\,\bar\rho)[1]\big) \cong \begin{cases} H^0(\Gamma, \mathrm{ad}^0\,\bar\rho) & \text{if } i = -1, \\ H^1(\Gamma, \mathrm{ad}\,\bar\rho) & \text{if } i = 0, \\ H^2(\Gamma, \mathrm{ad}\,\bar\rho) & \text{if } i = 1, \\ 0 & \text{otherwise.} \end{cases}$$

In particular, if $\bar\rho$ is absolutely irreducible, then

$$\mathrm{T}^i_k(\mathcal{D}_{\bar\rho}) \cong H^{i+1}(\Gamma, \mathrm{ad}\,\bar\rho).$$

The tangent complex of $\mathcal{D}^{\square}_{\bar\rho}$ at k is given by an animated k-module $\overline{C}^{\bullet}_{\mathrm{sc}}(\Gamma, \mathrm{ad}\,\bar\rho)[1]$ such that

$$\mathrm{T}^i_k(\mathcal{D}^{\square}_{\bar\rho}) \cong \pi_{-i}\big(\overline{C}^{\bullet}_{\mathrm{sc}}(\Gamma, \mathrm{ad}\,\bar\rho)[1]\big) \cong \begin{cases} Z^1(\Gamma, \mathrm{ad}\,\bar\rho) & \text{if } i = 0, \\ H^2(\Gamma, \mathrm{ad}\,\bar\rho) & \text{if } i = 1, \\ 0 & \text{otherwise.} \end{cases}$$

The analogous statements are true with $\xi \in \mathcal{X}^{\square}_{\Gamma,\mathbf{G}}(\mathcal{O})$, $\xi_r \in \mathcal{X}^{\square}_{\Gamma,\mathbf{G}}(\mathcal{O}_r)$ or $\rho \in \mathcal{X}^{\square}_{\Gamma,\mathbf{G}}(E)$ in place of $\bar\rho$.

Corollary 3.6.7. *Let $\Gamma = \mathrm{Gal}(F_S/F)$ or $\mathrm{Gal}(\overline{F}_v/F_v)$ and fix $\bar\rho \in \mathcal{X}^{\square}_{\Gamma,\mathbf{G}}(k)$. The framed derived deformation functor $\mathcal{D}^{\square}_{\bar\rho} = (\mathcal{X}^{\square}_{\Gamma,\mathbf{G}})_{\bar\rho}$ is representable and if $\bar\rho$ is Schur, the derived deformation functor $\mathcal{D}_{\bar\rho}$ is representable. The same statement is true for $\rho \in \mathcal{X}^{\square}_{\Gamma,\mathbf{G}}(E)$.*

Proof. This follows from Theorem 3.6.3 and Proposition 3.6.6. □

We deduce the same statement for deformations of type $\mathbf{S}$.

Proposition 3.6.8. *Let $\mathbf{S} = (\bar\rho, S, \{\mathcal{D}_v\}_{v\in S})$ be a deformation problem and let $T \subseteq S$ be a subset. Suppose $\bar\rho$ is Schur. Then the functors $\mathcal{D}_{\mathbf{S}}$ and $\mathcal{D}^T_{\mathbf{S}}$ are representable.*

3.7. **Localisation at a characteristic 0 point.** In this section, we prove two results which generalise results of Kisin, relating the deformation ring of a characteristic 0 representation to that of its reduction mod p [Kis09, Lem. 2.3.3, Prop. 2.3.5].

In the following definition, which is a direct analogue of [Kis09, §2.3.1], we use the functor $\mathcal{I}ntB \to (\mathcal{A}lg_{\mathcal{O}/})_{/\mathcal{O}}$ of Lemma 3.2.7 to view $A \in \mathcal{I}ntB$ as an object of $(\mathcal{A}lg_{\mathcal{O}/})_{/\mathcal{O}}$.

Definition 3.7.1. Let $\mathcal{D}\colon \mathcal{A}rt_k \to \mathcal{A}ni$ be a functor and $\xi \in \mathcal{D}(\mathcal{O}) = \lim \mathcal{D}(\mathcal{O}_r)$. We define

$$\begin{aligned} \mathcal{D}_{(\xi)}\colon \mathcal{A}rt_E &\to \mathcal{A}ni \\ (B \to E) &\mapsto \operatorname*{colim}_{A\in\mathcal{I}ntB} \mathcal{D}(A) \times_{\mathcal{D}(\mathcal{O})} \{\xi\}. \end{aligned}$$

As before, we extend $\mathcal{D}_{(\xi)}$ to $\mathrm{Pro}(\mathcal{A}rt_E)$ by defining $\mathcal{D}_{(\xi)}(\{A_i\}) \coloneqq \lim \mathcal{D}_{(\xi)}(A_i)$.

Proposition 3.7.2. *Let $\mathcal{X}^{\square}\colon \mathcal{A}lg_{\mathcal{O}/} \to \mathcal{A}ni$ be a functor commuting with filtered colimits, finite limits and limits of the form $A = \lim A/^{\mathbf{L}}\varpi^r$ when $\pi_*(A)$ is a finitely generated $\mathcal{O}$-module. Fix $\bar\rho \in \mathcal{X}^{\square}(k)$ and $\xi \in \mathcal{X}^{\square}_{\bar\rho}(\mathcal{O})$, and let ρ be the image of ξ in $\mathcal{X}^{\square}(E)$. Then there is a natural weak equivalence $(\mathcal{X}^{\square}_{\bar\rho})_{(\xi)} \simeq \mathcal{X}^{\square}_{\rho}$ in $\mathcal{F}un(\mathcal{A}rt_E, \mathcal{A}ni)$.*

Proof. Let $(B \to E) \in \mathcal{A}rt_E$. By definition,

$$(\mathcal{X}^{\square}_{\bar\rho})_{(\xi)}(B) = \operatorname*{colim}_{A \in \mathcal{I}ntB} (\mathcal{X}^{\square}_{\bar\rho}(A) \times_{\mathcal{X}_{\bar\rho}(\mathcal{O})} \{\xi\}).$$

Now, $A \simeq \lim A/^{\mathbf{L}}\varpi^r$ and $A/^{\mathbf{L}}\varpi^r \in \mathcal{A}rt_k$ by Lemma 3.2.6(3,4) Hence, if we let $\xi_r \in \mathcal{X}^{\square}(\mathcal{O}_r)$ denote the $\bmod \varpi^r$ reduction of ξ, we have by definition that

$$\mathcal{X}^{\square}_{\bar\rho}(A) \times_{\mathcal{X}^{\square}_{\bar\rho}(\mathcal{O})} \{\xi\} \simeq \lim_r \mathcal{X}^{\square}_{\bar\rho}(A/^{\mathbf{L}}\varpi^r) \times_{\mathcal{X}^{\square}_{\bar\rho}(\mathcal{O}_r)} \{\xi_r\} \simeq \mathcal{X}^{\square}(A) \times_{\mathcal{X}^{\square}(\mathcal{O})} \{\xi\}.$$

The category $\mathcal{I}ntB$ is filtered by Proposition 3.2.2 and filtered colimits commute with finite limits. Hence, by the assumption on $\mathcal{X}^{\square}$ and Lemma 3.2.6(1),

$$\operatorname*{colim}_{A \in \mathcal{I}ntB} (\mathcal{X}_{\bar\rho}(A) \times_{\mathcal{X}_{\bar\rho}(\mathcal{O})} \{\xi\}) \simeq \mathcal{X}^{\square}(B \times_E \mathcal{O}) \times_{\mathcal{X}^{\square}(\mathcal{O})} \{\xi\} \simeq \mathcal{X}^{\square}(B) \times_{\mathcal{X}^{\square}(E)} \{\rho\} = \mathcal{X}^{\square}_{\rho}(B),$$

as required. □

Theorem 3.7.3. *Let $\mathcal{D}\colon \mathcal{A}rt_k \to \mathcal{A}ni$ be a formally cohesive functor represented by $\mathcal{R} \in \mathcal{CNL}_k$, and $\xi \in \mathcal{D}(\mathcal{O})$. Then $\mathcal{D}_{(\xi)}\colon \mathcal{A}rt_E \to \mathcal{A}ni$ is represented by*

$$(\mathcal{R} \otimes_{\mathcal{O}} E)^{\wedge}_{\xi} \coloneqq \lim_s (\mathcal{R} \otimes_{\mathcal{O}} E)/^{\mathbf{L}} I^s_{\xi},$$

the derived completion of $\mathcal{R} \otimes_{\mathcal{O}} E$ at the maximal ideal $I_\xi = \ker(\pi_0(\mathcal{R}) \otimes_{\mathcal{O}} E \to E)$ defined by ξ.

Proof. Let $(B \to E) \in \mathcal{A}rt_E$ and let $\xi_r \in \mathcal{D}(\mathcal{O}_r)$ denote the $\bmod \varpi^r$ reduction of ξ. By definition and Lemma 3.2.6(3),

$$\mathcal{D}_{(\xi)}(B) = \operatorname*{colim}_{A \in \mathcal{I}ntB} (\mathcal{D}(A) \times_{\mathcal{D}(\mathcal{O})} \{\xi\}) \simeq \operatorname*{colim}_{A \in \mathcal{I}ntB} \big(\lim_r \mathcal{D}(A/^{\mathbf{L}}\varpi^r) \times_{\mathcal{D}(\mathcal{O}/^{\mathbf{L}}\varpi^r)} \{\xi_r\}\big).$$

Now, by representability of $\mathcal{D}$, we have

$$\lim_r \big(\mathcal{D}(A/^{\mathbf{L}}\varpi^r) \times_{\mathcal{D}(\mathcal{O}/^{\mathbf{L}}\varpi^r)} \{\xi_r\}\big) \simeq \lim_r \operatorname{Map}_{/k}(\mathcal{R}, A/^{\mathbf{L}}\varpi^r) \times_{\operatorname{Map}_{/k}(\mathcal{R}, \mathcal{O}/^{\mathbf{L}}\varpi^r)} \{\xi_r\},$$

and from Lemma 3.2.6(4) and commutativity of limits, we obtain

$$\lim_r \operatorname{Map}_{/k}(\mathcal{R}, A/^{\mathbf{L}}\varpi^r) \times_{\operatorname{Map}_{/k}(\mathcal{R}, \mathcal{O}/^{\mathbf{L}}\varpi^r)} \{\xi_r\} \simeq \operatorname{Map}_{/\mathcal{O}}(\mathcal{R}, A).$$

For any $A \in \mathcal{I}ntB$, the map $\pi_0(A) \to \pi_0(B)$ is injective and hence, for any $\mathcal{R} \to A$ in $(\mathcal{A}lg_{\mathcal{O}/})_{/\mathcal{O}}$, the induced map $\pi_0(\mathcal{R}) \to \pi_0(A)$ factors through a quotient $\pi_0(\mathcal{R})/(I^\circ_\xi)^s$ for some $s \geq 1$, where $I^\circ_\xi \subset \pi_0(\mathcal{R})$ is the inverse image of $I_\xi = \ker(\pi_0(\mathcal{R}) \otimes_{\mathcal{O}} E \to E)$. Proposition 2.2.4 implies

$$\operatorname{Map}_{/\mathcal{O}}(\mathcal{R}, A) \simeq \operatorname*{colim}_s \operatorname{Map}_{/\mathcal{O}} \big(\mathcal{R}/^{\mathbf{L}}(I^\circ_\xi)^s, A\big).$$

For any $A \in \mathcal{I}ntB$, $\pi_*(A)$ injects into $\pi_*(B)$, which is truncated. Hence, A is n-truncated for n sufficiently large depending only on B. Now, $\mathcal{R}/^{\mathbf{L}}(I^\circ_\xi)^s$ is almost of finite presentation over $\mathcal{O}$, since $\pi_0(\mathcal{R}/^{\mathbf{L}}(I^\circ_\xi)^s) \cong \pi_0(\mathcal{R})/(I^\circ_\xi)^s$ is a finitely generated $\mathcal{O}$-module and $\mathcal{R}/^{\mathbf{L}}(I^\circ_\xi)^s$ is a derived tensor product of noetherian animated algebras (Proposition 2.2.3). Moreover, $\mathcal{I}ntB$ is filtered (Proposition 3.2.2), and hence

$$\mathcal{D}_{(\xi)}(B) \simeq \operatorname*{colim}_{A \in \mathcal{I}ntB} \operatorname*{colim}_s \operatorname{Map}_{/\mathcal{O}}(\mathcal{R}/^{\mathbf{L}}(I^\circ_\xi)^s, A) \simeq \operatorname*{colim}_s \operatorname{Map}_{/\mathcal{O}}(\mathcal{R}/^{\mathbf{L}}(I^\circ_\xi)^s, \operatorname*{colim}_{A \in \mathcal{I}ntB} A).$$

By Lemma 3.2.6(1), $\operatorname{colim} A \simeq B \times_E \mathcal{O}$ and the space above is equivalent to

$$\operatorname*{colim}_s \operatorname{Map}_{/\mathcal{O}} \big((\mathcal{R}/^{\mathbf{L}}(I^\circ_\xi)^s, B \times_E \mathcal{O}\big) \simeq \operatorname*{colim}_s \operatorname{Map}_{/E} \big((\mathcal{R} \otimes_{\mathcal{O}} E)/^{\mathbf{L}} I^s_\xi, B\big).$$

Thus, $\mathcal{D}_{(\xi)}(B \to E) \simeq \operatorname{Map}_{/E} \big((\mathcal{R} \otimes_{\mathcal{O}} E)^{\wedge}_{\xi}, B\big)$ as required. □

The construction behaves well with respect to finite limits, as the following simple observation shows.

Lemma 3.7.4. *Suppose given a pullback of formally cohesive functors*

$$\mathcal{D}' \times_{\mathcal{D}} \mathcal{D}'' \in \mathcal{F}un(\mathcal{A}rt_k, \mathcal{A}ni),$$

and let $\tilde{\xi} \in (\mathcal{D}' \times_{\mathcal{D}} \mathcal{D}'')(\mathcal{O})$. *Then*

$$(\mathcal{D}' \times_{\mathcal{D}} \mathcal{D}'')_{(\tilde{\xi})} \simeq \mathcal{D}'_{(\xi')} \times_{\mathcal{D}_{(\xi)}} \mathcal{D}''_{(\xi'')}$$

where ξ, ξ', ξ'' *denote the images of* $\tilde{\xi}$ *in* $\mathcal{D}(\mathcal{O}), \mathcal{D}'(\mathcal{O})$ *and* $\mathcal{D}''(\mathcal{O})$, *respectively.*

Proof. This follows from the fact that filtered colimits commute with finite limits. Indeed, for any $B \in \mathcal{A}rt_E$,

$$(\mathcal{D}' \times_{\mathcal{D}} \mathcal{D}'')_{(\tilde{\xi})}(B) \simeq \operatorname*{colim}_{A \in \mathcal{I}ntB} (\mathcal{D}' \times_{\mathcal{D}} \mathcal{D}'')(A) \simeq \mathcal{D}'_{(\xi')}(B) \times_{\mathcal{D}_{(\xi)}(B)} \mathcal{D}''_{(\xi'')}(B),$$

as needed. □

Recall that we defined derived deformation functors $\mathcal{D}^{\square}_{\bar{\rho}}, \mathcal{D}_{\bar{\rho}}, \mathcal{D}^{\square}_{\rho}, \mathcal{D}_{\rho}$ in Definition 3.5.2. The following is a typical application of Proposition 3.7.2 and Theorem 3.7.3.

Corollary 3.7.5. *Let* $\mathcal{X}^{\square}_{\Gamma,\mathbf{G}}$ *be the derived moduli stack of framed* $\mathbf{G}$*-valued* Γ*-representations (see Definition 3.4.2). Suppose* $\bar{\rho} \in \mathcal{X}^{\square}_{\Gamma,\mathbf{G}}(k)$, *let*

$$\mathcal{D}^{\square}_{\bar{\rho}} = (\mathcal{X}^{\square}_{\Gamma,\mathbf{G}})_{\bar{\rho}}$$

and fix $\xi \in \mathcal{D}^{\square}_{\bar{\rho}}(\mathcal{O})$ *with image* $\rho = \xi \otimes E \in \mathcal{D}^{\square}_{\bar{\rho}}(E)$. *Then there is a natural weak equivalence of functors*

$$(\mathcal{D}^{\square}_{\bar{\rho}})_{(\xi)} \simeq \mathcal{D}^{\square}_{\rho}.$$

The same statement holds for the functor $\mathcal{D}_{\bar{\rho}}$ *of unframed deformations (see* **??***).*

Proof. The statement for the framed functors follows from Lemma 3.4.4 and Proposition 3.7.2. To prove the analogous statement for the unframed functor, we let $(B \to E) \in \mathcal{A}rt_E$. The proof of Proposition 3.7.2 comprises an equivalence

$$(\mathcal{D}^{\square}_{\bar{\rho}})_{(\xi)}(B) = \operatorname*{colim}_{A \in \mathcal{I}ntB} \mathcal{X}^{\square}_{\Gamma,\mathbf{G}}(A) \times_{\mathcal{X}^{\square}_{\Gamma,\mathbf{G}}(\mathcal{O})} \{\xi\} \simeq \mathcal{X}^{\square}_{\Gamma,\mathbf{G}}(B) \times_{\mathcal{X}^{\square}_{\Gamma,\mathbf{G}}(E)} \{\rho\},$$

which is equivariant for the action of

$$\operatorname*{colim}_{A \in \mathcal{I}ntB} \widehat{\mathbf{PG}}_k(A) \times_{\widehat{\mathbf{PG}}_k(\mathcal{O})} \{\mathrm{id}\} \simeq \mathbf{PG}(A) \times_{\mathcal{O}} \{\mathrm{id}\} \simeq \mathbf{PG}(B) \times_{\mathbf{PG}(E)} \{\mathrm{id}\} \simeq \widehat{\mathbf{PG}}_E(B).$$

where we have subscripted $\widehat{\mathbf{PG}}$ to distinguish the functors defined on $\mathcal{A}rt_k$ resp. $\mathcal{A}rt_E$. Hence, taking geometric realisations now prove the claim. □

Proposition 3.7.6. *Let* $\mathcal{D} \colon \mathcal{A}rt_k \to \mathcal{A}ni$ *be a formally cohesive functor such that* $\mathrm{T}^i_k(\mathcal{D})$ *is finite for every* i, *and fix* $\xi \in \mathcal{D}(\mathcal{O})$. *Then* $\mathcal{D}_{(\xi)} \colon \mathcal{A}rt_E \to \mathcal{A}ni$ *has tangent complex*

$$\mathrm{T}^{\bullet}_E(\mathcal{D}_{(\xi)}) = \left(\lim_r \mathrm{T}^{\bullet}_{\mathcal{O}_r}(\mathcal{D})\right) \otimes_{\mathcal{O}} E.$$

Proof. It suffices to prove that, for every perfect connective $V \in \mathcal{M}od_{\mathcal{O}}$, we have an equivalence

$$\mathcal{D}(\mathcal{O} \oplus V) \times_{\mathcal{D}(\mathcal{O})} \{\xi\} \simeq \tau_{\geq 0}\big(\lim_r \mathrm{T}^{\bullet}_{\mathcal{O}_r}(\mathcal{D}) \otimes^{\mathbf{L}}_{\mathcal{O}} V\big) \simeq \lim_r \tau_{\geq 0}\big(\mathrm{T}^{\bullet}_{\mathcal{O}_r}(\mathcal{D}) \otimes^{\mathbf{L}}_{\mathcal{O}} V\big).$$

Arguing as in Proposition 3.6.4, one sees that $\mathrm{T}^i_{\mathcal{O}_r}(\mathcal{D})$ is finitely generated for every i. Thus, if $V_r \coloneqq V/^{\mathbf{L}}\varpi^r$, and ξ_r is the image of ξ in $\mathcal{D}(\mathcal{O}_r)$, then

$$\begin{aligned}\tau_{\geq 0}\big(\lim_r \mathrm{T}^{\bullet}_{\mathcal{O}_r}(\mathcal{D}) \otimes^{\mathbf{L}}_{\mathcal{O}} V\big) \simeq \lim_r \tau_{\geq 0}\big(\mathrm{T}^{\bullet}_{\mathcal{O}_r}(\mathcal{D}) \otimes^{\mathbf{L}}_{\mathcal{O}_r} V_r\big) &\simeq \lim_r \mathcal{D}(\mathcal{O}_r \oplus V_r) \times_{\mathcal{D}(\mathcal{O}_r)} \{\xi_r\} \\ &\simeq \mathcal{D}(\mathcal{O} \oplus V) \times_{\mathcal{D}(\mathcal{O})} \{\xi\},\end{aligned}$$

where we have used that $\mathrm{T}^{\bullet}_{\mathcal{O}_r}(\mathcal{D}) \otimes^{\mathbf{L}}_{\mathcal{O}_r} V_r$ is almost perfect, so that the inverse limit has no higher derived functors. □

Proposition 3.7.7. *Let $R = \pi_0(R) \in \mathcal{CNL}_k$ be static and suppose $\xi\colon R \to \mathcal{O}$ is a morphism such that $\mathfrak{p} = \ker(R \otimes_{\mathcal{O}} E \to E)$ is a smooth point. Then, for any $r \geq 1$ and $i \geq 1$, the $\mathcal{O}$-module $\mathrm{T}^i_{\mathcal{O}_r}(R)$ has finite length bounded independently of r.*

Proof. Let $i, r \geq 1$. Consider the short exact sequence $0 \to \mathcal{O} \to \mathcal{O} \to \mathcal{O}_r \to 0$ given by multiplication by ϖ^r, and the induced exact sequence

$$\mathrm{T}^i_{\mathcal{O}}(R) \to \mathrm{T}^i_{\mathcal{O}_r}(R) \to \mathrm{T}^{i+1}_{\mathcal{O}}(R).$$

By Lemma 3.6.5 and Proposition 3.6.4, these are finitely generated $\mathcal{O}$-modules. Since ξ defines a smooth point, $\mathrm{T}^i_{\mathcal{O}}(R)$ is torsion for $i \geq 1$. It follows that

$$\mathrm{length}_{\mathcal{O}}\left(\mathrm{T}^i_{\mathcal{O}_r}(R)\right) \leq \mathrm{length}_{\mathcal{O}}\left(\mathrm{T}^i_{\mathcal{O}}(R)\right) + \mathrm{length}_{\mathcal{O}}\left(\mathrm{T}^{i+1}_{\mathcal{O}}(R)\right)$$

which is finite and independent of r. □

Proposition 3.7.8. *Let $\mathcal{D}_{\mathbf{S}}$ be the derived deformation functor of the mod p deformation problem*

$$\mathbf{S} = (\bar{\rho}, S, \{\mathrm{Map}_{/k}(R^D_{\bar{\rho}_v}, -)\}_{v \in S}).$$

Suppose $\xi \in \mathcal{D}_{\mathbf{S}}(\mathcal{O})$ and let $\rho = \xi[1/p]\colon \Gamma_{F,S} \to \mathrm{GL}_n(E)$. Then $(\mathcal{D}_{\mathbf{S}})_{(\xi)} \simeq \mathcal{D}_{\mathbf{S}(\rho)}$, where $\mathbf{S}(\rho)$ is the characteristic 0 deformation problem

$$\mathbf{S}(\rho) \coloneqq (\rho, S, \{\mathrm{Map}_{/E}((R^D_{\bar{\rho}_v} \otimes_{\mathcal{O}} E)^{\wedge}_{\xi_v}, -)\}_{v \in S})$$

and $(R^D_{\bar{\rho}_v} \otimes_{\mathcal{O}} E)^{\wedge}_{\xi_v}$ denotes the completion of $R^D_{\bar{\rho}_v} \otimes_{\mathcal{O}} E$ at the ideal $I_{\xi_v} \subset \pi_0(R^D_{\bar{\rho}_v} \otimes E \to E)$ defined by $\xi_v = \xi|_{\Gamma_{F_v}}$, as in Theorem 3.7.3.

Proof. Let $\mathbf{S} = (\bar{\rho}, S, \{\mathcal{D}_v\}_{v \in S})$ and $(B \to E) \in \mathcal{A}rt_E$. By definition, the functor of framed liftings of type $\mathbf{S}$ is given by the pullback $\mathcal{D}^{\square}_{\mathbf{S}} = \mathcal{D}^{\square}_{\bar{\rho}} \times_{\prod_{v \in S} \mathcal{D}^{\square}_{\bar{\rho}_v}} \prod_{v \in S} \mathcal{D}_v$. Filtered colimits commute with finite limits and $\mathcal{I}ntB$ is filtered (Proposition 3.2.2). Hence, if we let $\xi_v \coloneqq \xi|_{\Gamma_{F_v}} \in \mathcal{D}^{\square}_{\bar{\rho}_v}(\mathcal{O})$, we have

$$(\mathcal{D}^{\square}_{\mathbf{S}})_{(\xi)} \simeq \left(\mathcal{D}^{\square}_{\bar{\rho}} \times_{\prod_{v \in S} \mathcal{D}^{\square}_{\bar{\rho}_v}} \prod_{v \in S} \mathcal{D}_v\right)_{(\xi)} \simeq (\mathcal{D}^{\square}_{\bar{\rho}})_{(\xi)} \times_{\prod_{v \in S} (\mathcal{D}^{\square}_{\bar{\rho}_v})_{(\xi_v)}} \prod_{v \in S} (\mathcal{D}_v)_{(\xi_v)}.$$

Applying Theorem 3.7.3 and Proposition 3.7.2 now proves a conjugation-equivariant equivalence between $(\mathcal{D}^{\square}_{\mathbf{S}})_{(\xi)}$ and the framed deformation functor of the characteristic 0 deformation problem displayed above. Taking geometric realisations now completes the proof. □

Corollary 3.7.9. *Let $\mathcal{D}_{\mathbf{S}}$ be the deformation functor of the deformation problem*

$$\mathbf{S} = (\bar{\rho}, S, \{\mathcal{D}_{\bar{\rho}_v}^{\lambda_v, \preceq \tau_v}\}_{v \in S_p} \cup \{\mathcal{D}_{\bar{\rho}_v}^{\square}\}_{v \in S \setminus S_p})$$

where $\bar{\rho}$ is absolutely irreducible, and suppose $\xi \in \mathcal{D}_{\mathbf{S}}(\mathcal{O})$ is a point corresponding to a representation $\rho\colon \Gamma_{F,S} \to \mathrm{GL}_n(E)$ such that the Weil–Deligne representations $\mathrm{WD}(\rho|_{\Gamma_{F_v}})$ are generic for every $v \in S$. Then we have a weak equivalence $(\mathcal{D}_{\mathbf{S}})_{(\xi)} \simeq \mathcal{D}_{\rho}^{\mathrm{ss}}$, where $\mathcal{D}_{\rho}^{\mathrm{ss}}$ is the deformation functor of the characteristic 0 deformation problem

$$(\rho, S, \{\mathcal{D}_{\rho_v}^{\lambda_v}\}_{v \in S_p} \cup \{\mathcal{D}_{\rho_v}^{\square}\}_{v \in S^p}).$$

Moreover, the tangent complex of $(\mathcal{D}_{\mathbf{S}})_{(\xi)}$ is concentrated in degrees $[0,1]$ where we have natural identifications

$$\mathrm{T}_E^0\left((\mathcal{D}_{\mathbf{S}})_{(\xi)}\right) \cong H_g^1(\Gamma_{F,S}, \mathrm{ad}\,\rho), \quad \mathrm{T}_E^1\left((\mathcal{D}_{\mathbf{S}})_{(\xi)}\right) \cong H_g^1(\Gamma_{F,S}, \mathrm{ad}\,\rho(1))^\vee.$$

Proof. By Proposition 3.7.8, we have an equivalence $(\mathcal{D}_{\mathbf{S}})_{(\xi)} \simeq \mathcal{D}_{\mathbf{S}(\rho)}$, where $\mathbf{S}(\rho)$ has local conditions at $v \in S \setminus S_p$ given by the unrestricted local framed deformation rings $R_{\rho_v}^{\square}$. At $v \in S_p$, the local conditions are defined by

$$(R_{\bar{\rho}_v}^{\lambda_v, \preceq \tau_v} \otimes_{\mathcal{O}} E)_{\xi}^{\wedge} \simeq \varprojlim_s (R_{\bar{\rho}_v}^{\lambda_v, \preceq \tau_v} \otimes_{\mathcal{O}} E)/I_{\xi}^s \simeq R_{\rho_v}^{\lambda_v},$$

where we have used Theorem 3.5.13(3) and the assumption on $\mathrm{WD}(\rho_v)$, and $\varprojlim$ denotes the 1-categorical inverse limit in the category of rings. Note that the local conditions are defined by smooth algebras by Proposition 3.5.11 and Theorem 3.5.13(3). To describe the tangent complex, we use the argument in [GV18, Thm. 9.2, App. B] which proves the corresponding statement for mod p deformations. The same argument, applied to $\mathcal{D}_{\rho}^{\mathrm{ss}}$, identifies the tangent complex with the complex with local conditions of Corollary 2.4.2; the result follows. □

The calculation of the tangent complex $\mathrm{T}_E^{\bullet}(\mathcal{R}_{\rho}^{\mathrm{ss}})$ allows us to identify the graded ring $\pi_*(\mathcal{R}_{\rho}^{\mathrm{ss}})$ under the assumption that the adjoint Bloch–Kato Selmer group vanishes.

Corollary 3.7.10. *Suppose $H_g^1(\Gamma_{F,S}, \mathrm{ad}\,\rho) = 0$. Then there is a natural isomorphism*

$$\pi_*(\mathcal{R}_{\rho}^{\mathrm{ss}}) \cong \bigwedge H_g^1(\Gamma_{F,S}, \mathrm{ad}\,\rho(1)).$$

Proof. By Corollary 3.7.9 and the assumed vanishing of $H_g^1(\Gamma_{F,S}, \mathrm{ad}\,\rho)$, $\mathcal{R}_{\rho}^{\mathrm{ss}}$ is quasi-smooth with vanishing tangent space. It follows from [AG15, Cor. 2.1.6] that $\pi_*(\mathcal{R}_{\rho}^{\mathrm{ss}})$ is an exterior algebra on $\pi_*(\mathcal{R}_{\rho}^{\mathrm{ss}})$. Thus, it suffices to show that

$$\pi_1(\mathcal{R}_{\rho}^{\mathrm{ss}}) \cong H_g^1(\Gamma_{F,S}, \mathrm{ad}\,\rho(1)).$$

These are vector spaces of dimension l_0, hence it suffices to constuct a natural surjective map between them. By Corollary 3.7.9, we have an isomorphism

$$H_g^1(\Gamma_{F,S}, \mathrm{ad}\,\rho(1))^\vee \cong \mathrm{T}_E^1(\mathcal{R}_{\rho}^{\mathrm{ss}}) = \pi_0\Big(\mathrm{Map}_{/E}\left(\mathcal{R}_{\rho}^{\mathrm{ss}}, E \oplus E[1]\right)\Big).$$

Any class $[f]$ in this π_0 induces a morphism $\pi_1(\mathcal{R}_{\rho}^{\mathrm{ss}}) \to E$. This defines a map

$$\varphi\colon \mathrm{T}_E^1(\mathcal{R}_{\rho}^{\mathrm{ss}}) \to \pi_1(\mathcal{R}_{\rho}^{\mathrm{ss}})^\vee.$$

We claim that φ is injective. Indeed, suppose $\varphi([f]) = 0$, i.e. f is a morphism $\mathcal{R}_\rho^{\mathrm{ss}} \to E \oplus E[1]$ such that its induced map on π_1 is trivial. Repeated application of [Lur04, Prop. 3.3.3] shows that the space of such maps is equivalent to

$$\mathrm{Map}_{/E}\left(\pi_0(\mathcal{R}_\rho^{\mathrm{ss}}), E \oplus E[1]\right) \hookrightarrow \mathrm{Map}_{/E}\left(\mathcal{R}_\rho^{\mathrm{ss}}, E \oplus E[1]\right).$$

Now, $\pi_0(\mathcal{R}_\rho^{\mathrm{ss}}) \cong E$ and thus, this space is contractible and $[f]$ is trivial. Passing to duals, we obtain a surjection $\varphi^\vee \colon \pi_1(\mathcal{R}_\rho^{\mathrm{ss}}) \cong H^1_g(\Gamma_{F,S}, \mathrm{ad}\,\rho(1))$, as required. This completes the proof. □

4. Patching and Venkatesh's conjecture

In this section, we prove the following result.

Theorem 4.0.1. *Let Π be a regular algebraic cuspidal automorphic representation of $\mathrm{GL}_n(\mathbb{A}_F)$, cohomological with respect to an algebraic representation $\mathcal{V}_\lambda$ where $\lambda \in (\mathbb{Z}^n)^{\mathrm{Hom}(F,E)}$, and let $\rho = \rho_{\Pi,\iota} \colon \Gamma_{F,S} \to \mathrm{GL}_n(E)$ denote the Galois representation attached to (Π, ι) where $\iota \colon \overline{\mathbb{Q}}_p \xrightarrow{\sim} \mathbb{C}$ and $E/\mathbb{Q}_p$ is a finite extension containing all embeddings $F \to \overline{\mathbb{Q}}_p$. Suppose the following:*

1. *$p \nmid 2n$.*
2. *Π is tempered at almost all finite places.*
3. *$\zeta_p \notin F$ and $\bar\rho|_{\Gamma_{F(\zeta_p)}}$ has enormous image* [ACC+23, Def. 6.2.28].
4. *$\bar\rho$ is absolutely irreducible and decomposed generic* [ACC+23, Def. 4.3.1].
5. *F contains an imaginary quadratic subfield in which p splits.*
6. *For every pair $\bar v \neq \bar v'$ of places of F^+ above p, we have (with S_p^+ the places of F^+ above p)*
$$\sum_{\bar w \in S_p^+ \setminus \{\bar v, \bar v'\}} [F^+_{\bar w} \colon \mathbb{Q}_p] > \frac{1}{2}[F^+ \colon \mathbb{Q}].$$
7. *For every $v \in S$, the Weil-Deligne representation $\mathrm{WD}(\rho|_{\Gamma_{F_v}})$ is generic.*

Let $\mathcal{R}_\rho^{\mathrm{ss}}$ denote the derived deformation ring representing the characteristic 0 deformation problem $(\rho, S, \{\mathcal{D}_{\rho_v}^{\lambda_v}\}_{v\in S_p} \cup \{\mathcal{D}_{\rho_v}^{\square}\}_{v \in S \setminus S_p})$. Then

$$\pi_*(\mathcal{R}_\rho^{\mathrm{ss}}) \cong \bigwedge H^1_g(\Gamma_{F,S}, \mathrm{ad}\,\rho(1)),$$

and there is an action of $\mathcal{R}_\rho^{\mathrm{ss}}$ on $R\Gamma(K, \mathcal{V}_\lambda)_{\mathfrak{p}}$ inducing a free action

$$\pi_*(\mathcal{R}_\rho^{\mathrm{ss}}) \circlearrowright H_*(X_K, \mathcal{V}_\lambda)_\Pi.$$

Remark 4.0.2. We briefly describe the roles played by the assumptions in Theorem 4.0.1 and what is currently known.

- *Taylor–Wiles method:* We require (1)-(4) to prove the existence of Taylor–Wiles primes in Proposition 4.1.2 below. We do not expect (1) to be essential. The Ramanujan conjecture predicts that (2) always holds; the case $n = 2$ is known by [BCG+25] and the polarisable case holds by [Car12, Car14]. One might expect to be able to replace the condition in (3) on $\bar\rho|_{\Gamma_{F(\zeta_p)}}$ with a 'big image' assumption on the characteristic 0 representation, and to remove the 'decomposed generic' part of (4).
- *Local-global compatibility:* Assumptions (5) and (6) are necessary to apply Hevesi's local-global compatibility result [Hev24, Thm. 1.4] which implies that ρ_v has the correct Hodge–Tate weights, in particular that $\mathbf{S}$ is a well-defined deformation problem; this compatibility is expected to hold in general – when stronger results are available in the literature, we expect to be able to relax assumptions (5) and (6).

- *Vanishing of the Bloch–Kato Selmer group:* All of the assumptions (1)-(7) are required to apply the abstract patching lemma of A'Campo [A'C24, Lem. 2.6.4], though our assumptions are somewhat weaker than those in *loc.cit.*, thanks to Hevesi's local-global compatibility result [Hev24]. Assumption (7) would follow from a strong form of local-global compatibility at the ramified places. In the conjugate self-dual case, it follows from the main results of [Car14, Car12].

Remark 4.0.3. A priori, the action of $\mathcal{R}^{\mathrm{ss}}_{\rho}$ depends on non-canonical choices made in the Taylor–Wiles argument below. With stronger local assumptions on $\bar{\rho}$ at $v \mid p$, the naturality is established in [GV18, §15] by making a comparison between the characteristic p derived deformation ring $\mathcal{R}_{\mathbf{S}}$ and the derived Hecke algebra introduced in [Ven19].

From now on, we fix Π and ρ as in Theorem 4.0.1, and an $\mathcal{O}$-lattice $\xi\colon \Gamma_{F,S} \to \mathrm{GL}_n(\mathcal{O})$ of ρ with reductions $\xi_r\colon \Gamma_{F,S} \to \mathrm{GL}_n(\mathcal{O}_r)$. The main result of [Hev24] then implies the following.

Theorem 4.0.4. [Hev24] *With assumptions as in Theorem 4.0.1,*

$$\mathbf{S} = (\bar{\rho}, S, \{\mathcal{D}_{\bar{\rho}_v}^{\lambda_v, \preceq \tau_v}\}_{v \in S_p} \cup \{\mathcal{D}_{\bar{\rho}_v}^{\square}\}_{v \in S \setminus S_p})$$

is a well-defined deformation problem.

4.1. **Taylor–Wiles primes.** The basis of our patching argument later on is a delicate choice of Taylor–Wiles places, detailed in this subsection.

Definition 4.1.1. Let $\mathbf{S} = (\bar{\rho}, S, \{\mathcal{D}_v\}_{v \in S})$ be a deformation problem. A Taylor–Wiles datum of level N for $\mathbf{S}$ is a tuple $(Q_N, \{(\alpha_{v,1}, \dots, \alpha_{v,n})\}_{v \in Q_N})$ where:

(1) Q_N is a finite set of places of F, disjoint from S, such that, for every $v \in Q_N$, $v \equiv 1 \bmod p^N$.
(2) For every $v \in Q_N$, $(\alpha_{v,1}, \dots, \alpha_{v,n}) \in k^n$ are the eigenvalues of $\bar{\rho}(\mathrm{Frob}_v)$, assumed to be pairwise distinct and k-rational.

We generally omit $(\alpha_{v,i})$ and write simply Q_N for Taylor–Wiles data. Our main result Theorem 4.0.1 contains the assumption (3) that $\bar{\rho}|_{\Gamma_{F(\zeta_{p^\infty})}}$ has enormous image. In [NT23, Def. 2.23], a notion of 'enormous image' for characteristic 0 representations is defined, which is implied by the 'enormous image' property of its reduction $\bar{\rho}$.

Proposition 4.1.2. *Let $\mathbf{S}' = (\bar{\rho}, S, \{\mathcal{D}_{\bar{\rho}_v}^{\square}\}_{v \in S})$ be the deformation problem with no conditions at $v \in S$, and suppose that*

- *$p \nmid 2n$*
- *$\zeta_p \notin F$ and $\bar{\rho}|_{\Gamma_{F(\zeta_p)}}$ has enormous image.*
- *Π is tempered at almost all finite places.*

Then there exist integers $q, l \geq 0$ such that, for any $N \geq 1$, there exists a Taylor–Wiles datum Q_N for $\mathbf{S}'$ such that

(1) $|Q_N| = q$
(2) *For every $v \in Q_N$, $v \equiv 1 \bmod p^N$ and the rational prime below v splits in F_0.*
(3) *There is a local $\mathcal{O}$-algebra surjection $R^{S,\mathrm{loc}}_{\mathbf{S}}[[X_1, \dots, X_g]] \twoheadrightarrow R^{S}_{\mathbf{S}_{Q_N}}$.*
(4) *For every $r \leq N$, with $SQ_N := S \cup Q_N$ we have*

$$\mathrm{length}_{\mathcal{O}} H^1(\Gamma_{F,SQ_N}, \mathrm{ad}\, \xi_r(1)) \leq l,$$

hence if $\mathcal{L}_S = \{\mathcal{L}_{v,r}\}_{v\in S}$ *is a Selmer system for* $\operatorname{ad}\xi_r$ *and*

$$\mathcal{L}_{SQ_N} = \mathcal{L}_S \cup \{H^1(\Gamma_{F_v}, \operatorname{ad}\xi_r)\}_{v\in Q_N},$$

then

$$\operatorname{length}_{\mathcal{O}} H^1_{\mathcal{L}^\perp_{SQ_N}}(\Gamma_{F,SQ_N}, \operatorname{ad}\xi_r(1)) \le l.$$

Proof. By [A'C24, Prop. 6.2.32], for q large enough we can find, for every $N \ge 1$, a Taylor–Wiles datum $(Q_N, \{(\alpha_{v,1}, \dots, \alpha_{v,n})\}_{v\in Q_N})$ such that (1)-(3) hold. To achieve (4), we will enlarge the sets Q_N by adding suitable Taylor–Wiles primes, following the proof of [NT23, Lem. 2.30]. Thus, in (1) we replace if necessary q by a larger number. The splitting condition in (2) defines a set of Dirichlet density 1, so we may restrict our attention to places satisfying the condition. Enlarging Q_N does not affect (3), as the statement is equivalent to the vanishing of

$$H^1(\Gamma_{F,SQ_N}, \operatorname{ad}\bar\rho(1)) = \ker\left(H^1(\Gamma_{F,S}, \operatorname{ad}\bar\rho(1)) \to \prod_{v\in Q_N} H^1(\Gamma_{F_v}, \operatorname{ad}\bar\rho(1))\right)$$

and adding more factors to the product over v preserves the vanishing of this kernel. The second statement in (4) follows from the first, since we have an inclusion

$$H^1_{\mathcal{L}^\perp_{SQ_N}}(\Gamma_{F,SQ_N}, \operatorname{ad}\xi_r(1)) \hookrightarrow H^1(\Gamma_{F,SQ_N}, \operatorname{ad}\xi_r(1)).$$

Let $\operatorname{ad}_{E/\mathcal{O}} = \operatorname{ad}\xi \otimes_{\mathcal{O}} E/\mathcal{O}$ and similarly for $\operatorname{ad}_{\mathcal{O}}, \operatorname{ad}_E, \operatorname{ad}_{\mathcal{O}_r}$ so that $\operatorname{ad}_{E/\mathcal{O}} = \operatorname{colim}\operatorname{ad}_{\mathcal{O}_r}$. Fix an integer

$$s \ge \operatorname{corank}_{\mathcal{O}} H^1(\Gamma_{F,S}, \operatorname{ad}_{E/\mathcal{O}}(1)) \coloneqq \operatorname{rank}_{\mathcal{O}} \operatorname{Hom}_{\mathcal{O}}\left(H^1(\Gamma_{F,S}, \operatorname{ad}_{E/\mathcal{O}}(1)), E/\mathcal{O}\right).$$

Our goal is to find Frobenius elements $\sigma_1, \dots, \sigma_s \in G_{F(\zeta_{p^\infty})}$ such that:

(i) $\rho(\sigma_i)$ has n distinct eigenvalues in E.
(ii) The kernel of the map

$$H^1(\Gamma_{F,S}, \operatorname{ad}_{E/\mathcal{O}}(1)) \to \bigoplus_{i=1}^{s} H^1(\langle\sigma_i\rangle, \operatorname{ad}_{E/\mathcal{O}}(1)) \cong \bigoplus_{i=1}^{s} \operatorname{ad}_{E/\mathcal{O}}/(\sigma_i - 1)\operatorname{ad}_{E/\mathcal{O}}$$

is of finite length as an $\mathcal{O}$-module.

Given such $\sigma_i \in G_{F(\zeta_{p^\infty})}$, for every $r \le N$, consider the commutative diagram with exact columns

$$\begin{array}{ccc}
0 & & 0 \\
\downarrow & & \downarrow \\
H^0(\Gamma_{F,S}, \operatorname{ad}_{E/\mathcal{O}}(1))/\varpi^r & \longrightarrow & \oplus_{i=1}^s H^0(\langle\sigma_i\rangle, \operatorname{ad}_{E/\mathcal{O}}(1))/\varpi^r \\
\downarrow & & \downarrow \\
H^1(\Gamma_{F,S}, \operatorname{ad}_{\mathcal{O}_r}(1)) & \longrightarrow & \oplus_{i=1}^s H^1(\langle\sigma_i\rangle, \operatorname{ad}_{\mathcal{O}_r}(1)) \\
\downarrow & & \downarrow \\
H^1(\Gamma_{F,S}, \operatorname{ad}_{E/\mathcal{O}}(1))[\varpi^r] & \longrightarrow & \oplus_{i=1}^s H^0(\langle\sigma_i\rangle, \operatorname{ad}_{E/\mathcal{O}}(1))[\varpi^r] \\
\downarrow & & \downarrow \\
0 & & 0.
\end{array}$$

Note that $H^0(\Gamma_{F,S}, \mathrm{ad}_{E/\mathcal{O}}(1))$ has finite length by irreducibility of ρ. It follows from the snake lemma that the kernel of

$$H^1(\Gamma_{F,S}, \mathrm{ad}_{\mathcal{O}_r}(1)) \to \bigoplus_{i=1}^{s} H^1(\langle \sigma_i \rangle, \mathrm{ad}_{\mathcal{O}_r}(1))$$

has length bounded independently of r and N. By the Chebotarev density theorem, we can then find places $v_1, \dots v_s$ of F with Frobenius elements $\mathrm{Frob}_{v_1}, \dots, \mathrm{Frob}_{v_s}$ approximating $\sigma_1, \dots, \sigma_s$, and thus we obtain statement (4) of the proposition. Now, we have an isomorphism

$$H^1(\Gamma_{F,S}, \mathrm{ad}_{E/\mathcal{O}}(1)) \cong (E/\mathcal{O})^{s_0} \oplus M$$

where M is a finite length $\mathcal{O}$-module and $s_0 \leq s$ is the corank defined above. Thus, the problem reduces to finding, for every non-zero

$$f \colon E/\mathcal{O} \to H^1(\Gamma_{F,S}, \mathrm{ad}_{E/\mathcal{O}}(1)),$$

an element $\sigma \in G_{F(\zeta_{p^\infty})}$ such that $\rho(\sigma)$ has n distinct eigenvalues and the composition

$$E/\mathcal{O} \to H^1(\Gamma_{F,S}, \mathrm{ad}_{E/\mathcal{O}}(1)) \to H^1(\langle \sigma \rangle, \mathrm{ad}_{E/\mathcal{O}}(1)) \cong \frac{\mathrm{ad}_{E/\mathcal{O}}(1)}{(\sigma - 1)\,\mathrm{ad}_{E/\mathcal{O}}(1)}$$

is non-zero. Let $F_\infty = F(\zeta_{p^\infty})$ and $L_\infty = \overline{F}^{\ker \mathrm{ad}_E(1)}$. Then $F_\infty \subseteq L_\infty$ since $E(1) \subset \mathrm{ad}_E(1)$. The short exact sequence

$$0 \to \mathrm{ad}_{\mathcal{O}} \to \mathrm{ad}_E \to \mathrm{ad}_{E/\mathcal{O}} \to 0$$

induces an exact sequence (with $\Gamma(L_\infty/F) = \mathrm{Gal}(L_\infty/F)$)

$$H^1\big(\Gamma(L_\infty/F), \mathrm{ad}_E(1)\big) \to H^1\big(\Gamma(L_\infty/F), \mathrm{ad}_{E/\mathcal{O}}(1)\big) \to H^2\big(\Gamma(L_\infty/F), \mathrm{ad}_{\mathcal{O}}(1)\big).$$

The first term vanishes by [Kis04, Lem. 6.2] and the assumption that Π is tempered at all finite places. The third term is a finitely generated $\mathcal{O}$-module. It follows that the middle term is annihilated by a power of ϖ. Thus, the exactness of the inflation-restriction sequence

$$0 \to H^1\big(\Gamma(L_\infty/F), \mathrm{ad}_{E/\mathcal{O}}(1)\big) \to H^1\big(\Gamma_{F,S}, \mathrm{ad}_{E/\mathcal{O}}(1)\big) \to H^1\big(\Gamma(F_S/L_\infty), \mathrm{ad}_{E/\mathcal{O}}(1)\big)^{\Gamma(L_\infty/F)}$$

implies that the composition

$$\begin{aligned} E/\mathcal{O} \xrightarrow{f} H^1\big(\Gamma_{F,S}, \mathrm{ad}_{E/\mathcal{O}}(1)\big) &\to H^1\big(\Gamma(F_S/L_\infty), \mathrm{ad}_{E/\mathcal{O}}(1)\big)^{\Gamma(L_\infty/F)} \\ &\cong \mathrm{Hom}_{\Gamma(F_S/F)}\left(\Gamma(L_{\infty,S_{L_\infty}}/L_\infty), \mathrm{ad}_{E/\mathcal{O}}(1)\right) \end{aligned}$$

is non-zero. Let $W \subset \mathrm{ad}_{E/\mathcal{O}}(1)$ be the $\mathcal{O}$-submodule spanned by $\{f(x)(\sigma) \mid x \in E/\mathcal{O}, \sigma \in G_{L_\infty}\}$. Then W is non-zero and divisible. By [NT23, Lem. 2.22] there exists $\sigma \in G_{F_\infty}$ such that $\rho(\sigma)$ has n distinct eigenvalues in E and $W \not\subset (\sigma - 1)\,\mathrm{ad}_{E/\mathcal{O}}(1)$. Thus, for some $m \geq 0$ and $\tau \in G_{L_\infty}$ we have

$$f(1/\varpi^m)(\tau) \notin (\sigma - 1)(\mathrm{ad}_{E/\mathcal{O}}(1)).$$

If $f(1/\varpi^m)(\sigma) \notin (\sigma - 1)(\mathrm{ad}_{E/\mathcal{O}}(1))$ then we are done. If not, then $\tau\sigma$ has the sought property, i.e.

$$E/\mathcal{O} \to H^1(F_S, \mathrm{ad}_{E/\mathcal{O}}(1)) \to \frac{\mathrm{ad}_{E/\mathcal{O}}(1)}{(\tau\sigma - 1)\,\mathrm{ad}_{E/\mathcal{O}}(1)}$$

is non-zero, since τ acts trivially on $\mathrm{ad}_E(1)$. □

4.2. **Patching.** In this section, we set up the patching argument, basically following [A'C24] and [GV18] but with some adjustments. Define deformation problems

$$\mathbf{S} \coloneqq (\bar{\rho}, S, \{\mathcal{D}_{\bar{\rho}_v}^{\lambda_v, \preceq \tau_v}\}_{v \in S_p} \cup \{\mathcal{D}_{\bar{\rho}_v}^{\square}\}_{v \in S \setminus S_p})$$
$$\mathbf{S}' \coloneqq (\bar{\rho}, S, \{\mathcal{D}_{\bar{\rho}_v}^{\square}\}_{v \in S})$$

We let $\mathcal{R}_{\mathbf{S}}, \mathcal{R}_{\mathbf{S}'} \in \mathcal{CNL}_k$ denote the corresponding derived deformation rings.

For a Taylor–Wiles-datum Q, we define an augmented deformation problem

$$\mathbf{S}_Q = (\bar{\rho}, S \cup Q, \{\mathcal{D}_{\bar{\rho}_v}^{\lambda_v, \preceq \tau_v}\}_{v \in S_p} \cup \{\mathcal{D}_{\bar{\rho}_v}^{\square}\}_{v \in S \setminus S_p} \cup \{\mathcal{D}_{\bar{\rho}_v}^{\square}\}_{v \in Q}).$$

and similarly for $\mathbf{S}'_Q$. Applying Proposition 4.1.2, we obtain $q, l \geq 0$ and for every $N \geq 1$ a Taylor–Wiles datum Q_N satisfying the conclusions of the proposition. We let

$$\mathcal{R}_N \coloneqq \mathcal{R}^S_{\mathbf{S}_{Q_N}} \text{ and } \mathcal{R}'_N \coloneqq \mathcal{R}^S_{\mathbf{S}'_{Q_N}}$$

be the S-framed type-$\mathbf{S}_{Q_N}$ and type-$\mathbf{S}'_{Q_N}$ derived deformation rings. We also let $R_N \coloneqq \pi_0(\mathcal{R}_N)$ and $R'_N \coloneqq \pi_0(\mathcal{R}'_N)$ denote the underlying classical deformation rings. These objects and the ones introduced below depend on the specific choice of Q_N and not just N, but in our applications, we will fix a choice of Q_N for every $N \geq 1$, motivating our simplified notation.

Remark 4.2.1. In [GV18], the authors consider deformations of $\bar{\rho}_v$ for $v \in Q_N$ valued in the diagonal torus $\mathbf{T} \subset \mathbf{G}$. The local $\mathbf{T}$-valued deformation functors at Taylor–Wiles places are proved in [GV18, Lem. 8.3] to be equivalent to their $\mathbf{G}$-valued counterparts, and for typographical convenience we will not make a distinction.

We let $\mathcal{T} \coloneqq \mathcal{O}[[T_1, \dots, T_{n^2|S|-1}]]$ be the power series ring in the 'S-frame variables' and define, for every $v \in Q_N$, $N_v \coloneqq \max\{r \in \mathbb{Z} \mid v \equiv 1 \bmod p^r\} \geq N$ and

$$\mathcal{S}_v \coloneqq \mathcal{T}[[Y_1, \dots, Y_n]]/\langle (1+Y_i)^{p^{N_v}} - 1 \rangle \text{ and } \mathcal{S}_N \coloneqq \widehat{\bigotimes}_{\mathcal{T}, v \in Q_N} \mathcal{S}_v$$

The ring $\mathcal{S}_N$ is static; to emphasise this we sometimes write $S_N \coloneqq \pi_0(\mathcal{S}_N) \simeq \mathcal{S}_N$.

Lemma 4.2.2. *Let $r \leq N$. Then*

$$\mathrm{T}^1_{\mathcal{O}_r}(\mathcal{S}_N) \cong \prod_{v \in Q_N} H^2(\Gamma_{F_v}, \operatorname{ad} \xi_r)$$

Proof. By [GV18, Eq. 8.14], the static ring $\mathcal{S}_N \otimes^{\mathbf{L}}_{\mathcal{T}} \mathcal{O}$ is the framed deformation ring of the trivial representation of the product of inertia subgroups $\prod_{v \in Q_N} I_v$. Hence, the result follows from Proposition 3.6.6. □

Lemma 4.2.3. [GV18, Eq. 11.4] *For every $N \geq 1$, we have natural maps $\mathcal{S}_N \to \mathcal{R}'_N \to \mathcal{R}_N$ and R'_N and R_N are local S_N-algebras. Moreover, there are weak equivalences*

$$\mathcal{R} \simeq \mathcal{R}_N \otimes^{\mathbf{L}}_{\mathcal{S}_N} \mathcal{O} \text{ and } \mathcal{R}' \simeq \mathcal{R}'_N \otimes^{\mathbf{L}}_{\mathcal{S}_N} \mathcal{O}.$$

As usual in patching arguments, we will approximate $\mathcal{R}$ by artinian quotients which, 'in the limit', are of a simple form. We let

$$\begin{aligned} R_\infty &\coloneqq R_{\mathbf{S}}^{S,\mathrm{loc}}[[X_1,\dots,X_{nq-n^2[F^+:\mathbb{Q}]}]],\\ S_\infty &\coloneqq \mathcal{T}[[Y_1,\dots,Y_{qn}]],\\ \mathfrak{a} &= (T_1,\dots,T_{n^2|S|-1},Y_1,\dots,Y_{qn}) \subset S_\infty,\\ \mathfrak{a}_N &= \langle (1+Y_i)^{p^{N_{v_i}}}-1, (1+T_i)^{p^{N_{v_i}}}-1\rangle \subset S_\infty,\\ \mathfrak{c}_N &= \left(p^N, \mathfrak{a}_{Q_N}\right) \subset S_\infty, \end{aligned}$$

so that $S_N \cong S_\infty/\mathfrak{a}_N$. Then, for every $N \geq 1$, R_N is a local S_N-algebra, we have $R_N/\mathfrak{a} \cong \pi_0(\mathcal{R}_{\mathbf{S}})$, and by Proposition 4.1.2 we may fix a surjection $R_\infty \twoheadrightarrow R_N$. Note that $\mathfrak{c}_N \subset S_\infty$ is an open ideal and

$$\mathfrak{m}_{S_\infty}^{p^N} \subset \mathfrak{c}_N \subset \mathfrak{m}_{S_\infty}^N,$$

so that $S_\infty \cong \varprojlim_N S_\infty/\mathfrak{c}_N$ as topological $\mathcal{O}$-algebras. We define artinian quotients

$$\begin{aligned} \overline{R}_N &\coloneqq R^S_{\mathbf{S}_{Q_N}}/\mathfrak{m}^N_{R_\infty},\\ \overline{S}_N &\coloneqq S_\infty/\mathfrak{c}_N \cong S_N/\varpi^N. \end{aligned}$$

Then $\overline{R}_N$ is a local $\overline{S}_N$-algebra.

Theorem 4.2.4. [A'C24] *With the same notation and assumptions as Theorem 4.0.1, we have:*

(i) *There exists an infinite subset $\mathbb{N}^* \subseteq \mathbb{N}$ and isomorphisms of R_∞-algebras $\overline{R}_N/\mathfrak{m}^M_{R_\infty} \cong \overline{R}_M$ for $M \leq N$ in $\mathbb{N}^*$, such that we have an isomorphism*

$$\widehat{R}_{\infty,\mathfrak{p}} \to (\varprojlim_{\mathbb{N}^*} \overline{R}_N)^\wedge_{\mathfrak{p}},$$

where the transition maps in the inverse system are given by the compositions

$$\overline{R}_N \twoheadrightarrow \overline{R}_N/\mathfrak{m}^M_{R_\infty} \xrightarrow{\sim} \overline{R}_M,$$

and $\mathfrak{p} = \ker(R_\infty \to \mathcal{O})$ corresponds to ξ.

(ii) *There is an R_∞-module M_∞ such that $\widehat{M}_{\infty,\mathfrak{p}}$ is free over $\widehat{R}_{\infty,\mathfrak{p}}$ and*

$$\widehat{M}_{\infty,\mathfrak{p}} \otimes^{\mathbf{L}}_{\widehat{S}^\circ_{\infty,\mathfrak{a}}} E \simeq R\Gamma(K,\mathcal{V}_\lambda)^\vee_{\mathfrak{p}}.$$

Consequently, the graded ring

$$\pi_*\big(\widehat{R}_{\infty,\mathfrak{p}} \otimes^{\mathbf{L}}_{\widehat{S}^\circ_{\infty,\mathfrak{a}}} E\big) \cong \mathrm{Tor}_*^{\widehat{S}^\circ_{\infty,\mathfrak{a}}}\big(\widehat{R}_{\infty,\mathfrak{p}}, E\big)$$

acts freely on the graded module

$$\pi_*\big(\widehat{M}_{\infty,\mathfrak{p}} \otimes^{\mathbf{L}}_{\widehat{S}^\circ_{\infty,\mathfrak{a}}} E\big) \cong H_*(K,\mathcal{V}_\lambda)_\Pi$$

(iii) *The geometric Bloch–Kato Selmer group vanishes:*

$$H^1_g(\Gamma_{F,S}, \mathrm{ad}\,\rho) = 0.$$

Proof. All three essentially follow from the main result of [A'C24]. Since our local conditions at $v \mid p$ differ slightly, we spell out the proof which is a straightforward application of the abstract patching lemma [A'C24, Lem. 2.6.4]. For $N \geq 1$, one defines (see [A'C24, Prop. 5.3.3] for details):

- An open compact subgroup $K_0(Q_N) = \prod_v K_{0,v}(Q_N) \subset K$ where $K_{0,v}(Q_N) = K_v$ for $v \notin Q_N$ and $K_{0,v}(Q_N) = \mathrm{Iw}_v$ is the matrices which are upper triangular modulo ϖ_v for $v \in Q_N$.
- An open compact subgroup $K_1(Q_N) = \prod_v K_{1,v}(Q_N) \subset K$ where $K_{1,v}(Q_N) = K_v$ for $v \notin Q_N$ and $K_{1,v}(Q_N) = \mathrm{Iw}_{1,v} \subset \mathrm{Iw}_v$ is the matrices which are upper triangular and unipotent modulo ϖ_v for $v \in Q_N$.
- For any object M with a $\mathbb{T}^S$-action, an enlarged (faithful) Hecke algebra $\mathbb{T}^{S\cup Q_N}_{Q_N}(M)$ obtained by adjoining Hecke operators at the places $v \in Q_N$.
- Maximal ideals $\mathfrak{m}_1, \mathfrak{m}_0$ such that, if $\mathfrak{m} \subset \mathbb{T}^S(R\Gamma(X_K, \mathcal{V}_{\lambda,\tau}))$ is the ideal corresponding to Π, we have morphisms

$$\mathbb{T}^{S\cup Q_N}_{Q_N}\left(R\Gamma(X_{K_1(Q_N)}, \mathcal{V}_{\lambda,\tau})\right)_{\mathfrak{m}_1} \to \mathbb{T}^{S\cup Q_N}\left(R\Gamma(X_{K_0(Q_N)}, \mathcal{V}_{\lambda,\tau})\right)_{\mathfrak{m}_0} \xrightarrow{\sim} \mathbb{T}^S\left(R\Gamma(X_K, \mathcal{V}_{\lambda,\tau})\right)_{\mathfrak{m}}.$$

  where the first is given by taking $K_0(Q_N)/K_1(Q_N)$-coinvariants, and corresponding quasi-isomorphisms of complexes

$$R\Gamma\left(K_0(Q_N)/K_1(Q_N), R\Gamma(X_{K_1(Q_N)}, \mathcal{V}_{\lambda,\tau})_{\mathfrak{m}_1}\right) \xrightarrow{\sim} R\Gamma(X_{K_0(Q_N)}, \mathcal{V}_{\lambda,\tau})_{\mathfrak{m}_0} \xrightarrow{\sim} R\Gamma(X_K, \mathcal{V}_{\lambda,\tau})_{\mathfrak{m}}.$$

In the numbering of [A'C24, Lem. 2.6.4], we have 7 statements to verify.

(1) R_∞ is a complete noetherian local $\mathcal{O}$-algebra.
(2) R_N is a local S_N-algebra and a quotient of R_∞.
(3) Define complexes (or animated modules)

$$C_0 := R\,\mathrm{Hom}_{\mathcal{O}}\left(R\Gamma(X_K, \mathcal{V}_{\lambda,\tau})_{\mathfrak{m}}, \mathcal{O}\right)$$
$$C_N := R\,\mathrm{Hom}_{\mathcal{O}}\left(R\Gamma(X_{K_1(Q_N)}, \mathcal{V}_{\lambda,\tau})_{\mathfrak{m}_1}, \mathcal{T}\right)/\mathfrak{a}_N$$

  where the right-hand sides denote minimal representatives in the derived category of $\mathcal{O}$ and S_N, respectively. Then $C^\bullet_N/\mathfrak{a} \simeq C^\bullet_0$ by the fourth bullet point above.
(4) By Proposition 2.1.1, $C_0[1/\varpi]$ has homotopy concentrated in degrees $[q_0, q_0 + l_0]$ and is not exact. Moreover, $\dim(S_\infty)_{\mathfrak{a}} = n^2|S| - 1 + qn$ and using Proposition 3.5.11 and Theorem 3.5.13(2) we deduce

$$\dim R_\infty[1/\varpi] \le n^2|S| + qn - n[F^+ : \mathbb{Q}] = \dim(S_\infty)_{\mathfrak{a}} - l_0.$$

(5) For $N \ge 1$, we let

$$T_N := \mathbb{T}^{S\cup Q_N}_{Q_N}(C_N)_{\mathfrak{m}_1} \cdot \mathcal{T} \subset \mathrm{End}_{\mathbf{D}(S_N)}(C_N),$$

  where $\mathfrak{m}_1, \mathfrak{m}_0$ are maximal ideals as in [A'C24, Prop. 5.3.3], so that the image of T_N in $\mathrm{End}_{\mathbf{D}(S_0)}(C_0)$ equals

$$\mathbb{T}^{S\cup Q_N}(C_0) \subset \mathbb{T}^S(C_0) =: T_0.$$

(6) Using Proposition 3.5.10 and [Hev24, Thm. 2.24], we obtain for every N an S_N-algebra surjection (after taking a tensor product with $\mathcal{T}$)

$$R_N \to T_N/I_N$$

  where $I_N^\delta = 0$ for some δ independent of N, obtaining a commutative square

$$\begin{array}{ccc} R_N & \twoheadrightarrow & T_N/I_N \\ \downarrow & & \downarrow \\ R_0 & \twoheadrightarrow & T_0/I_0 \end{array}$$

(7) Since the Weil–Deligne representations $\mathrm{WD}(\rho_v)$ are generic for every $v \in S$, the representation ρ defines a smooth point of $R_\infty[1/\varpi] \cong R_{\mathbf{S}}^{S,\mathrm{loc}}[[X_1, \dots, X_{nq-n^2[F^+:\mathbb{Q}]}]]$.

Now, [A'C24, Lem. 2.6.4, Lem. 2.3.3] provide a subsequence $\mathbb{N}^* \subseteq \mathbb{N}$, morphisms $\overline{R}_N \to \overline{R}_M$ for every $M \leq N$ in $\mathbb{N}^*$ as in the statement, and an isomorphism as in (i):

$$\widehat{R}_{\infty,\mathfrak{p}} \cong \mathbf{P}_{R_\infty}(R_N)^\wedge_{\mathfrak{p}} \cong (\varprojlim_N \overline{R}_N)^\wedge_{\mathfrak{p}}.$$

Statement (ii) follows from [A'C24, Thm. 5.3.5], and statement (iii) follows from [A'C24, Lem. 2.6.4]. □

Remark 4.2.5. The subset $\mathbb{N}^* \subseteq \mathbb{N}$ and the maps $\overline{R}_N \to \overline{R}_M$ depends on a choice of non-principal ultrafilter on $\mathbb{N}$. As such, the isomorphism provided by (1) is non-canonical. This ambiguity will not play a role in the sequel; we fix from now on a subset $\mathbb{N}^*$ and maps as in the theorem.

In light of part (ii) of Theorem 4.2.4, we see that to define an action as in Theorem 4.0.1, it suffices to prove a weak equivalence

$$(\mathcal{R}_{\mathbf{S}} \otimes_{\mathcal{O}} E)^\wedge_\xi \simeq \widehat{R}_{\infty,\mathfrak{p}} \otimes^{\mathbf{L}}_{\widehat{S}^\circ_{\infty,\mathfrak{a}}} E,$$

and to identify the right-hand side with the exterior algebra on the Bloch–Kato Selmer group.

4.3. **Tangent complex calculations.** Our proof of Theorem 4.0.1 will rely on a comparison of tangent complexes. In this section, we collect some preliminary calculations for easy reference later.

For every $M \leq N$ in $\mathbb{N}^*$, Theorem 4.2.4(1) provides a commutative diagram

$$\begin{array}{ccccc} \overline{R}_N & \longleftarrow & \overline{S}_N & \longrightarrow & \mathcal{O}_N \\ \downarrow & & \downarrow & & \downarrow \\ \overline{R}_M & \longleftarrow & \overline{S}_M & \longrightarrow & \mathcal{O}_M. \end{array}$$

We let $\mathcal{C}_N := \overline{R}_N \otimes^{\mathbf{L}}_{\overline{S}_N} \mathcal{O}_N$ and denote by $e_{N,M} \colon \mathcal{C}_N \to \mathcal{C}_M$ the induced map on derived tensor products.

Lemma 4.3.1. *With notation as above, we have the following:*

(1) *The limit $\mathcal{C}_\infty = \lim_N \mathcal{C}_N$ is noetherian.*
(2) *The tangent complex of the completion $(\mathcal{C}_\infty \otimes_{\mathcal{O}} E)^\wedge_\xi \cong \mathcal{C}^\wedge_{\infty,\mathfrak{p}}$ has cohomology concentrated in degrees $[0,1]$ and*

$$\dim_E \mathrm{T}^0_E(\mathcal{C}^\wedge_{\infty,\mathfrak{p}}) - \dim_E \mathrm{T}^1_E(\mathcal{C}^\wedge_{\infty,\mathfrak{p}}) = -l_0.$$

Proof. (1) By Theorem 3.6.3, it suffices to prove that $\mathrm{T}^i_k(\mathcal{C}_\infty) = \operatorname{colim} \mathrm{T}^i_k(\mathcal{C}_N)$ is finite-dimensional for every $i \geq 0$. Consider the fibre sequence

$$\mathrm{T}^\bullet_k(\mathcal{C}_N) \to \mathrm{T}^\bullet_k(\overline{R}_N) \oplus \mathrm{T}^\bullet_k(\mathcal{O}_N) \to \mathrm{T}^\bullet_k(\overline{S}_N) \xrightarrow{+1}.$$

Filtered colimits are exact, whence we have a sequence

$$\mathrm{T}^\bullet_k(\mathcal{C}_\infty) \to \mathrm{T}^\bullet_k(\varprojlim_N \overline{R}_N) \to \mathrm{T}^\bullet_k(S_\infty) \xrightarrow{+1}.$$

For every $N \geq 0$, we have a surjection $R_\infty \twoheadrightarrow \overline{R}_N$, and thus $R_\infty \twoheadrightarrow \lim_N \overline{R}_N \simeq \varprojlim_N \overline{R}_N$, where $\varprojlim$ denotes the 1-categorical inverse limit. Since R_∞ and S_∞ are noetherian, it follows from Theorem 3.6.3 and the fibre sequence above that $\mathrm{T}^i_k(\mathcal{C}_\infty)$ is finite-dimensional for every $i \geq 0$.

(2) For any $r \leq N$, we have a fibre sequence

$$\mathrm{T}^\bullet_{\mathcal{O}_r}(\mathcal{C}_N) \to \mathrm{T}^\bullet_{\mathcal{O}_r}(\overline{R}_N) \oplus \mathrm{T}^\bullet_{\mathcal{O}_r}(\mathcal{O}_N) \to \mathrm{T}^\bullet_{\mathcal{O}_r}(\overline{S}_N) \xrightarrow{+1}.$$

Passing to the colimit over $N \geq r$ and limit over r, then inverting p yields a fibre sequence

$$\big(\lim_r \operatorname*{colim}_{N \geq r} \mathrm{T}^\bullet_{\mathcal{O}_r}(\mathcal{C}_N)\big) \otimes_{\mathcal{O}} E \to \mathrm{T}^\bullet_E(\widehat{R}_{\infty,\mathfrak{p}}) \to \mathrm{T}^\bullet_E(\widehat{S}^\circ_{\infty,\mathfrak{a}}) \xrightarrow{+1},$$

The first term is identified with $\mathrm{T}^\bullet_E(\mathcal{C}^\wedge_{\infty,\mathfrak{p}})$ by Proposition 3.7.6. Now, $\widehat{R}_{\infty,\mathfrak{p}}$ and $\widehat{S}^\circ_{\infty,\mathfrak{a}}$ are smooth E-algebras and therefore upon taking cohomology we obtain an exact sequence of E-vector spaces

$$0 \to \mathrm{T}^0_E(\mathcal{C}^\wedge_{\infty,\mathfrak{p}}) \to \mathrm{T}^0_E(\widehat{R}_{\infty,\mathfrak{p}}) \to \mathrm{T}^0_E(\widehat{S}^\circ_{\infty,\mathfrak{a}}) \to \mathrm{T}^1_E(\mathcal{C}^\wedge_{\infty,\mathfrak{p}}) \to 0.$$

Using the smoothness of $\widehat{R}_{\infty,\mathfrak{p}}$ and $\widehat{S}^\circ_{\infty,\mathfrak{a}}$ again, we deduce that

$$\dim_E \mathrm{T}^0_E(\mathcal{C}^\wedge_{\infty,\mathfrak{p}}) - \dim_E \mathrm{T}^1_E(\mathcal{C}^\wedge_{\infty,\mathfrak{p}}) = \dim \widehat{R}_{\infty,\mathfrak{p}} - \dim \widehat{S}^\circ_{\infty,\mathfrak{a}} = -l_0.$$

□

Proposition 4.3.2. *Let $r \leq N$. With assumptions and notation as above, the kernel*

$$\ker\big(\mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R}_N) \to \mathrm{T}^1_{\mathcal{O}_r}(\mathcal{S}_N)\big)$$

is a finite length $\mathcal{O}$-module with length bounded independently of r and N.

Proof. We begin by proving the same statement for $\mathcal{R}'_N$, the derived deformation ring with trivial local conditions at all $v \in S$. That is, we let $r \leq N$ and prove that the kernel

$$\ker\big(\mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R}'_N) \to \mathrm{T}^1_{\mathcal{O}_r}(\mathcal{S}_N)\big)$$

is a finite length $\mathcal{O}$-module with length bounded independently of r and N. By Proposition 3.6.6 and Lemma 4.2.2(4), we have isomorphisms

$$\mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R}'_N) \cong H^2(\Gamma_{F,SQ_N}, \operatorname{ad}\xi_r)$$
$$\mathrm{T}^1_{\mathcal{O}_r}(\mathcal{S}_N) \cong \prod_{v \in Q_N} H^2(\Gamma_{F_v}, \operatorname{ad}\xi_r)$$

We factor the map $\mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R}'_N) \to \mathrm{T}^1_{\mathcal{O}_r}(S_N)$ as $\beta \circ \alpha$ where

$$H^2(\Gamma_{F,SQ_N}, \operatorname{ad}\xi_r) \xrightarrow{\alpha} \prod_{v \in SQ_N} H^2(\Gamma_{F_v}, \operatorname{ad}\xi_r) \xrightarrow{\beta} \prod_{v \in Q_N} H^2(\Gamma_{F_v}, \operatorname{ad}\xi_r)$$

to obtain an exact sequence

$$0 \to \ker(\alpha) \to \ker(\beta \circ \alpha) \to \ker(\beta).$$

To prove the proposition, it suffices to show that $\ker(\alpha)$ and $\ker(\beta)$ have finite length bounded independently of r and N. To see this, consider the Selmer system

$$\{\mathcal{L}_{v,r}\}_{v \in SQ_N} = \{H^1(\Gamma_{F_v}, \operatorname{ad}\xi_r)\}_{v \in SQ_N} \cup \{H^1_{\mathrm{ur}}(\Gamma_{F_v}, \operatorname{ad}\xi_r)\}_{v \notin SQ_N}.$$

It follows from the Poitou–Tate sequence that we have an exact sequence

$$H^1(\Gamma_{F,SQ_N}, \operatorname{ad}\xi_r(1)) \to H^2(\Gamma_{F,SQ_N}, \operatorname{ad}\xi_r) \xrightarrow{\alpha} \prod_{v \in SQ_N} H^2(\Gamma_{F_v}, \operatorname{ad}\xi_r).$$

In particular, we have

$$\operatorname{length}_{\mathcal{O}} \ker(\alpha) \leq \operatorname{length}_{\mathcal{O}} H^1(\Gamma_{F,SQ_N}, \operatorname{ad}\xi_r(1)),$$

which is bounded from above independently of r, N by Proposition 4.1.2(4). Now,

$$\ker(\beta) = \prod_{v \in S} H^2(\Gamma_{F_v}, \operatorname{ad} \xi_r) = \mathrm{T}^1_{\mathcal{O}_r}(R^{S,\mathrm{loc}}_{\mathbf{S}'})$$

and since the characteristic 0 point of $\operatorname{Spec} R^{S,\mathrm{loc}}_{\mathbf{S}'}$ corresponding to ξ is smooth by assumption, the $\mathcal{O}$-module $\ker(\beta)$ has finite length bounded independently of r (and N) by Proposition 3.7.7. This completes the proof of the initial claim.

We now consider the kernel

$$\ker\left(\mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R}_N) \to \mathrm{T}^1_{\mathcal{O}_r}(\mathcal{S}_N)\right)$$

in the statement of the proposition. The equivalence

$$\mathcal{R}_N \simeq \mathcal{R}'_N \otimes^{\mathbf{L}}_{R^{S,\mathrm{loc}}_{\mathbf{S}'}} R^{S,\mathrm{loc}}_{\mathbf{S}}$$

induces a commutative diagram

$$\begin{array}{ccccc} 0 & \longrightarrow & \operatorname{coker}^0_{r,N} & \longrightarrow & \mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R}_N) \longrightarrow \mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R}'_N) \oplus \mathrm{T}^1_{\mathcal{O}_r}(R^{S,\mathrm{loc}}_{\mathbf{S}}) \\ & & & & \downarrow{\scriptstyle \gamma} \qquad\qquad \downarrow \\ & & & & \mathrm{T}^1_{\mathcal{O}_r}(\mathcal{S}_N) \longleftarrow \mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R}'_N) \end{array}$$

where

$$\operatorname{coker}^0_{r,N} \coloneqq \operatorname{coker}\left(\mathrm{T}^0_{\mathcal{O}_r}(\mathcal{R}'_N) \oplus \mathrm{T}^0_{\mathcal{O}_r}(R^{S,\mathrm{loc}}_{\mathbf{S}}) \to \mathrm{T}^0_{\mathcal{O}_r}(R^{S,\mathrm{loc}}_{\mathbf{S}'})\right).$$

The top row is exact and the right vertical map is the projection onto the first coordinate. Thus, we see that $\operatorname{coker}^0_{r,N} \hookrightarrow \ker(\gamma)$, so that γ factors as

$$\mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R}_N) \twoheadrightarrow \mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R}_N)/\operatorname{coker}^0_{r,N} \xrightarrow{\tilde{\gamma}} \mathrm{T}^1_{\mathcal{O}_r}(\mathcal{S}_N),$$

and there is a short exact sequence

$$0 \to \operatorname{coker}^0_{r,N} \to \ker(\gamma) \to \ker(\tilde{\gamma}) \to 0.$$

By Proposition 4.1.2(4), $\operatorname{coker}^0_{r,N}$ is annihilated by a power of ϖ which is independent of r and N. Thus, it suffices to establish the same property for $\ker(\tilde{\gamma})$. By the first diagram, we have

$$\begin{aligned} \ker(\tilde{\gamma}) &\subseteq \ker\left(\mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R}'_N) \oplus \mathrm{T}^1_{\mathcal{O}_r}(R^{S,\mathrm{loc}}_{\mathbf{S}}) \twoheadrightarrow \mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R}'_N) \to \mathrm{T}^1_{\mathcal{O}_r}(\mathcal{S}_N)\right) \\ &\cong \mathrm{T}^1_{\mathcal{O}_r}(R^{S,\mathrm{loc}}_{\mathbf{S}}) \oplus \ker\left(\mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R}'_N) \to \mathrm{T}^1_{\mathcal{O}_r}(\mathcal{S}_N)\right), \end{aligned}$$

whereby it suffices to show that $\mathrm{T}^1_{\mathcal{O}_r}(R^{S,\mathrm{loc}}_{\mathbf{S}})$ and $\ker(\mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R}'_N) \to \mathrm{T}^1_{\mathcal{O}_r}(\mathcal{S}_N))$ are annihilated by powers of ϖ that are independent of r and N. The former statement follows from Proposition 3.7.7 together with the genericity assumption on $\mathrm{WD}(\rho_v)$, Proposition 3.5.11 and Theorem 3.5.13; the latter statement is the claim we proved in the beginning of this proof. □

4.4. **Proof of the main theorem.**

Theorem 4.4.1. *With $\mathcal{C}_N = \overline{R}_N \otimes^{\mathbf{L}}_{\overline{S}_N} \overline{S}^{\mathrm{ur}}_N$ as above, let*

$$\mathcal{F}_\infty = \operatorname*{colim}_N \operatorname{Map}_{/k}(\mathcal{C}_N, -) \colon \mathcal{A}rt_k \to \mathcal{A}ni.$$

Then there exists a morphism of formally cohesive functors

$$\mathcal{F}_\infty \to \operatorname{Map}_{/k}(\mathcal{R}_{\mathbf{S}}, -)$$

lifting $\xi\colon \mathcal{R}_{\mathbf{S}} \to \mathcal{O}$ *such that the induced morphism*

$$(\mathcal{F}_\infty)_{(\xi)} \to (\mathcal{D}_{\mathbf{S}})_{(\xi)}$$

is a weak equivalence in $\mathcal{CNL}_E$.

The notation might be confusing; the functor $\mathcal{F}_\infty$ is not of 'patched' level, but rather the limit as $N \to \infty$ of mod ϖ^N approximations $\mathcal{C}_N$ of $\mathcal{R}_{\mathbf{S}}$.

Remark 4.4.2. Recall that the inverse system $(\mathcal{C}_N)_{N\in\mathbb{N}^*}$ is is not canonically defined. The proof of the theorem above is also non-constructive and as such, the provided weak equivalence is a priori non-canonical and depends on our fixed choice of non-principal ultrafilter on $\mathbb{N}$. In [GV18, §15], the authors are able to prove, under their assumptions, that the equivalence is canonical and independent of any choices in the Taylor–Wiles process. It would be interesting to prove a similar result in characteristic 0 without introducing extra assumptions.

The proof of Theorem 4.4.1 is subdivided into separate results. The morphism of functors of the theorem is induced by a morphism of rings

$$\mathcal{R}_{\mathbf{S}} \to \mathcal{C}_\infty = \lim_N \mathcal{C}_N.$$

We note that, a priori, it is not clear whether the natural morphism

$$\operatorname*{colim}_N \operatorname{Map}_{/k}(\mathcal{C}_N, -) \to \operatorname{Map}_{/k}(\mathcal{C}_\infty, -)$$

is an equivalence. We begin by establishing that a homotopy class of maps $\mathcal{R} \to \mathcal{C}_\infty$ is simply a compatible sequence of homotopy classes of maps $\mathcal{R} \to \mathcal{C}_N$. We use the notation

$$[\mathcal{R}, -] = \pi_0\big(\operatorname{Map}_{/k}(\mathcal{R}, -)\big).$$

Lemma 4.4.3. *Let* $\mathcal{R} \in \mathcal{CNL}_k$ *and suppose* $(\mathcal{C}_N)_{N\geq 1}$ *is a projective system in* $\mathcal{CNL}_k$ *such that* $\tau_{\leq j}\mathcal{C}_N \in \mathcal{A}rt_k$ *for every* $j \geq 0$*, with (*∞*-categorical) limit* $\mathcal{C}_\infty = \lim \mathcal{C}_N$*. Then the natural maps*

$$[\mathcal{R}, \mathcal{C}_\infty] \to \varprojlim_N [\mathcal{R}, \mathcal{C}_N] \to \varprojlim_{j,N} [\mathcal{R}, \tau_{\leq j}\mathcal{C}_N]$$

are bijections (of sets), where $\varprojlim$ *denotes the 1-categorical inverse limit in* $\mathbf{Set}$.

Proof. By [GV18, Lem. A.9], there exists a short exact sequence of pointed sets (for any basepoint)

$$0 \to \varprojlim{}^1_{j,N} \pi_1\big(\operatorname{Map}_{/k}(\mathcal{R}, \tau_{\leq j}\mathcal{C}_N)\big) \to [\mathcal{R}, \mathcal{C}_\infty] \to \varprojlim_{j,N} [\mathcal{R}, \tau_{\leq j}\mathcal{C}_N] \to 0.$$

Since $\tau_{\leq j}\mathcal{C}_N \in \mathcal{A}rt_k$, Lemma 3.1.5 implies that $\operatorname{Map}_{/k}(\mathcal{R}, \tau_{\leq j}\mathcal{C}_N)$ has finite homotopy groups. Hence, by the Mittag–Leffler criterion, the $\varprojlim^1$-term in the sequence vanishes, and we obtain

$$[\mathcal{R}, \mathcal{C}_\infty] \cong \varprojlim_{j,N} [\mathcal{R}, \tau_{\leq j}\mathcal{C}_N].$$

It follows that $[\mathcal{R}, \mathcal{C}_\infty] \to \varprojlim_N [\mathcal{R}, \mathcal{C}_N]$ is injective. It is automatically surjective by the existence of the short exact sequence

$$0 \to \varprojlim{}^1_N \pi_1\big(\operatorname{Map}_{/k}(\mathcal{R}, \mathcal{C}_N)\big) \to [\mathcal{R}, \mathcal{C}_\infty] \to \varprojlim_N [\mathcal{R}, \mathcal{C}_N] \to 0,$$

also deduced from [GV18, Lem. A.9]. □

Proposition 4.4.4. *For an integer* $a \geq 0$*, let* $X_N(a) \subseteq [\mathcal{R}, \mathcal{C}_N]$ *denote the subset consisting of classes* $[f]$ *such that, for every* $r \leq M \leq N$*,*

- $\mathrm{T}^0_{\mathcal{O}_r}(e_{N,M} \circ f)$ *is an isomorphism,*

- $\varpi^a \operatorname{coker}(\mathrm{T}^1_{\mathcal{O}_r}(e_{N,M} \circ f)) = 0$,

where $e_{N,M} \colon \mathcal{C}_N \to \mathcal{C}_M$ is the morphism introduced at the beginning of Section 4.3 and

$$\mathrm{T}^i_{\mathcal{O}_r}(e_{N,M} \circ f) \colon \mathrm{T}^i_{\mathcal{O}_r}(\mathcal{C}_M) \to \mathrm{T}^i_{\mathcal{O}_r}(\mathcal{R})$$

is the induced homomorphism of $\mathcal{O}_r$-modules. Then we have the following:

$(X_N(a))_{N \in \mathbb{N}^*}$ is an inverse system of non-empty profinite sets with continuous transition maps induced by $e_{N,M}$, and there exists an $a \in \mathbb{Z}$ such that $X_N(a)$ is nonempty; in particular,

$$\varprojlim_N X_N(a) \subset [\mathcal{R}, \mathcal{C}_\infty]$$

is nonempty.

Proof. We subdivide the proof into four steps. In Steps 1-3, we show that $(X_N(a))$ forms an inverse system of profinite sets with continuous transition maps, which holds independently of choice of $a \in \mathbb{Z}$. Finally, in Step 4, we show that there exists an a such that $X_N(a) \neq \emptyset$. The theorem then follows from Tychonoff's theorem.

Step 1: $(X_N(a))$ is an inverse system. Let $[f] \in X_N(a)$. Then, for any $L \leq M \leq N$, we have

$$[e_{M,L} \circ e_{N,M} \circ f] = [e_{N,L} \circ f]$$

and hence $[e_{N,M} \circ f] \in X_M(a)$ and the transition maps are compatible. Hence, $(X_N(a))_{N \in \mathbb{N}^*}$ is an inverse system.

Step 2: $X_N(a)$ is profinite. By Lemma 3.1.5,

$$[\mathcal{R}, \mathcal{C}_N] = \varprojlim_j [\mathcal{R}, \tau_{\leq j}\mathcal{C}_N]$$

is profinite. Thus, to prove that $X_N(a)$ is profinite, we show that it is the inverse limit of a pro-subset of $([\mathcal{R}, \tau_{\leq j}\mathcal{C}_N])_{j \geq 0}$. To see this, note that $\mathrm{T}^i_{\mathcal{O}_r}(\mathcal{C}_M) = \operatorname{colim}_j \mathrm{T}^i_{\mathcal{O}_r}(\tau_{\leq j}\mathcal{C}_M)$ and therefore

$$\operatorname{coker}\left(\mathrm{T}^1_{\mathcal{O}_r}(e_{N,M} \circ f)\right) = \operatorname{colim}_j \operatorname{coker}\left(\mathrm{T}^1_{\mathcal{O}_r}(\tau_{\leq j}\mathcal{C}_M) \to \mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R})\right) =: \operatorname{colim}_j Q_{j,N,M}.$$

We can formulate the condition defining $X_N(a)$ degree-wise in j using the equivalence

$$\varpi^a \operatorname{colim}_j Q_{j,N,M} = 0 \iff \forall j \geq 0 : \ \varpi^a Q_j \subseteq \ker\left(Q_{j,N,M} \to \operatorname{colim}_j Q_{j,N,M}\right)$$

Thus, we define finite subsets $X^j_N(a) \subseteq [\mathcal{R}, \tau_{\leq j}\mathcal{C}_N]$ using the expression above and conclude that $X_N(a) = \varprojlim_j X^j_N(a)$ is profinite.

Step 3: $X_N(a) \to X_M(a)$ is continuous. Let $M \leq N$ and consider $e_{N,M} \colon \mathcal{C}_N \to \mathcal{C}_M$. For any j, we have commutative diagrams (the right one a consequence of the left one)

$$\begin{array}{ccc} \mathcal{C}_N & \longrightarrow & \mathcal{C}_M \\ \downarrow & & \downarrow \\ \tau_{\leq j}\mathcal{C}_N & \longrightarrow & \tau_{\leq j}\mathcal{C}_M \end{array} \qquad \begin{array}{ccc} [\mathcal{R}, \mathcal{C}_N] & \longrightarrow & [\mathcal{R}, \mathcal{C}_M] \\ \downarrow & & \downarrow \\ [\mathcal{R}, \tau_{\leq j}\mathcal{C}_N] & \longrightarrow & [\mathcal{R}, \tau_{\leq j}\mathcal{C}_M] \end{array}$$

It follows that $[\mathcal{R}, \mathcal{C}_N] \to [\mathcal{R}, \mathcal{C}_M]$ is continuous for the profinite topology, and hence the restriction $X_N(a) \to X_M(a)$ is also continuous.

Step 4: $X_N(a)$ is nonempty. Let $N \geq 1$ be arbitrary. We prove that there exists an $a \in \mathbb{Z}$, independent of N, such that $[f] \in X_N(a)$, where $f = f_N$ is the composite morphism

$$f \colon \mathcal{R} \simeq \mathcal{R}_N \otimes^{\mathbf{L}}_{\mathcal{S}_N} \mathcal{S}^{\mathrm{ur}}_N \to \overline{R}_N \otimes^{\mathbf{L}}_{\overline{S}_N} \overline{S}^{\mathrm{ur}}_N \simeq \mathcal{C}_N,$$

where the two equivalences are chosen arbitrarily; their existence follows from Lemma 4.2.3 and ###$\overline{R}_N \otimes^{\mathbf{L}}_{\overline{S}_N} \overline{S}^{\mathrm{ur}}_N \simeq \overline{R}_N \otimes^{\mathbf{L}}_{\overline{S}^\circ_N} \mathcal{O}_N = \mathcal{C}_N$. Note that we do not prove any compatibility between f_N's for varying N.

For any $r \leq M \leq N$, the map $e_{N,M} \circ f \colon \mathcal{R} \to \mathcal{C}_M$ induces a commutative diagram with exact rows

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathrm{T}^0_{\mathcal{O}_r}(\mathcal{C}_M) & \longrightarrow & \mathrm{T}^0_{\mathcal{O}_r}(\overline{R}_M) \oplus \mathrm{T}^0_{\mathcal{O}_r}(\overline{S}^{\mathrm{ur}}_M) & \longrightarrow & \mathrm{T}^0_{\mathcal{O}_r}(\overline{S}_M) \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathrm{T}^0_{\mathcal{O}_r}(\mathcal{R}) & \longrightarrow & \mathrm{T}^0_{\mathcal{O}_r}(\mathcal{R}_N) \oplus \mathrm{T}^0_{\mathcal{O}_r}(\mathcal{S}^{\mathrm{ur}}_N) & \longrightarrow & \mathrm{T}^0_{\mathcal{O}_r}(\mathcal{S}_N). \end{array}$$

The two rightmost vertical maps are isomorphisms, since $\mathfrak{c}_M$ is generated by elements whose $\bmod p^r$ reductions lie in the square of the maximal ideal, whereby

$$\mathrm{T}^0_{\mathcal{O}_r}(\mathcal{R}_N) \cong \mathrm{T}^0_{\mathcal{O}_r}(R_N) \cong \mathrm{T}^0_{\mathcal{O}_r}(R_N/\mathfrak{c}_M) = \mathrm{T}^0_{\mathcal{O}_r}(\overline{R}_M)$$

and similarly for $\mathcal{S}_N$ and $\mathcal{S}^{\mathrm{ur}}_N$. It follows from the five lemma that the leftmost vertical map is also an isomorphism. This establishes one of the two sought properties of f (irrespective of a). To prove the second property, we consider the continuation of the diagram above:

$$\begin{array}{ccccccccc} \mathrm{T}^0_{\mathcal{O}_r}(\overline{S}_M) & \longrightarrow & \mathrm{T}^1_{\mathcal{O}_r}(\mathcal{C}_M) & \longrightarrow & \mathrm{T}^1_{\mathcal{O}_r}(\overline{R}_M) \oplus \mathrm{T}^1_{\mathcal{O}_r}(\overline{S}^{\mathrm{ur}}_M) & \longrightarrow & \mathrm{T}^1_{\mathcal{O}_r}(\overline{S}_M) \\ \downarrow{\scriptstyle \cong} & & \downarrow{\scriptstyle \alpha} & & \downarrow & & \downarrow \\ \mathrm{T}^0_{\mathcal{O}_r}(\mathcal{S}_N) & \xrightarrow{\beta} & \mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R}) & \longrightarrow & \mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R}_N) \oplus \mathrm{T}^1_{\mathcal{O}_r}(\mathcal{S}^{\mathrm{ur}}_N) & \xrightarrow{\gamma} & \mathrm{T}^1_{\mathcal{O}_r}(\mathcal{S}_N). \end{array}$$

The claim to be proven is that $\varpi^a \operatorname{coker}(\alpha) = 0$ for an $a \geq 0$ which is independent of r, M and N. By the diagram, we have

$$\operatorname{coker}(\alpha) \twoheadleftarrow \operatorname{coker}(\beta) \cong \ker(\gamma),$$

so it suffices to prove that $\ker(\gamma)$ is annihilated by a power of ϖ which is independent of r, M, N. We recall that $\mathrm{T}^1_{\mathcal{O}_r}(\mathcal{S}^{\mathrm{ur}}_N) = 0$ by ###**ref here**, and that

$$\ker\left(\mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R}_N) \to \mathrm{T}^1_{\mathcal{O}_r}(\mathcal{S}_N)\right)$$

Has finite length bounded independently of N by Proposition 4.3.2. Thus, $\operatorname{coker}(\alpha)$ is annihilated by a power of ϖ which is independent of N. Thus, we have shown that there exists an absolute constant a such that $[f] \in X_N(a)$, and in particular, $X_N(a) \neq \emptyset$. □

Proof of Theorem 4.4.1. By Lemma 4.4.3 and Theorem 4.4.4, for $a \in \mathbb{Z}$ sufficiently large, there exists $[g] \in \varprojlim_N X_N(a)$ represented by a compatible sequence (g_N) of morphisms

$$g_N \colon \mathcal{R} \to \mathcal{C}_N$$

such that $\mathrm{T}^0_{\mathcal{O}_r}(g_N) \colon \mathrm{T}^0_{\mathcal{O}_r}(\mathcal{C}_N) \to \mathrm{T}^0_{\mathcal{O}_r}(\mathcal{R})$ is an isomorphism and the cokernel

$$Q_{r,N}(g) = \operatorname{coker}\left(\mathrm{T}^1_{\mathcal{O}_r}(\mathcal{C}_N) \to \mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R})\right)$$

is ϖ^a-torsion. We will prove that the morphism

$$(\mathcal{R} \otimes_{\mathcal{O}} E)^\wedge_\xi \to (\mathcal{C}_\infty \otimes_{\mathcal{O}} E)^\wedge_\xi$$

induced by g is a weak equivalence, by showing that it is an isomorphism on T^0_E and an epimorphism on T^1_E, then counting dimensions. The first statement follows from the fact that g_N induces isomorphisms on T^0. To prove the second statement, we note that ϖ^a-torsion modules are preserved under

filtered colimits and inverse limits, so that

$$\varpi^a Q_{r,N}(g) = 0 \implies \left(\varprojlim_r \operatorname*{colim}_N Q_{r,N}(g)\right) \otimes_{\mathcal{O}} E = 0.$$

Now, $\mathcal{C}_\infty$ is noetherian by Lemma 4.3.1(2), and thus $\mathrm{T}^1_{\mathcal{O}_r}(\mathcal{C}_\infty)$ is finite by Proposition 3.6.4. It follows from Proposition 3.7.6 that we have an isomorphism

$$\begin{aligned}\left(\varprojlim_r \operatorname*{colim}_N Q_{r,N}(g)\right) \otimes_{\mathcal{O}} E &\cong \operatorname{coker}\left(\left(\varprojlim_r \operatorname*{colim}_N \mathrm{T}^1_{\mathcal{O}_r}(\mathcal{C}_N)\right) \otimes_{\mathcal{O}} E \to \left(\varprojlim_r \mathrm{T}^1_{\mathcal{O}_r}(\mathcal{R})\right) \otimes_{\mathcal{O}} E\right) \\ &\cong \operatorname{coker}\left(\mathrm{T}^1_E\left((\mathcal{C}_\infty \otimes_{\mathcal{O}} E)^\wedge_\xi\right) \to \mathrm{T}^1_E\left((\mathcal{R} \otimes_{\mathcal{O}} E)^\wedge_\xi\right)\right)\end{aligned}$$

since colimits commute and $\varprojlim$ is exact on inverse systems of finite abelian groups. Thus, this cokernel vanishes and we have proved that the map

$$\mathrm{T}^*_E\left((\mathcal{C}_\infty \otimes_{\mathcal{O}} E)^\wedge_\xi\right) \to \mathrm{T}^*_E\left((\mathcal{R} \otimes_{\mathcal{O}} E)^\wedge_\xi\right)$$

induced by g is an isomorphism in degree 0 and an epimorphism in degree 1. Both are graded vector spaces concentrated in degrees $[0,1]$ and have the same dimensions by Lemma 4.3.1 and Proposition 3.7.8. It follows that $(\mathcal{R} \otimes_{\mathcal{O}} E)^\wedge_\xi \to (\mathcal{C}_\infty \otimes_{\mathcal{O}} E)^\wedge_\xi$ is a weak equivalence. □

We now turn to the problem of identifying the ring which represents the functor $(\mathcal{F}_\infty)_{(\xi)}$.

Lemma 4.4.5. *With $\mathcal{F}_\infty = \operatorname{colim}_N \operatorname{Map}_{/k}(\mathcal{C}_N, -)$ as above, the functor $(\mathcal{F}_\infty)_{(\xi)} \colon \mathcal{A}rt_E \to \mathcal{A}ni$ is represented by*

$$\widehat{R}_{\infty,\mathfrak{p}} \otimes^{\mathbf{L}}_{\widehat{S}^\circ_{\infty,\mathfrak{a}}} E \in \mathcal{CNL}_E.$$

Proof. We begin by proving that the natural morphism

$$\operatorname*{colim}_N \operatorname{Map}_{/k}(\overline{R}_N, -) \to \operatorname{Map}_{/k}(R_\infty, -)$$

becomes an equivalence upon passing to characteristic 0. We have a fibre sequence

$$\mathrm{T}^\bullet_{\mathcal{O}_r}(\mathcal{F}_N) \to \mathrm{T}^\bullet_{\mathcal{O}_r}(\overline{R}_N) \oplus \mathrm{T}^\bullet_{\mathcal{O}_r}(\mathcal{O}_N) \to \mathrm{T}^\bullet_{\mathcal{O}_r}(\overline{S}^\circ_N) \xrightarrow{+1},$$

to which we apply $(\lim_r \operatorname{colim}_N -) \otimes_{\mathcal{O}} E$ to obtain an exact sequence of cohomology groups:

$$0 \longrightarrow \mathrm{T}^0_E\left((\mathcal{F}_\infty)_{(\xi)}\right) \longrightarrow \left(\varprojlim_r \operatorname*{colim}_{N \geq r} \mathrm{T}^0_{\mathcal{O}_r}(\overline{R}_N)\right) \otimes_{\mathcal{O}} E \longrightarrow \mathrm{T}^0_E(\widehat{S}^\circ_{\infty,\mathfrak{a}})$$

$$\longrightarrow \mathrm{T}^1_E\left((\mathcal{F}_\infty)_{(\xi)}\right) \longrightarrow \left(\varprojlim_r \operatorname*{colim}_{N \geq r} \mathrm{T}^1_{\mathcal{O}_r}(\overline{R}_N)\right) \otimes_{\mathcal{O}} E \longrightarrow 0.$$

Here, we have used that the topology on S°_∞ defined by $(\mathfrak{a}_N)$ is profinite, so that

$$\operatorname*{colim}_N \operatorname{Map}_{/k}(\overline{S}^\circ_N, -) \simeq \operatorname{Map}_{/k}(S^\circ_\infty, -).$$

From Theorem 4.4.1 and Corollary 3.7.9, we know that the $T^i_E\big((\mathcal{F}_\infty)_{(\xi)}\big)$-terms have dimensions 0 and l_0, respectively. Now, since $R_\infty \twoheadrightarrow \overline{R}_N$ for every N, the induced map $\mathrm{T}^0_{\mathcal{O}_r}(\overline{R}_N) \to \mathrm{T}^0_{\mathcal{O}_r}(R_\infty)$ is injective, and hence the induced map

$$\left(\varprojlim_r \operatorname*{colim}_{N \geq r} \mathrm{T}^0_{\mathcal{O}_r}(\overline{R}_N)\right) \otimes_{\mathcal{O}} E \to \mathrm{T}^0_E(\widehat{R}_{\infty,\mathfrak{p}})$$

is injective. Writing $t^i = \dim_E \left(\lim \operatorname{colim} \mathrm{T}^0_{\mathcal{O}_r}(\overline{R}_N)\right) \otimes_{\mathcal{O}} E$, the exact sequence above implies

$$0 = t^0 - \dim_E \mathrm{T}^0_E(\widehat{S}^\circ_{\infty,\mathfrak{a}}) + l_0 - t^1 \leq \dim_E \mathrm{T}^0_E(\widehat{R}_{\infty,\mathfrak{p}}) - \dim_E \mathrm{T}^0_E(\widehat{S}^\circ_{\infty,\mathfrak{a}}) + l_0 - t^1 \leq -t^1 \leq 0,$$

and thus $t^1 = 0$ and

$$\left(\lim_r \operatorname*{colim}_{N \geq r} \mathrm{T}^\bullet_{\mathcal{O}_r}(\overline{R}_N)\right) \otimes_{\mathcal{O}} E \to \mathrm{T}^\bullet_E(\widehat{R}_{\infty,\mathfrak{p}})$$

is an equivalence. This proves that we have an equivalence of functors

$$\operatorname{Map}_{/k}(R_\infty, -)_{(\xi)} \simeq \left(\operatorname{colim}_N \operatorname{Map}_{/k}(\overline{R}_N, -)\right)_{(\xi)}.$$

Now, for every r, N with $r \leq N$, we have a natural morphism of tangent complexes

$$\mathrm{T}^\bullet_{\mathcal{O}_r}(\mathcal{C}_N) = \mathrm{T}^\bullet_{\mathcal{O}_r}(\mathcal{F}_N) \to \mathrm{T}^\bullet_{\mathcal{O}_r}(R_\infty \otimes^{\mathbf{L}}_{S^\circ_\infty} \mathcal{O}),$$

induced by the map $R_\infty \otimes^{\mathbf{L}}_{S^\circ_\infty} \mathcal{O} \to \overline{R}_N \otimes^{\mathbf{L}}_{\overline{S}^\circ_N} \mathcal{O}_N = \mathcal{C}_N$. It follows from the claim proven above (and the analogous claim for $S^\circ_\infty = \varprojlim \overline{S}^\circ_N$), that $(\mathcal{F}_\infty)_{(\xi)}$ is represented by

$$(R_\infty \otimes^{\mathbf{L}}_{S^\circ_\infty} \mathcal{O} \otimes_{\mathcal{O}} E)^\wedge_\xi \simeq \widehat{R}_{\infty,\mathfrak{p}} \otimes^{\mathbf{L}}_{\widehat{S}^\circ_{\infty,\mathfrak{a}}} E,$$

where the equivalence follows from Lemma 3.7.4 and Theorem 3.7.3. □

We are now in a position to deduce our main result.

Proof of Theorem 4.0.1. By Theorem 4.2.4, there exists a free $\widehat{R}_{\infty,\mathfrak{p}}$-module $\widehat{M}_{\infty,\mathfrak{p}}$ such that

$$\widehat{M}_{\infty,\mathfrak{p}} \otimes^{\mathbf{L}}_{\widehat{S}^\circ_{\infty,\mathfrak{a}}} E \simeq R\Gamma(K, \mathcal{V}_{\lambda,\tau})^\vee_{\mathfrak{p}}.$$

Theorem 4.4.1 together with Lemma 4.4.5 serve to define an action of $(\mathcal{R} \otimes^{\mathbf{L}}_{\mathcal{O}} E)^\wedge_\xi \simeq \mathcal{R}^{\mathrm{ss}}_\rho$ on this complex, and in particular, using Corollary 3.7.10, we obtain a free action

$$\bigwedge H^1_g(\Gamma_{F,S}, \operatorname{ad}\rho(1)) \cong \pi_*\left(\mathcal{R}^{\mathrm{ss}}_\rho\right) \circlearrowright H_*(K, \mathcal{V}_{\lambda,\tau})_\Pi.$$

□

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